AER 1515 – Perception for Robotics

Lecture 1 - Introduction

Prof. Steven Waslander



Institute for Aerospace Studies
UNIVERSITY OF TORONTO

Syllabus – Contact Info

- Contact Info:
  - Prof. Steven Waslander
    - Office: UTIAS - 184
    - steven.Waslander@robotics.utias.utoronto.ca
  - Can Xu
    - Assignments
    - Can.xu@robotics.utias.utoronto.ca



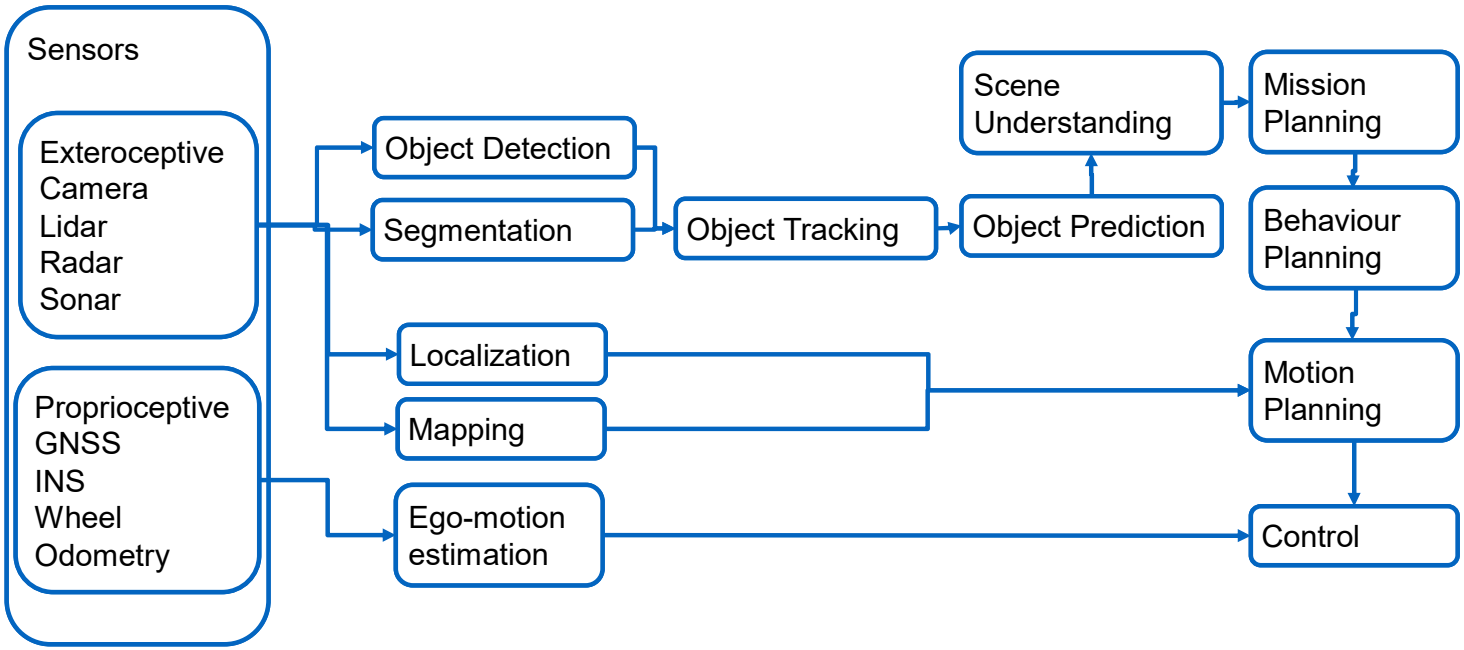
What is Perception for Robotics?

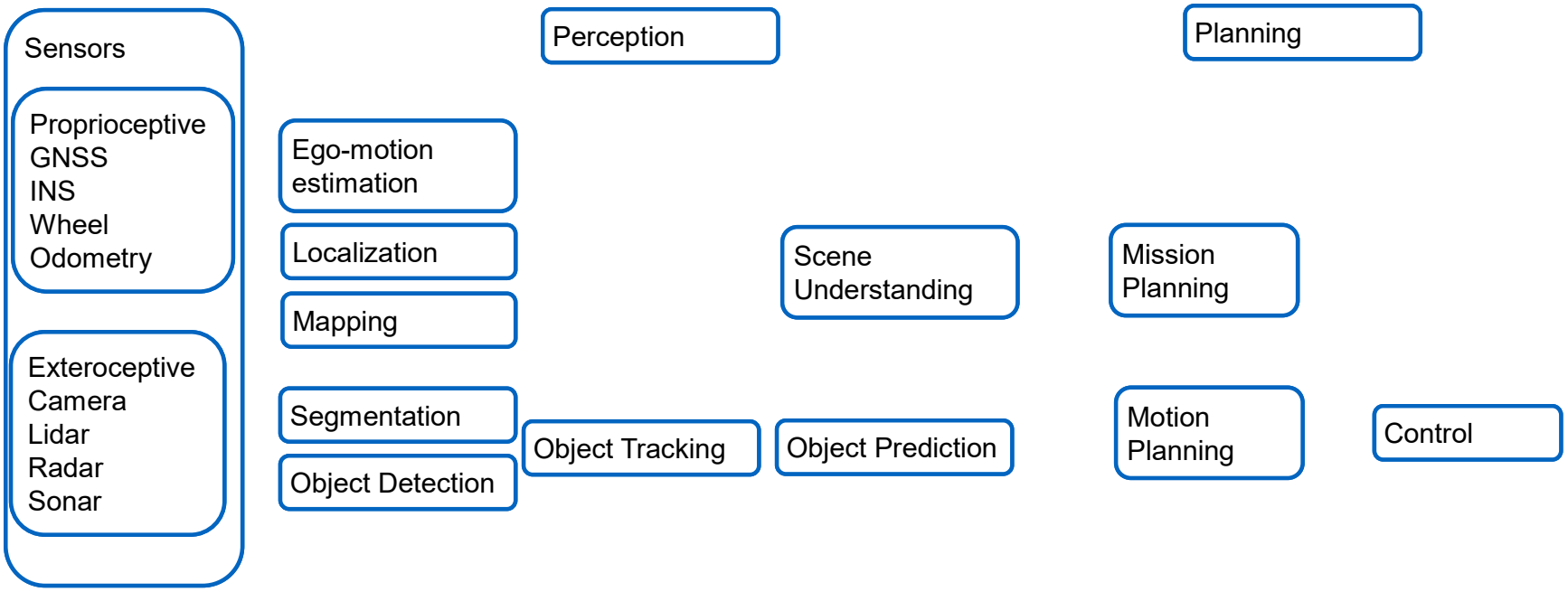


Magic

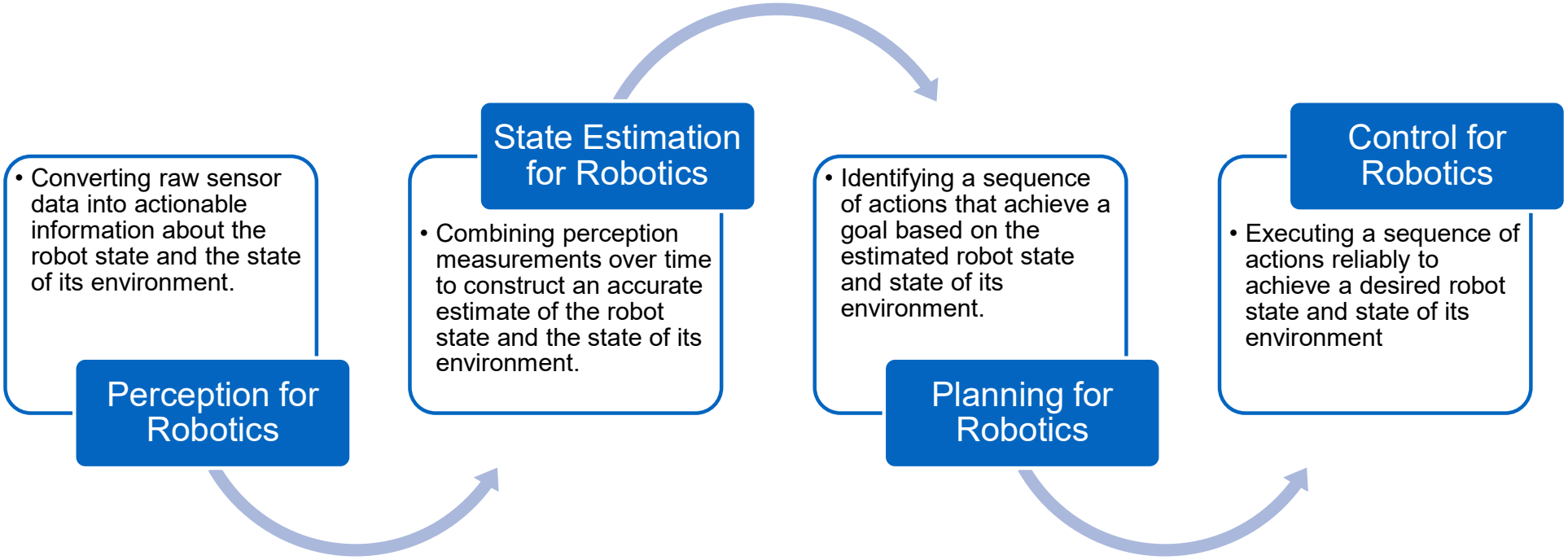


Modular Robotic Perception Architecture



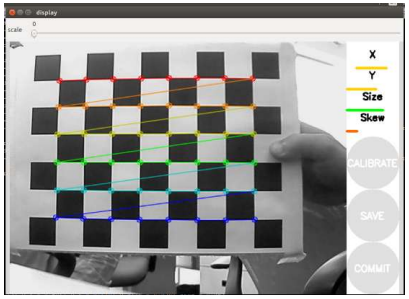

Sensors
Proprioceptive
GNSS
INS
Wheel
Odometry
Exteroceptive
Camera
Lidar
Radar
Sonar
Perception
Ego-motion estimation
Localization
Mapping
Segmentation
Object Detection
Object Tracking
Scene Understanding
Object Prediction
Planning
Mission Planning
Motion Planning
Control

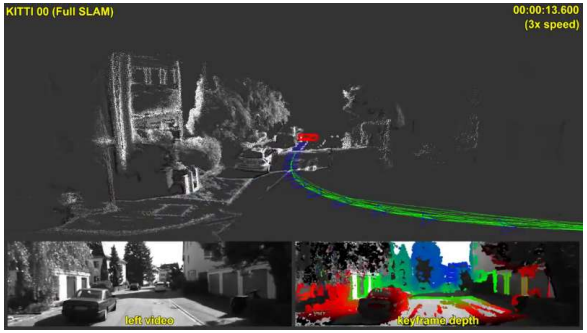
Course Sequence Overview

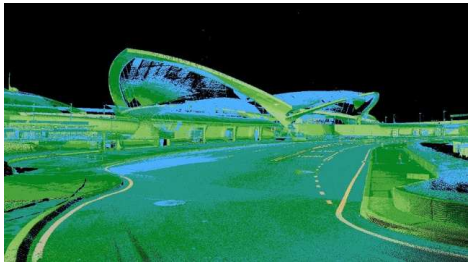


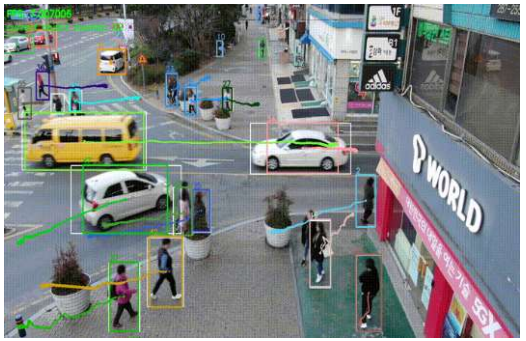
Perception’s Biggest Consumers

- Localization, Ego Motion Estimation
- Mapping, Reconstruction, Photogrammetry
- Planning & Control
- Calibration
- Safety Assurance



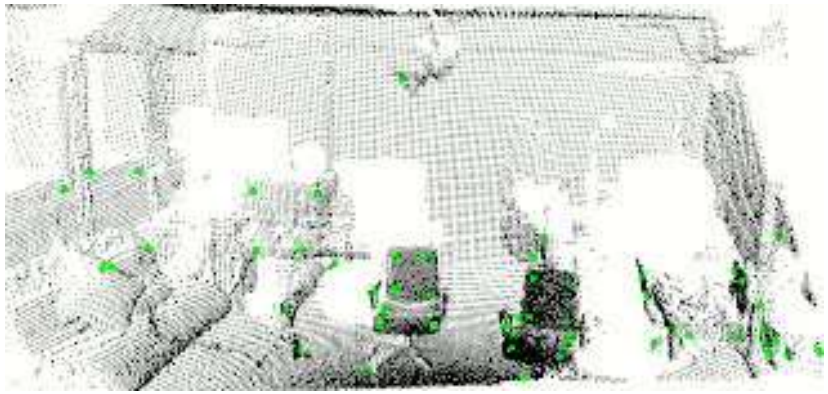




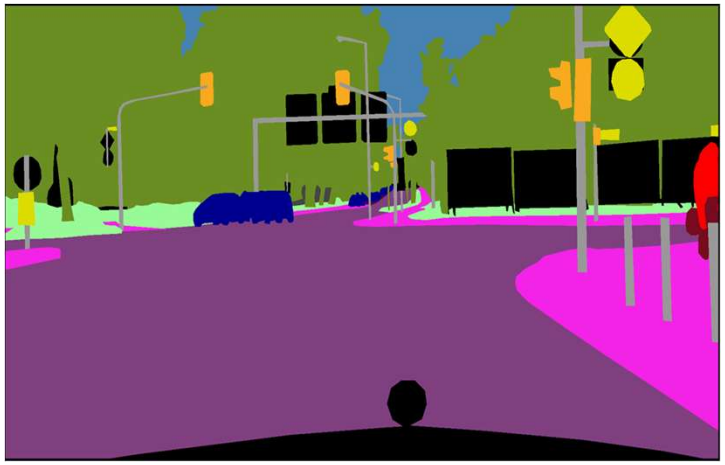


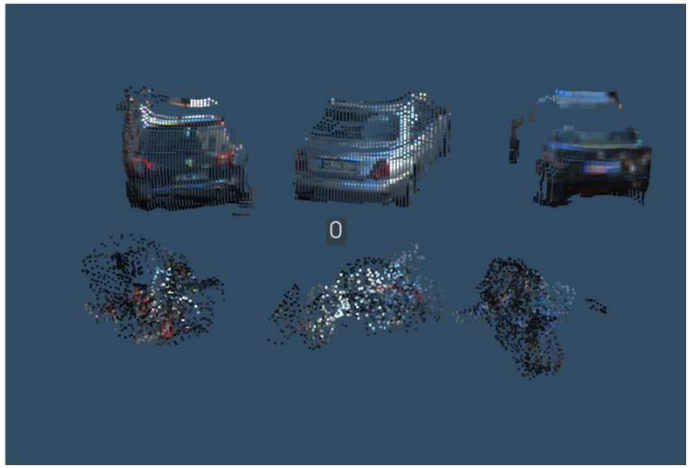


From humble points to glorious objects









Course Content

Lecture	Date	Title
Module 1: Foundations		
1	Sept. 10	Introduction
2	Sept. 17	Probability, Information and Optimization
3	Sept. 24	Machine Learning
-	Oct. 1	No lecture, Prof. Waslander Traveling
4	Oct. 8	Sensor Modeling
5	Oct. 15	Vision Sensor Modeling
Module 2: Static World		
6	Oct. 22	3D Point Clouds Registration and 3D Features
7	Oct. 29	Features – Detection, Description, Matching and Tracking
8	Nov. 5	Pose, Depth, Feature SLAM
9	Nov. 12	Direct Methods, Deep Methods and Place Recognition
Module 3: Dynamic World		
10	Nov. 19	2D Classification, Detection and Segmentation
11	Nov. 26	3D Detection and Segmentation
12	Dec. 3	3D Tracking and Prediction
13	Dec. 7	The Open World

Syllabus - Lectures

- Lectures:
  - Main Lecture - Thursdays 9:10 PM – 11:00PM, UTIAS Lecture Hall and Zoom

- Communication:
  - Announcements through Quercus
  - Slides, videos, code, resources posted to Quercus throughout semester
    - All corrections noted with revision number
  - Assignments posted on scheduled dates
    - Due 2-3 weeks later

Syllabus - Prerequisites

- Prerequisites
  - Linear Algebra
  - Probability Theory
  - Optimization theory
  - Python or Matlab

- Useful skills
  - Deep Learning
  - State Estimation theory
  - Statistics

Syllabus - Resources

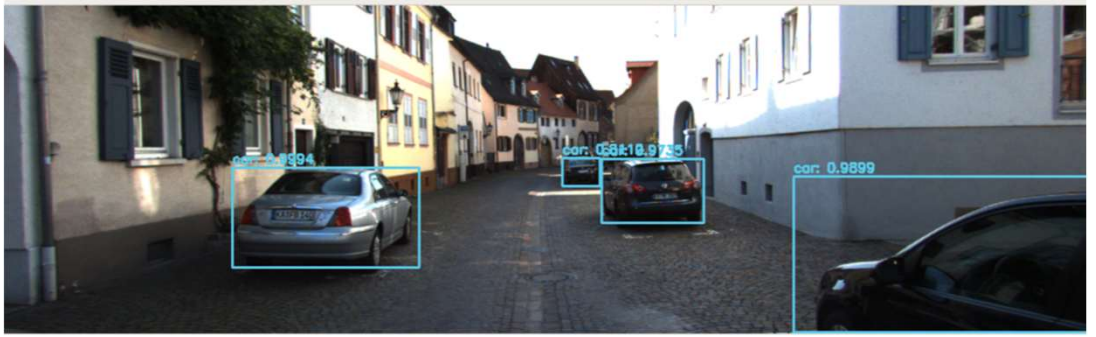
- Resources
  - Lecture Slides
    - Updated versions posted after lectures on Quercus

  - Recommended Texts (none required)
    - K. Murphy, **Probabilistic Machine Learning: An Introduction, MIT Press, 2022.**
    - S. Thrun, W. Burgard, and D. Fox. **Probabilistic Robotics**. MIT Press, 2005. 3rd Ed.
    - R. Szeliski, **Computer Vision: Algorithms and Applications**. 2nd. Ed., Springer-Nature., 2023.
    - Also **Multi-view Geometry**, Hartley and Zisserman, **Deep Learning,** Courville, Bengio, Goodfellow.

Syllabus – Assignments

- Assignments (Three total)
  - Three dataset exercises of different perception algorithms
    - Convnet Classification, Stereo Matching and Point Cloud Registration, 2D Object Detection and 3D Segmentation
    - Helpful to have good computational resources for final assignment (GPU!)
    - Independent write-up required, ok to work together in developing solution
      - Submit as pdf, more instruction to follow as part of assignment handout.
      - Do not include learning model in submissions!





Syllabus – Assignments

- Assignments (Four total)
  - One supplemental topic presentation – Paper review
    - Must be from 2019-2020 (ICRA, IROS, RSS, CoRL, CVPR, ICCV, ECCV, IJRR, JFR, T-RO)
    - 10 minute presentation in third hour of lecture
    - Select a slot based on topic of the day (try to match)
    - Sign up on quercus before Friday at midnight!
      - Name, Paper title
  - Independent work required

Presentation Rubric

- Paper description (20%): What is the core problem or method addressed and why is it challenging/interesting?

- Quality of Technical Content (50%): Are the concepts employed clearly understood, explained and applied correctly, are derivations correct, reasoning valid, content of use and well thought out? Is there an interesting example that demonstrates or explores the performance of the method.

- Presentation Quality (30%): Was the presentation clear, concise, did it follow a logical structure, could the presenter answer questions?

Example Presentation – PointPillars (CVPR 2019)

Point Pillars: Fast Encoders for Object Detection from Point Clouds

Presenter: Keenan Burnett
Course Instructor: Prof. Steven Waslander
2019/11/26

Syllabus - Projects

- Team Size: Groups of one or two
- Project must be related to one of the main topics of the course
  - From your own research
  - From a Kaggle competition or established benchmark
  - The application of existing methods to a new problem or dataset
  - An extensive literature review on a topic with some evaluation of methods

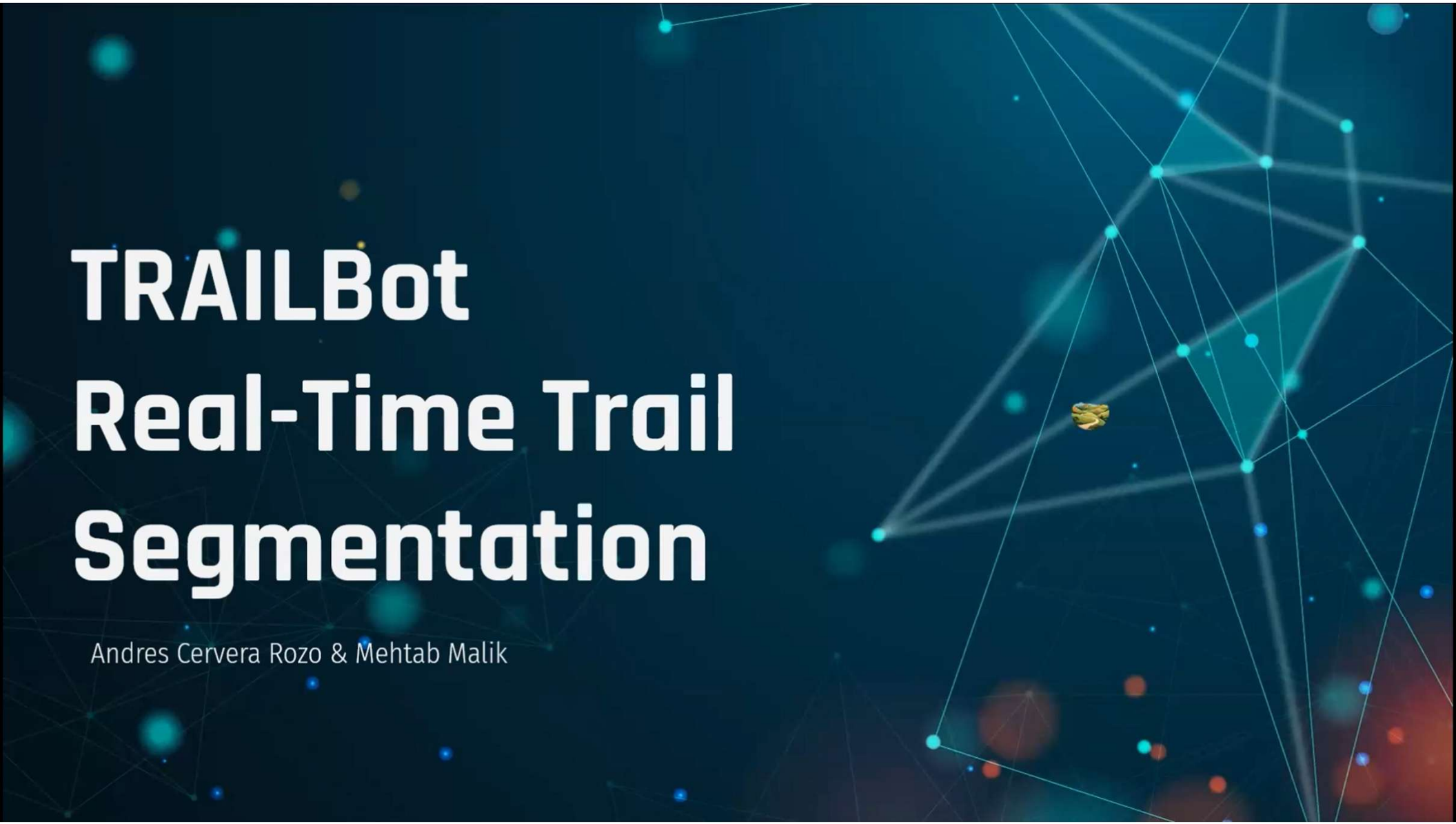
- Four deliverables: Proposal, Initial results, 5 – minute Video and 6 page IEEE 2 column format Report
- Must have some implemented and tested component to verify claims.
  - Not just literature review, or running of existing libraries
  - Can be existing code on a different dataset, or modified code on existing data
  - Some measure of novelty is important – aim for publication, most won't make it

Rubrics

- Video
  - Project Motivation/Applications (20%):
    - Is there a believable application for the work, does it relate to the course content, is there a need that is fullled by the project?
  - Quality of Technical Content (30%):
    - Are the concepts employed clearly understood, explained and applied correctly, is the work an extension of the theory presented in class, are derivations correct, reasoning valid, content of use and well thought out?
  - Overall Completion of Objectives (30%):
    - Did the team set reasonable goals and achieve those goals this semester, was sucient effort expended to achieve those goals?
  - Presentation Quality (20%):
    - Was the video clear, concise, did it document progress well, could the presenter answer questions?
- Report
  - Introduction (20%)
    - Motivation / Applications
    - Related Work
  - Quality of Technical Content (50%)
    - Problem Formulation
    - Algorithm Denition
    - Extensions to Course Knowledge
  - Completion of Objectives (20%)
    - Functional Simulations
    - Clear Presentation of Results
  - Report Quality (10%)
    - Formatting, spelling, grammar
    - Images, labeling of axes

TRAILBot Real-Time Trail Segmentation

Andres Cervera Rozo & Mehtab Malik



Final Report in IEEE Two-column format

BeVEL: Best View Estimation Using Learning

Archie Lee
Institute for Aerospace Studies
University of Toronto
Toronto, Canada
archie.lee@mail.utoronto.ca

***Abstract*—Cameras are heavily used in robotics for visual localization tasks, often within feature-based maps. The uncertainty on the final pose estimate partially depends on the information available from the robot's environment. Given the ability to determine the information available in different viewpoints, it is desirable to choose the viewpoint that will minimize the final pose estimate's uncertainty. This can often be time-consuming and computationally intensive. This project aims to examine the applicability of PointNet++-based neural network architectures in the best view estimation problem and whether a learned method reduces computation time. We compare our results against an expert performing exhaustive search within a set of simulated environments. Our neural network is able to achieve accurate results while performing best view estimation in a fraction of the time, making it useful for real-time motion planning problems.**

I. Introduction

Cameras are a common type of sensor used in many robotic applications. While the downside of cameras includes their

A potential solution to this problem is finding the viewpoint minimizes the robot's position uncertainty through feature uncertainty propagation. In fact, this problem can be formulated as an optimization problem subject to perception and motion constraints. However, the optimization problem is nonlinear and computationally complex. Additionally, searching a continuous space is computationally intensive. One can reduce the dimensionality of the problem by discretizing the search space, but this suffers from accuracy degradation for large discretizations. In this situation, applying machine learning techniques may be to reduce the computational cost at inference time and make the solution computationally tractable.

This project examines the performance of a neural network in inferring the best view at a given camera location in a map. Using the PointNet++ architecture as a foundation to process 3D point clouds made of visual features, we aim to classify the sector containing the best view angle. A dataset is collected using an expert policy which exhaustively searches for the best

Syllabus

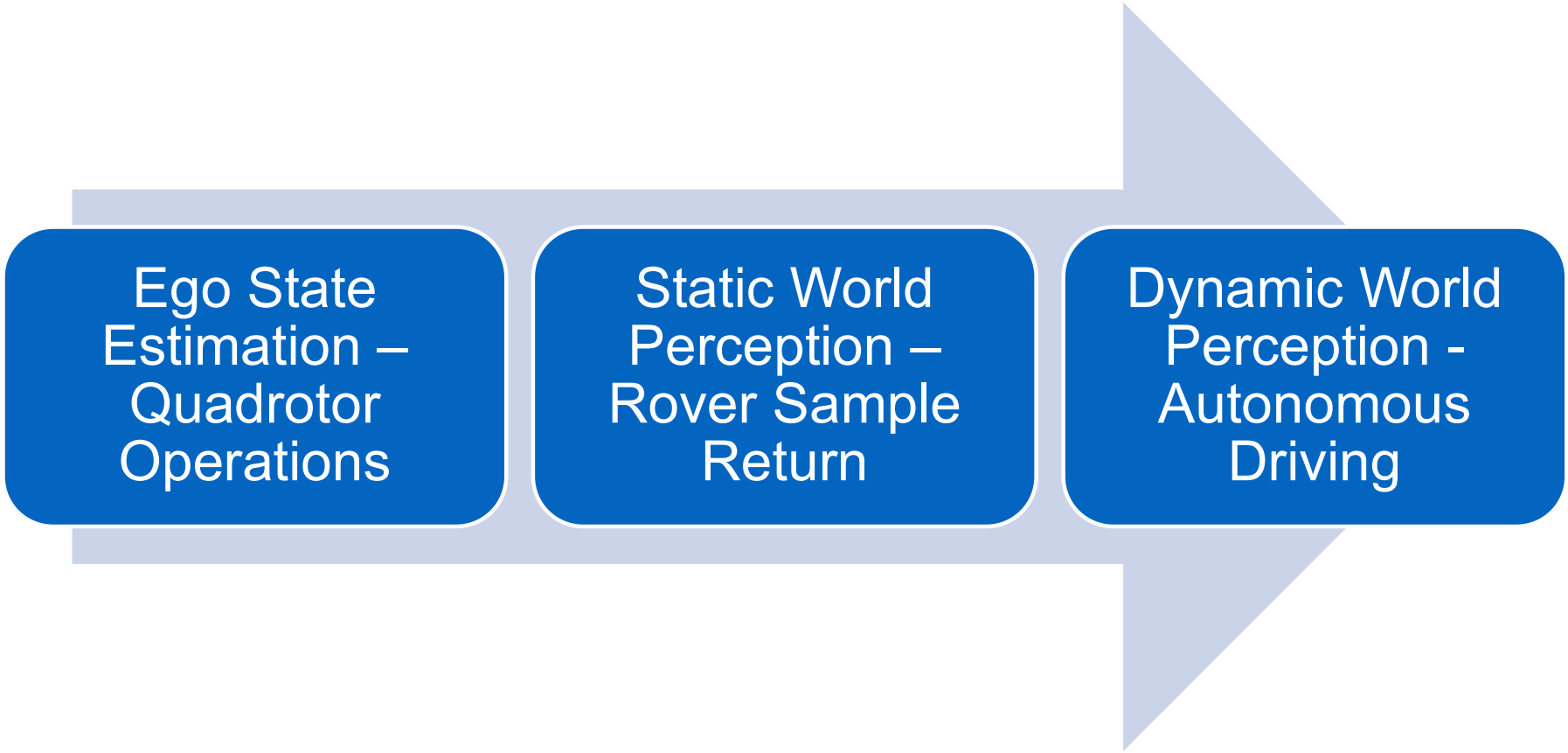
- AER 1515 Grading:
  - 40% Assignments – 13.333% per assignment
  - 60% Project –
    - 5% Initial Results
    - 20% Video
    - 35% Report

- Sitting in on lectures
  - No requirements, enjoy, ask questions!
  - Just ask me for access to Quercus materials

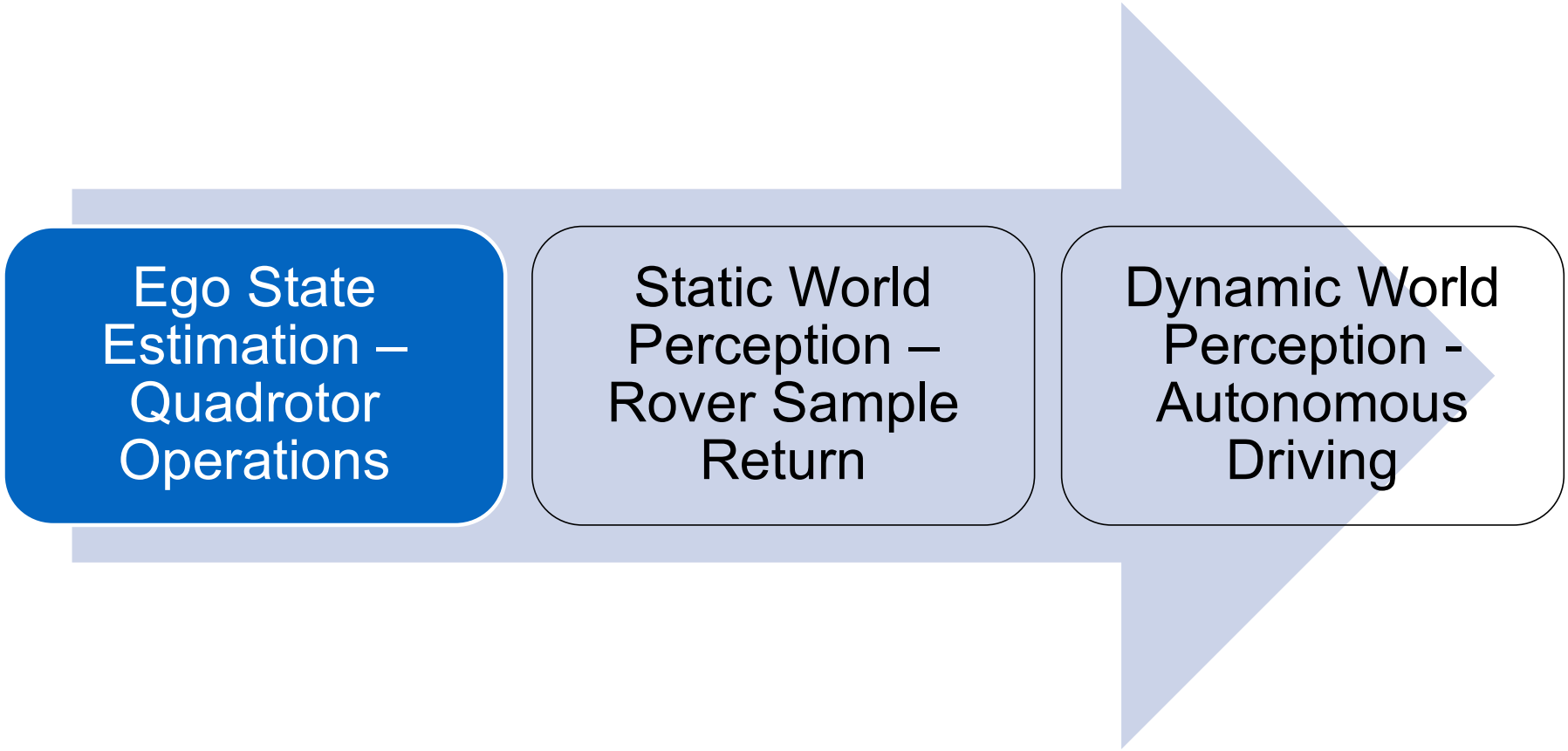
Class Survey

- How many of you have…
  - Used Matlab?
  - Coded in Python?
  - Used OpenCV?
  - Used Point Cloud Library
  - Used Pytorch or TensorFlow?
  - Worked with GNSS, INS or wheel odometry sensor data?
  - Worked with vision, lidar, radar or sonar sensor data?
  - Studied probability theory? Information theory? Optimization theory?
  - Defined a basic camera model?
  - Seen the Iterated Closest Point algorithm?
  - Corresponded features between images?
  - Performed object detection or semantic segmentation on image data? Lidar data?

A few stories from my work



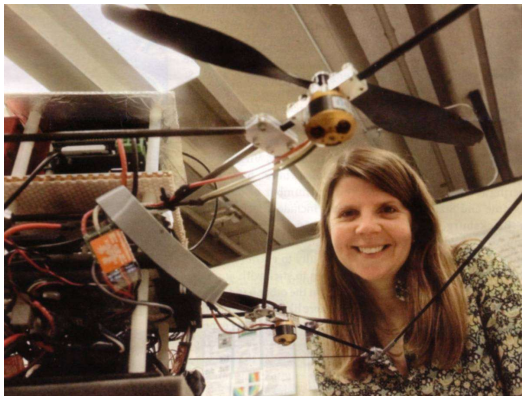
A few stories from my work



STARMAC Team 03-08



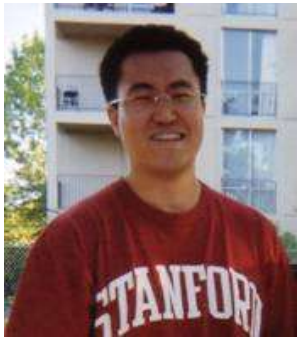
Gabe Hoffmann, Steven Waslander, Mike Vitus, Haomiao Huang, Vijay Pradeep



Prof. Claire Tomlin



Dev Rajnarayan



Jung Soon Jang

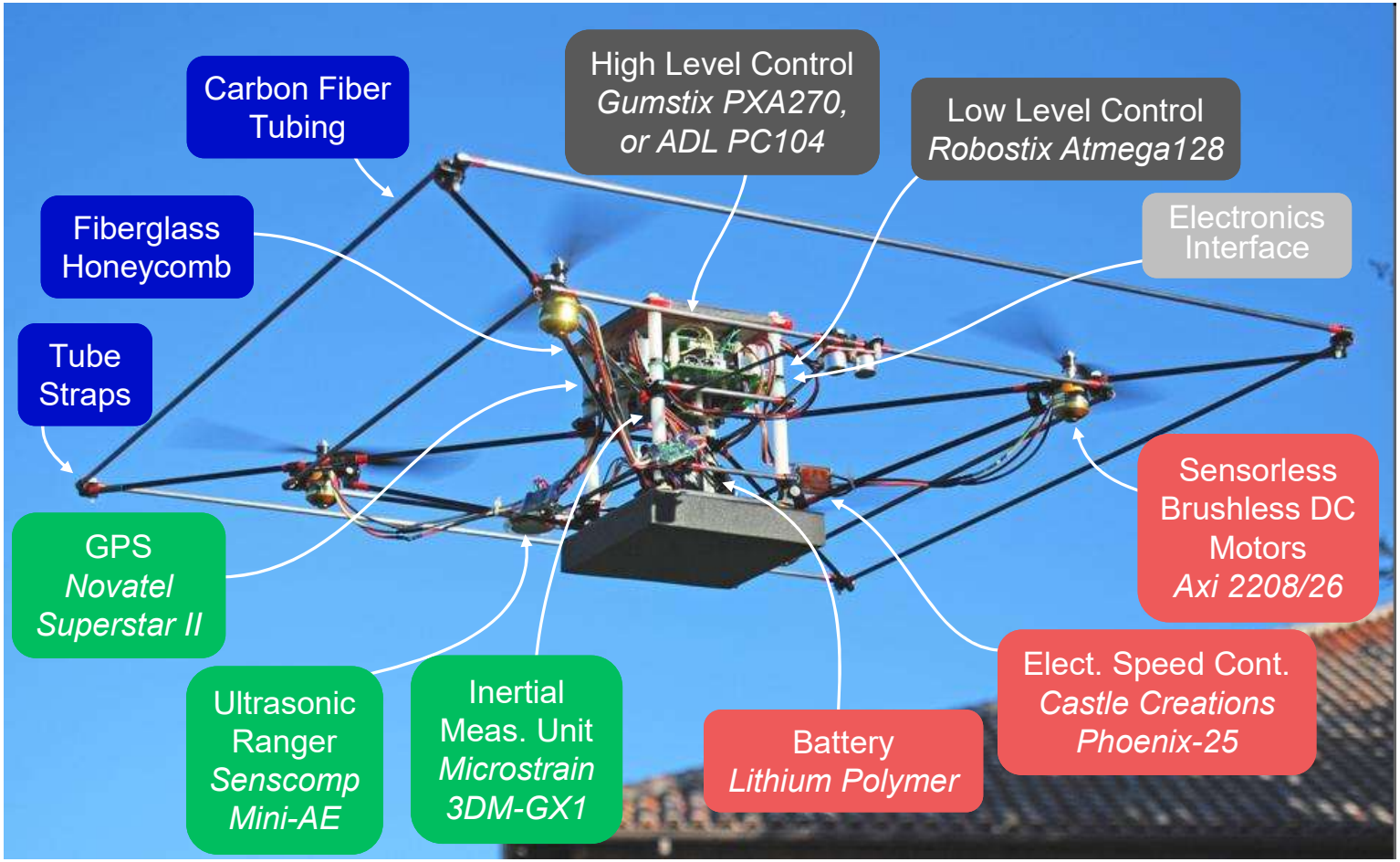


Dave Dostal



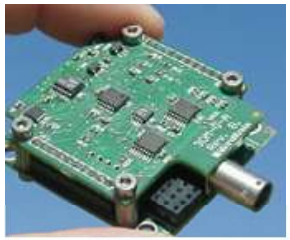
Jeremy Gillula

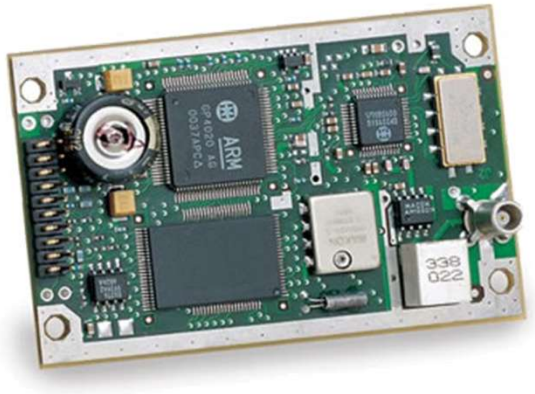
STARMAC V3 (2006 - 2009) - Vehicle



Main Sensors

- IMU
  - Microstrain 3DM-GX1
  - Attitude, attitude rate, acceleration @ 76 Hz
- SODAR
  - Senscomp Mini AE
  - Up to 50 Hz update
  - Up to 40 ft range
- GPS
  - Novatel Superstar II
  - Carrier Phase Position & velocity @10 Hz



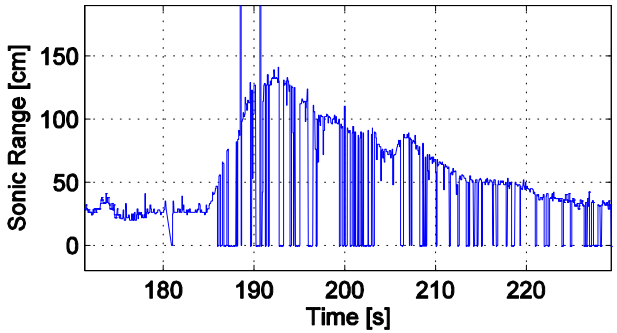


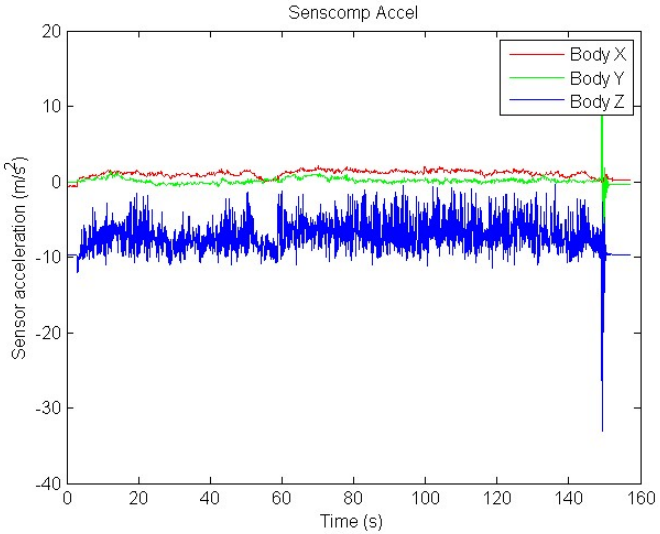


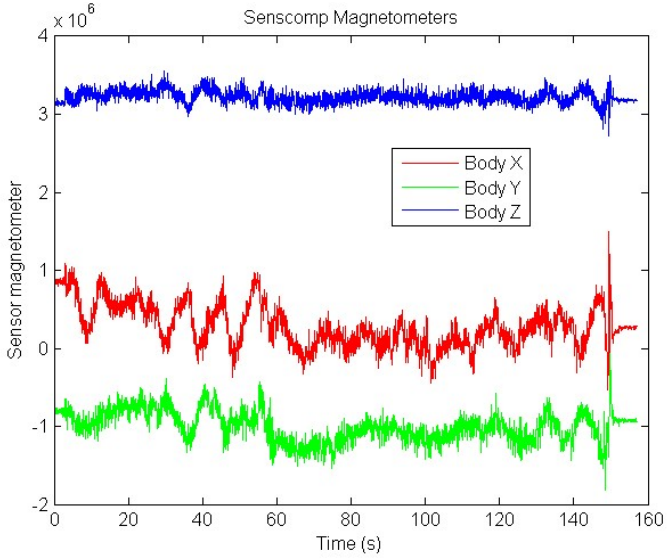


Sensor Data

- Aeryon Scout quadrotor in hover flight
  - Accelerometer Data
    - Low pass hardware filter added to raw sensor output
    - High vibration environment, particularly in z axis
  - Magnetometer Data
    - Relatively immune to vibration effects
    - Oscillation on similar time scale to vehicle motion
    - Large development effort to account for PCB current
  - Sonar Data (Exteroceptive)
    - Large reception cone leads to good performance over flat plane
    - Dropouts, false echoes quite common in cheap models





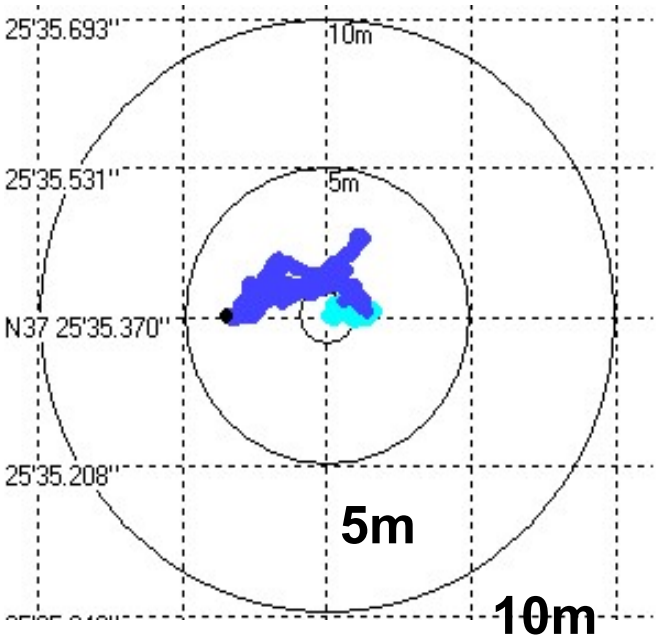


Sensor Data - GNSS

Coded Phase Solution: 25 Minutes

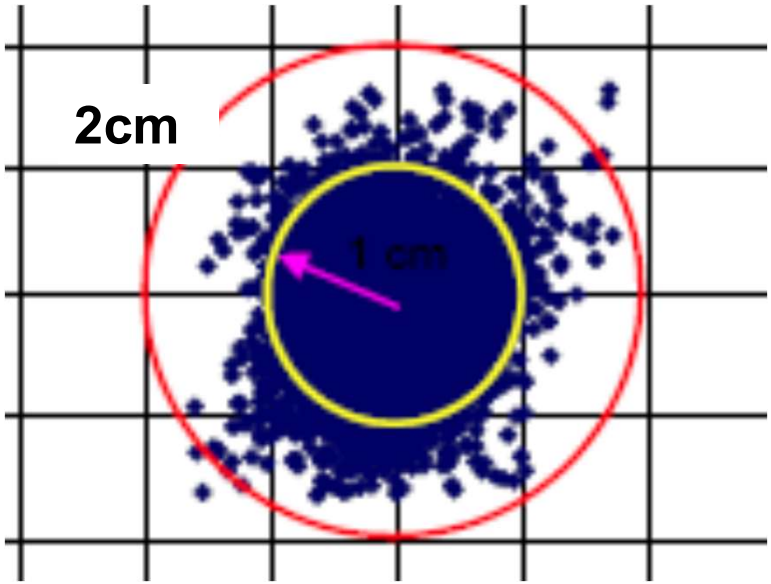
Standard (3-5m CEP)

With DGPS corrections (WAAS, PPP) (0.5 - 1.5m CEP)

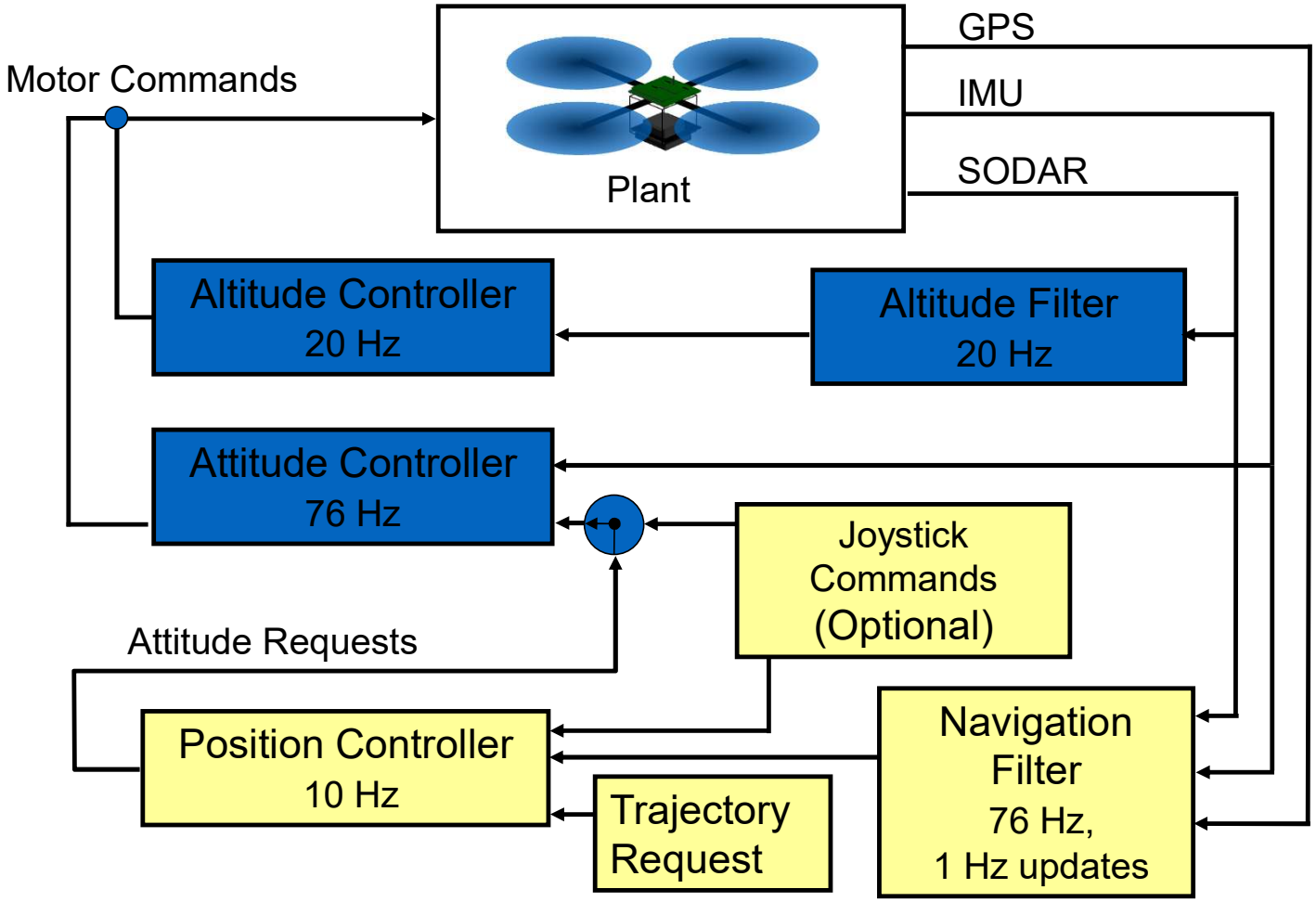


Carrier Phase Solution: 24 Hours

Double Difference Technique (1-2cm CEP)



Estimation and Control Structure



Estimation & Measurement model

- Altitude filter
  - Outlier rejection
  - Kalman Filter

- Standard EKF used for Navigation Filter
  - GPS – inertial position, velocity
  - Sodar, Barometer – inertial altitude
  - Gyros – body angular rates + biases
  - Magnetometers – body orientation vector + current calibration
  - Accelerometers – body accelerations – gravity

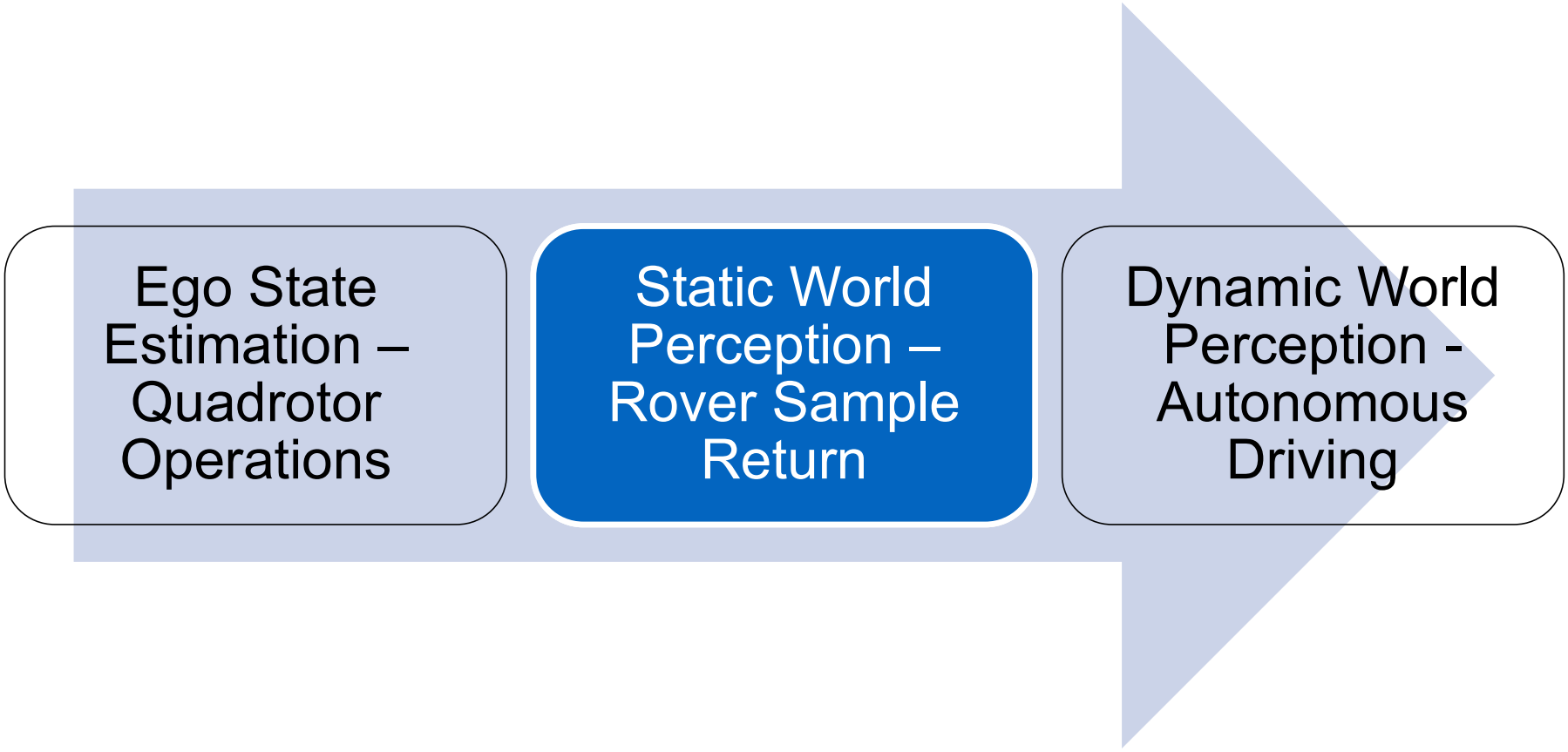
STARMAC in flight



Ego State Perception

- Others Cool vids
  - Dancing quads
  - Robot racing cars
  - Space X landing

A few stories from my work

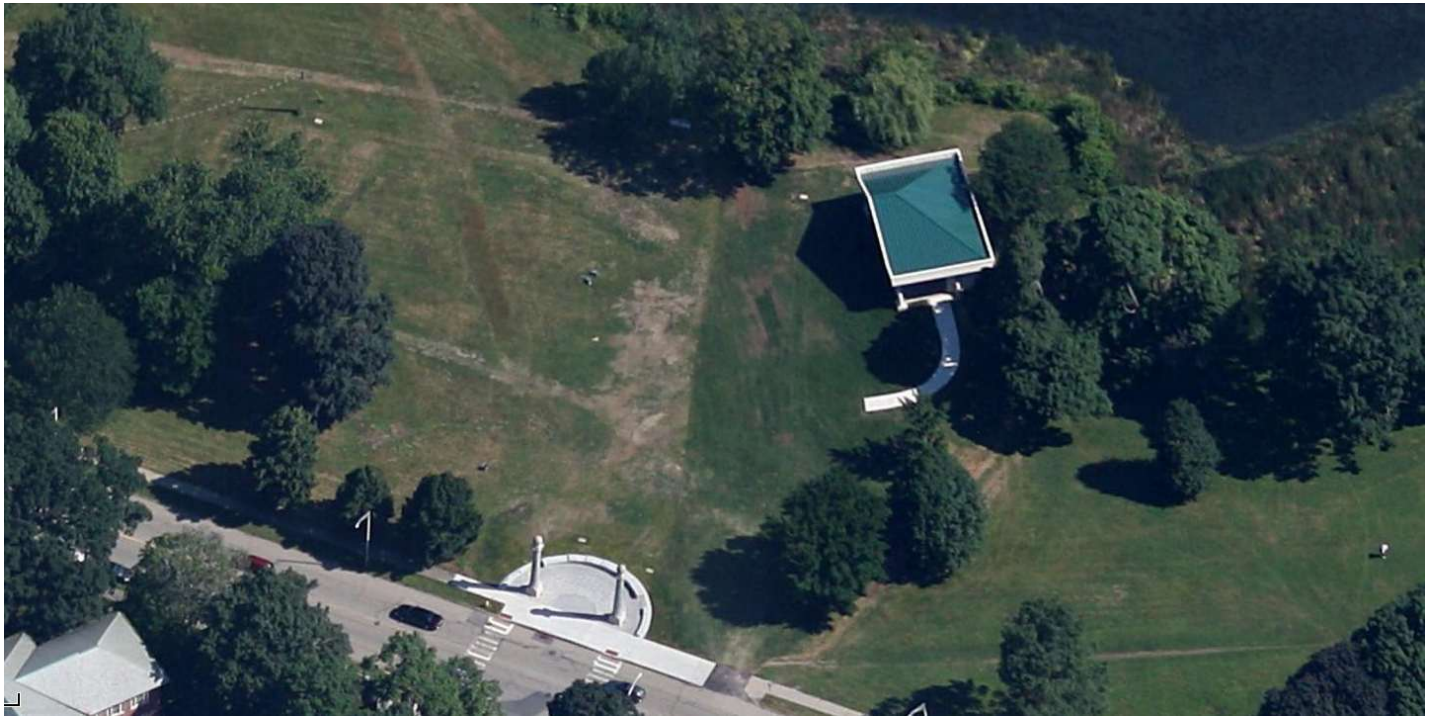


NASA Sample Return Robot Challenge Team (2012-2014)



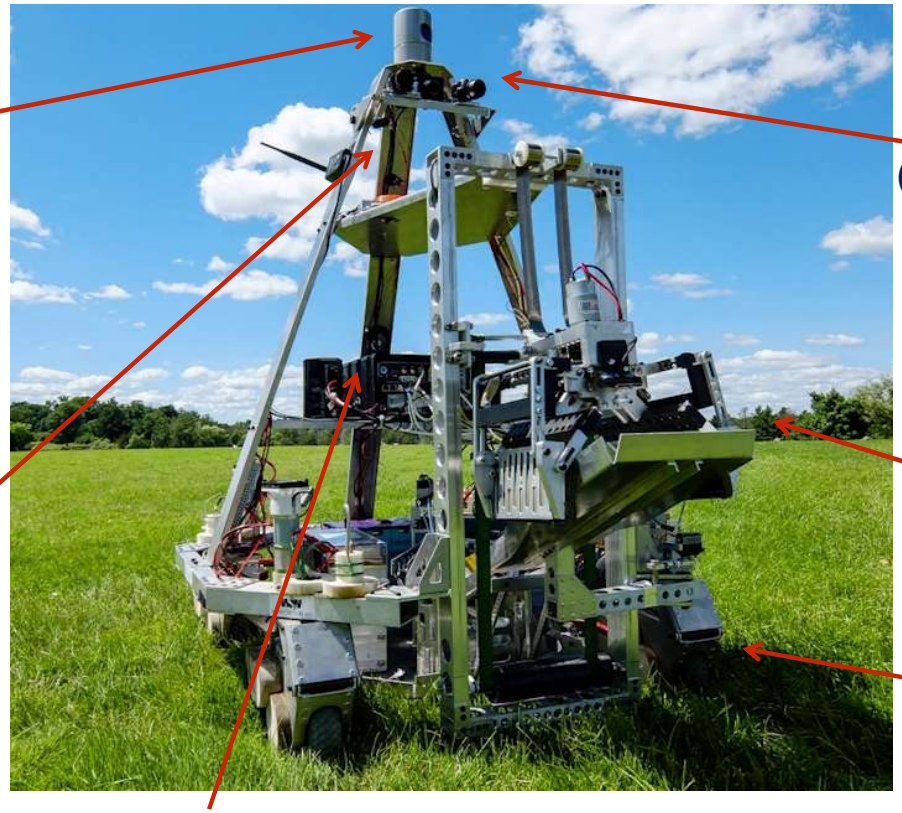


The NASA Sample Return Robot Challenge



Problem: Explore an 80,000 square meter area within 2 hours, collect samples of interest, return to 2m x 2m start platform

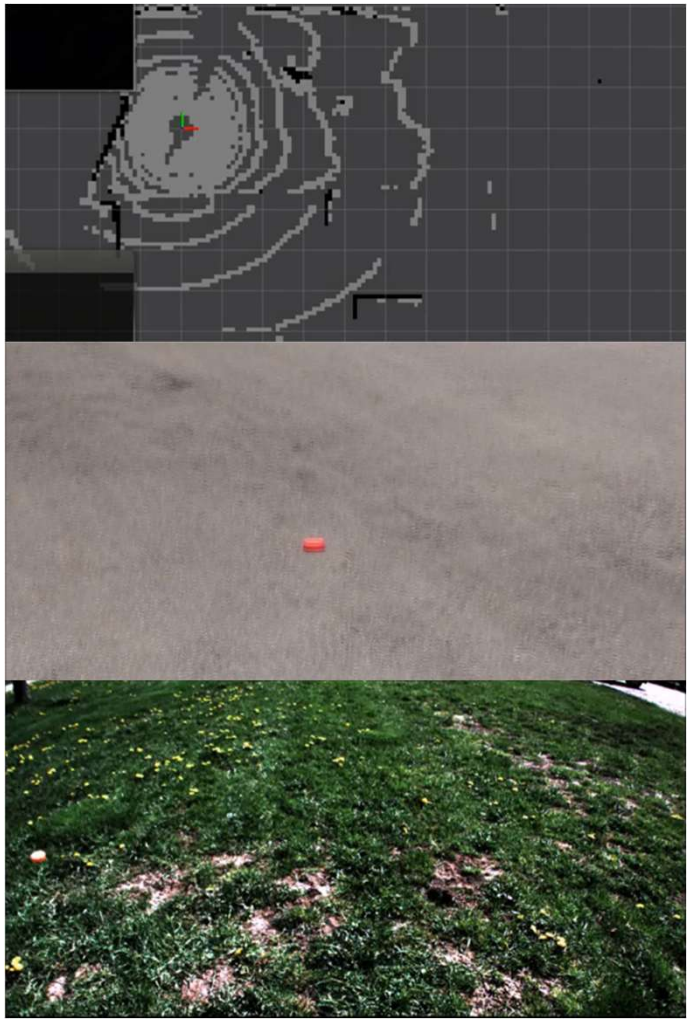
Vehicle and Sensors



Computation: Two embedded i7 computers

Major Algorithms

- Mapping: Velodyne HDL-32 SLAM
- Planning: Global coverage and local obstacle avoidance
- Sample Detection: detection and classification of salient objects



Mapping Overview

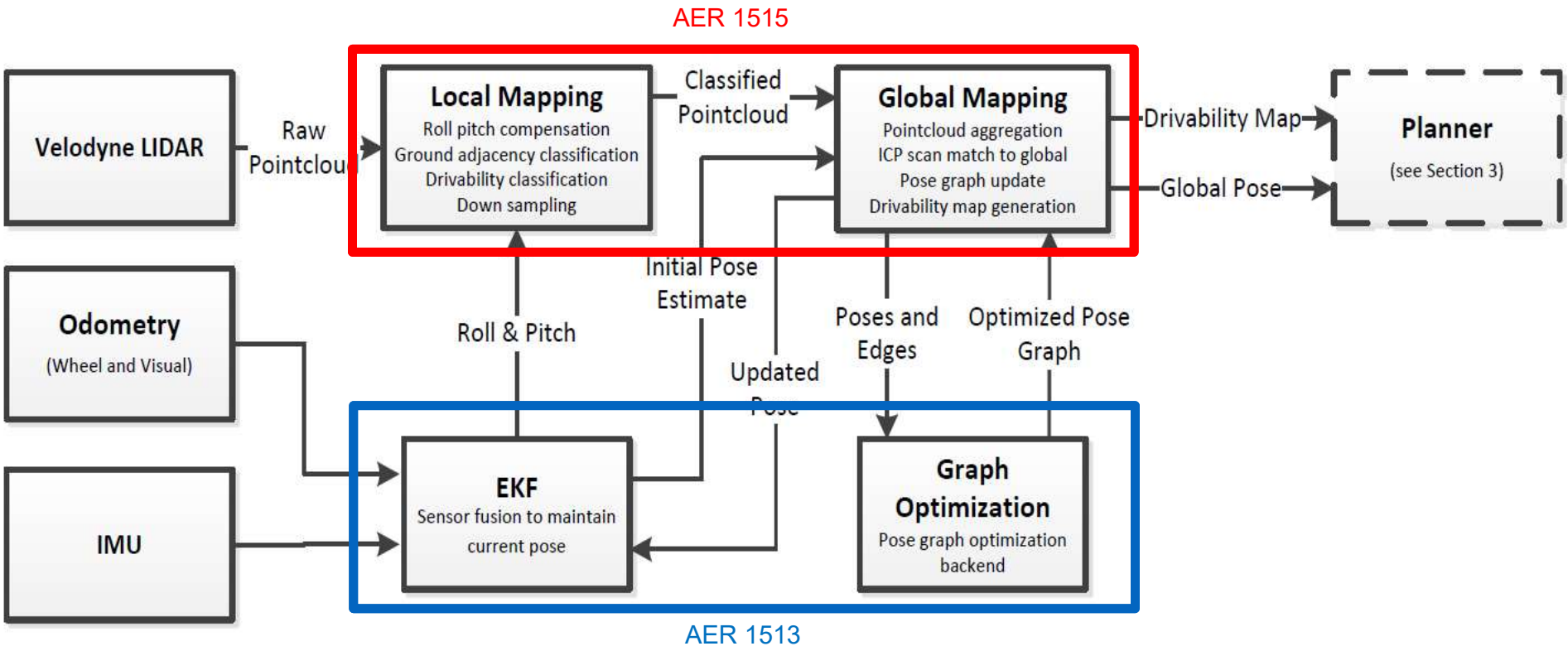
- Requirements:
  - Real-time performance (10Hz), on-board the robot moving 2 m/s
  - Metrically consistent maps
  - Determine obstacles

- Key Idea (1): Mapping in 3D not required
  - Vehicle moves in 2D, just need to determine drivable ground
  - Reliable dense, 3D mapping is slow, violates real-time requirements, not suitable for on-board hardware

- Key Idea (2): Remove the ground, use what remains to perform SLAM
  - “Natural features” in the environment

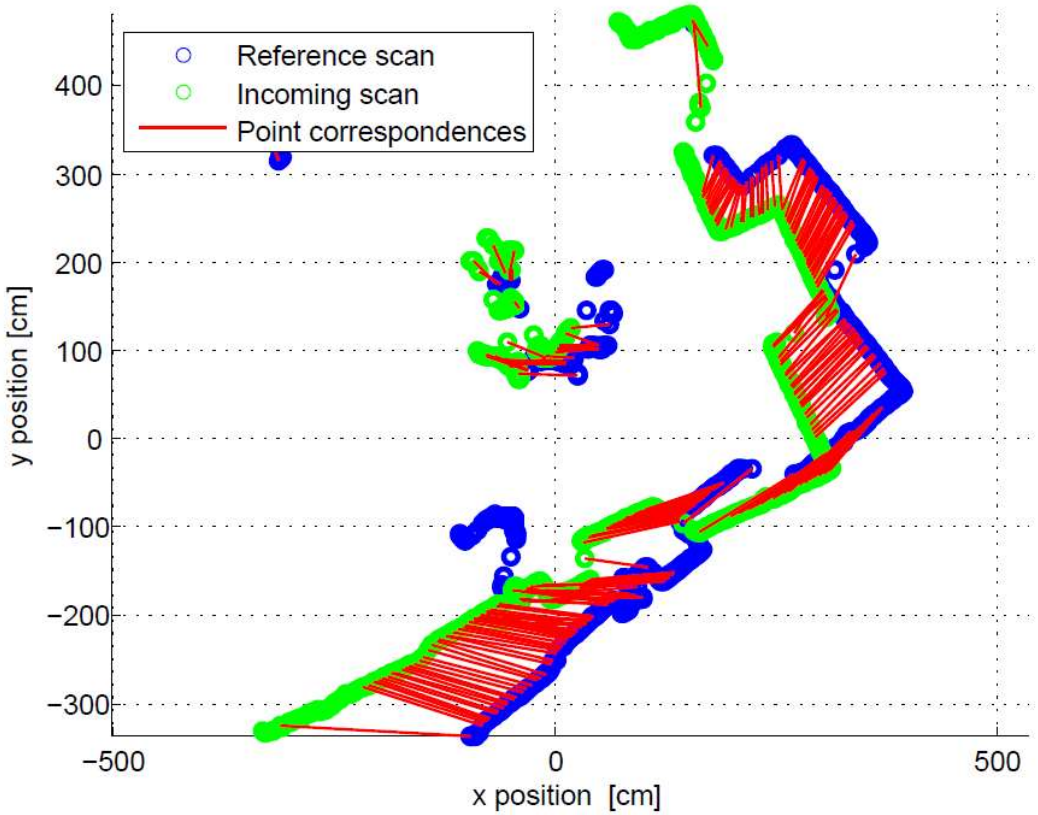
- Questions:
  - How do we remove the ground reliably?
  - How to perform reliable 2D SLAM using 3D data without “loss” of information
  - How do we keep the map metrically (globally) consistent?
  - How can all of this be performed at 10Hz on board the vehicle?

Mapping Overview



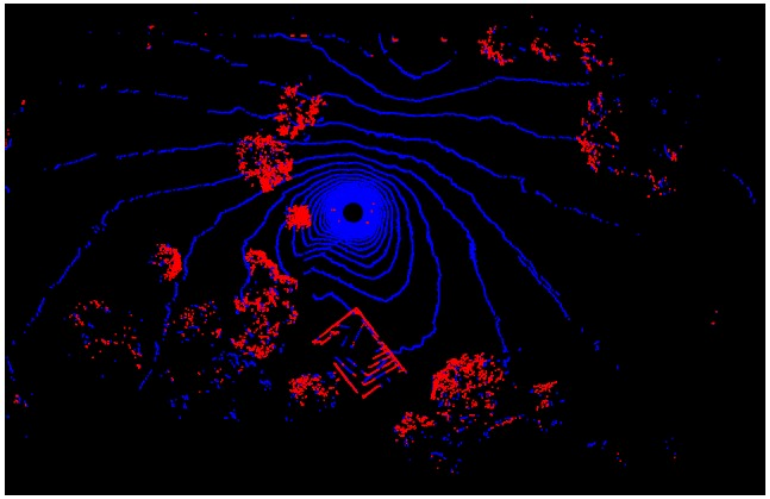
Iterated Closest Point Scan Registration

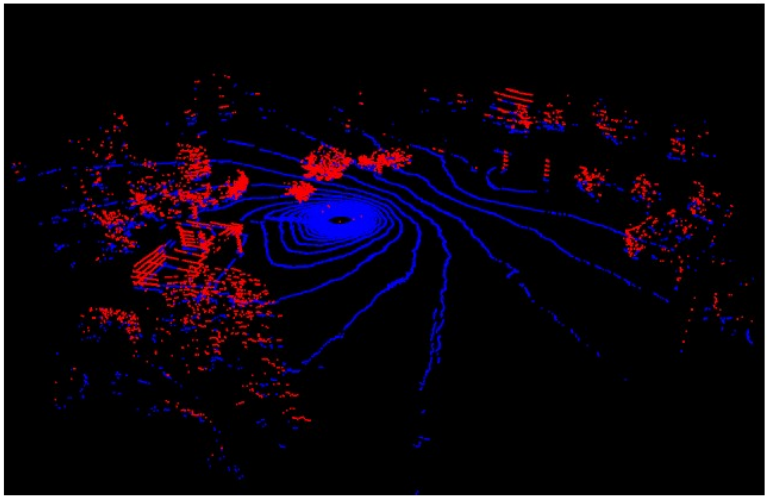
- Minimize the sum of Euclidean distances between corresponding points
  - Point to point
  - Point to plane
  - Generalized-ICP



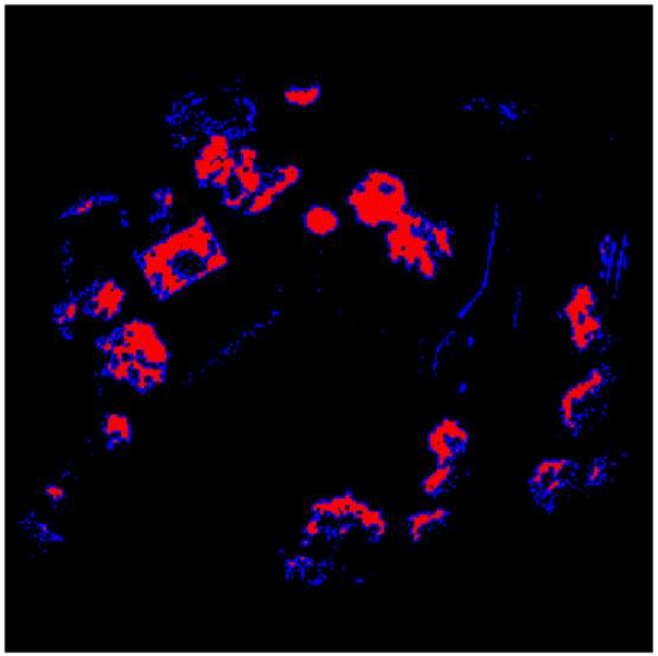
Ground Segmentation

- Performed using 1D Gaussian Process (GP) based approach [Tongtong et al., 2011]
- Works well for hills and undulating terrain

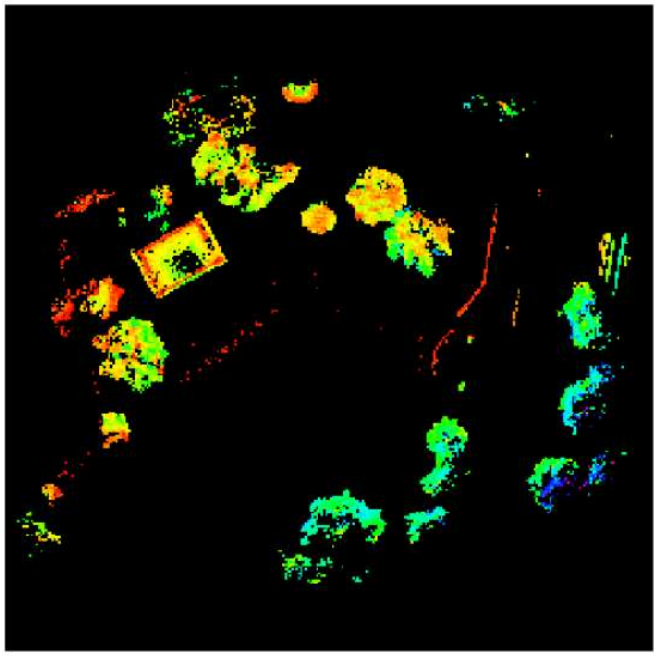




Point Classification

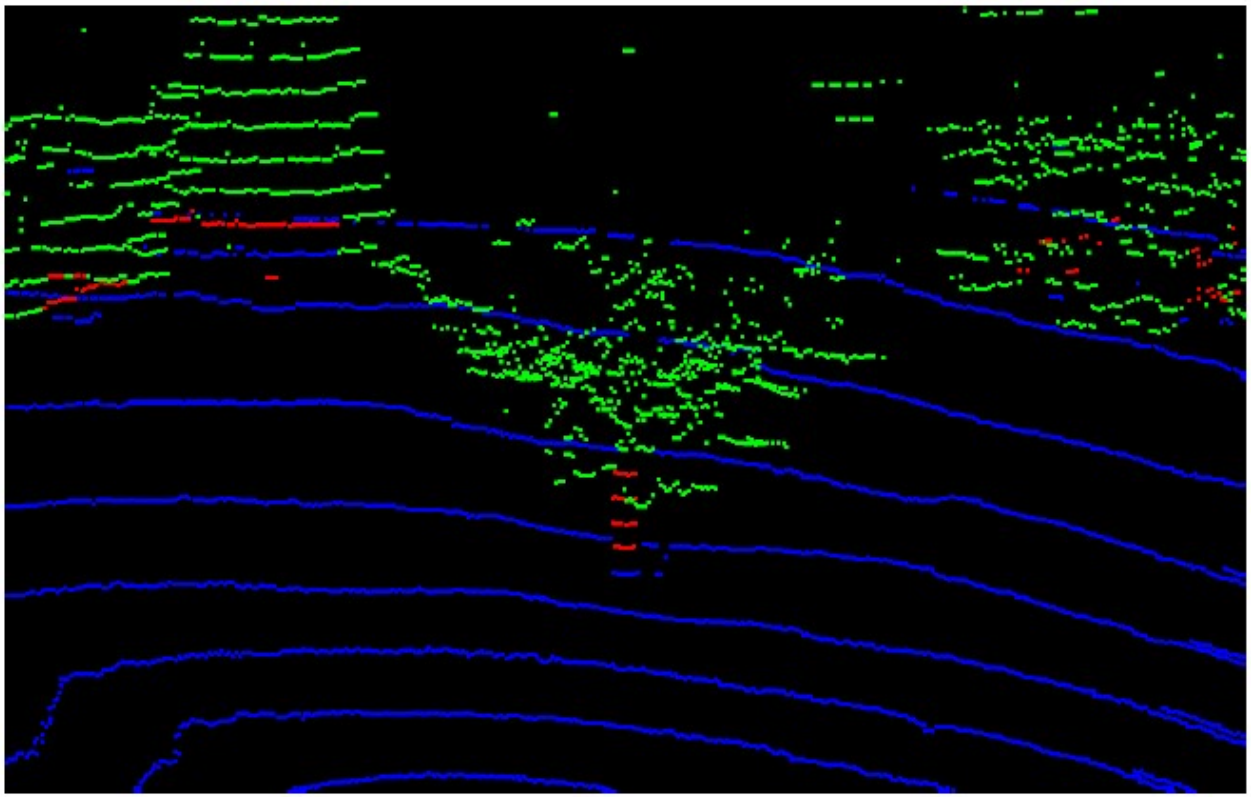


Ground adjacency classification



Height Based outlier rejection

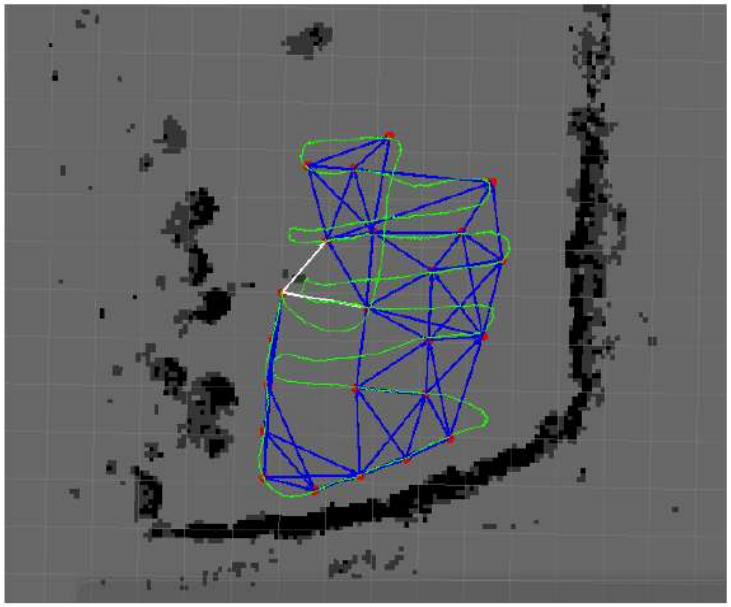
Drivability Analysis

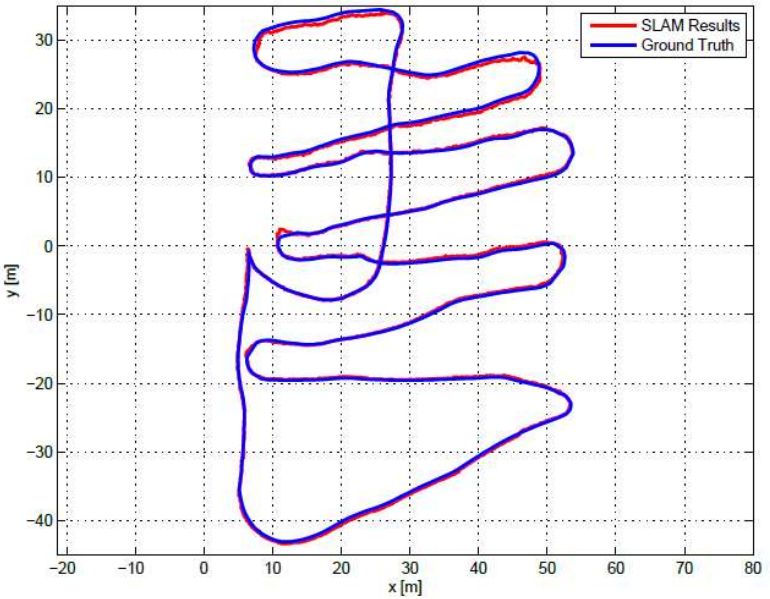


Robot height known, can drive under overhanging obstacles!

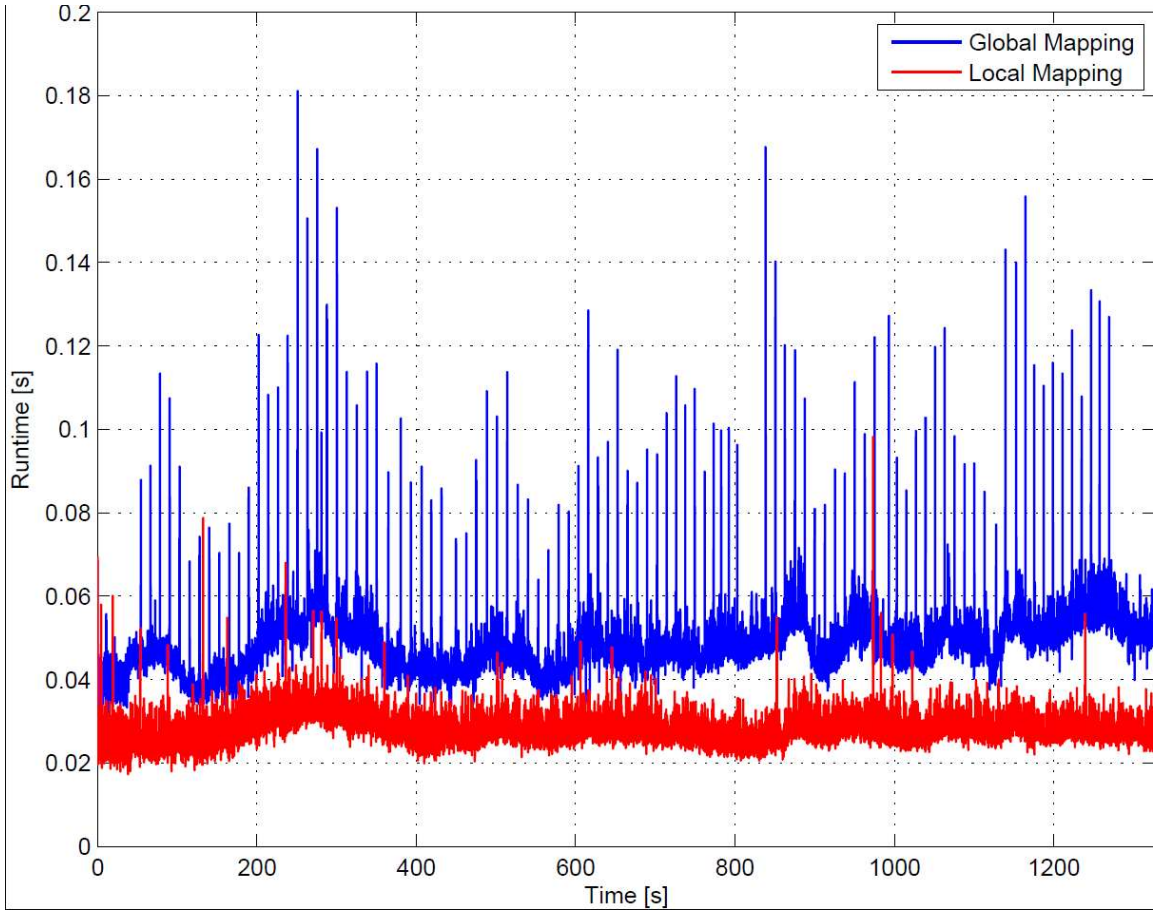
Results (Waterloo Park)

- Left: pose graph and vehicle path. Right: Vehicle path and ground truth collected using Robotic Total Station (mean error: 0.274 m, max error: 1.264 m)





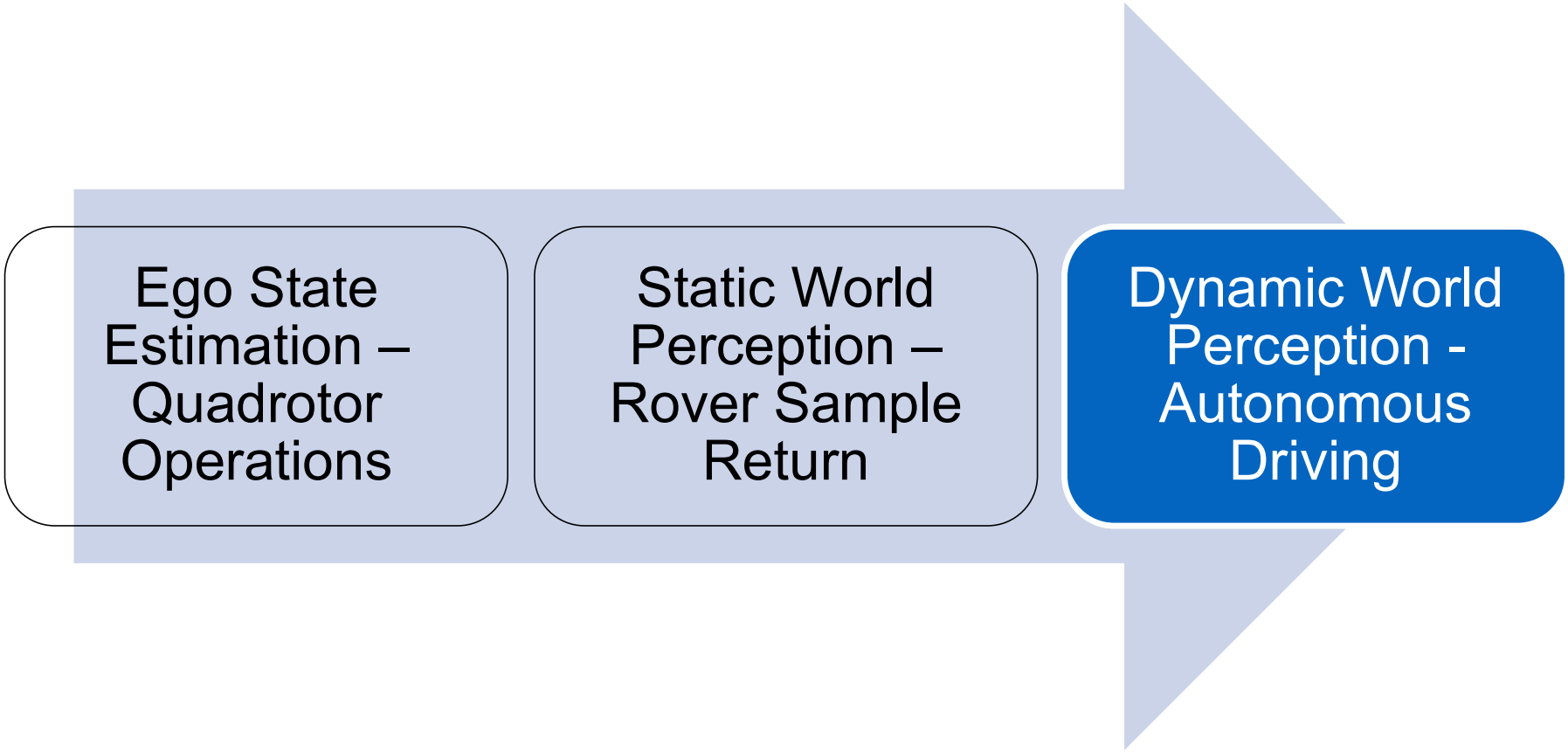
Results (Run Time)

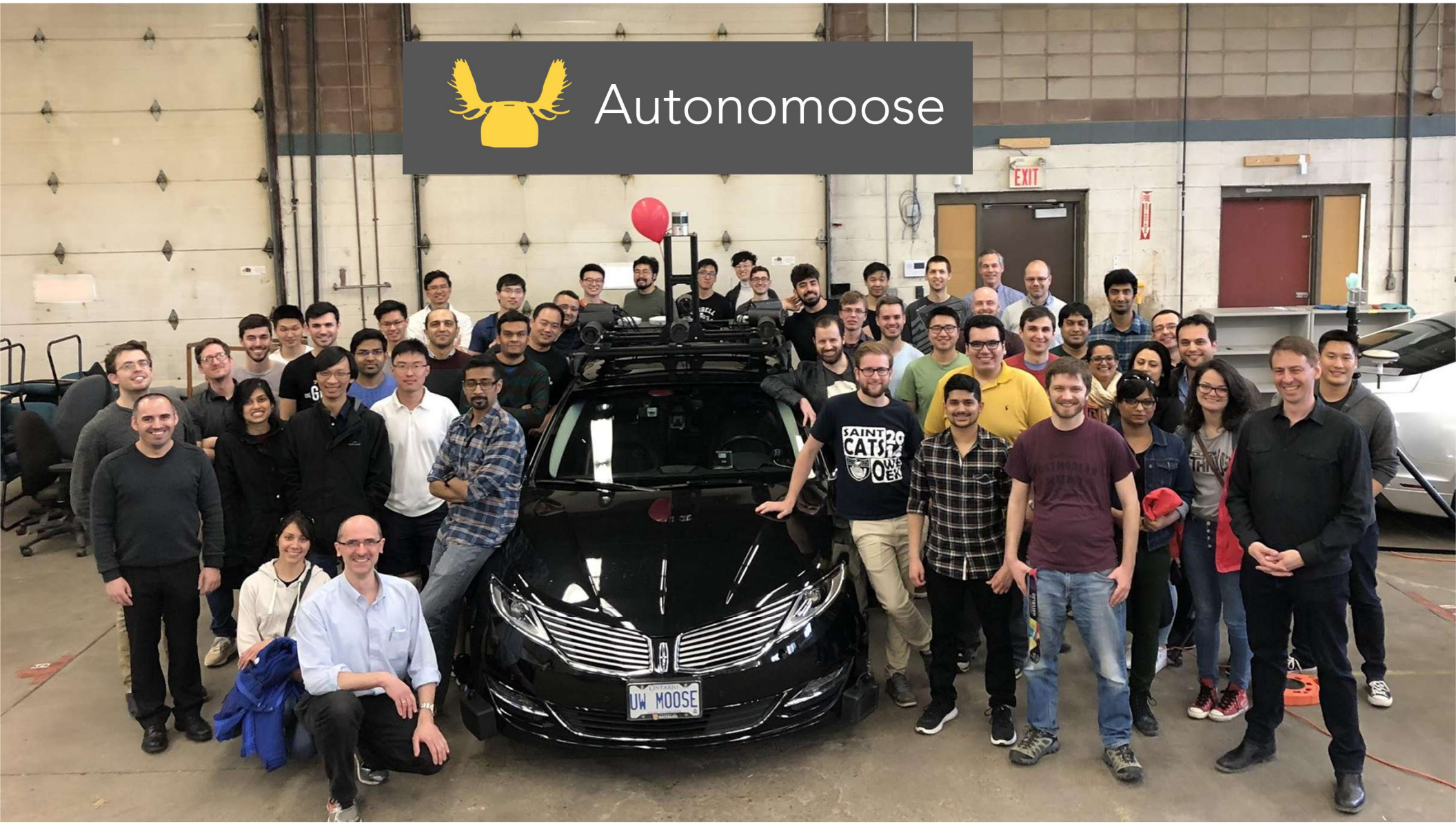


Static World Perception - Exteroception

- Cool Vids
  - Rovers
  - VT&R
  - Manipulator cal

A few stories from my work




Autonomoose
EXIT
UW MOOSE

Biggest Achievements

- First Canadian made autonomous vehicle to drive 100KM on public roads
- Top of Kitti 3D Object Detector
- Extreme Weather Capability
- 1000 KM Dataset with more images and more diverse driving than anything publically available
- 10cm accurate localization engine capable of running without GPS
- Unique Rule Engine Based Behaviour Planner
- Extremely Efficient Local Motion Planning
- Extensive, Integrated Safety Assessment Procedure Developed at Waterloo





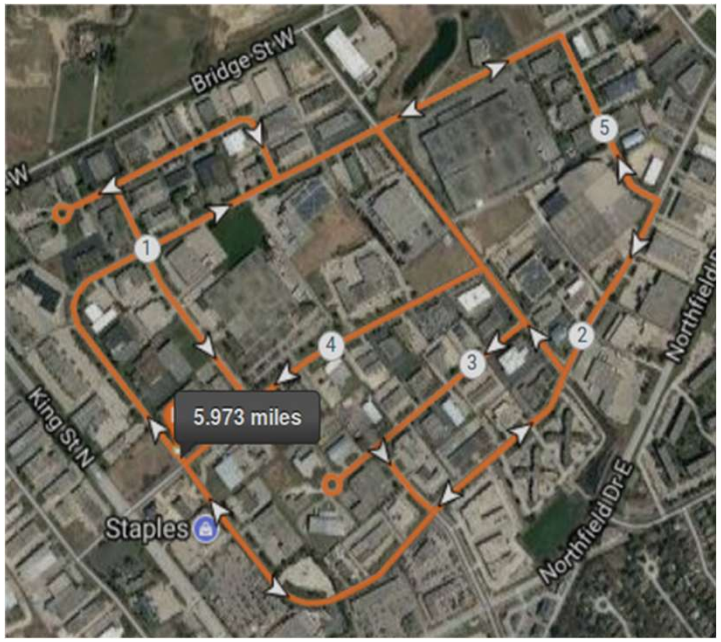
Bathurst Rd Route, Waterloo

- 9 unique intersections
- 15 intersection crosses
- 2 roundabouts
- ~4.07 Km

The public drive challenge

- Goal: 100 km on public roads with MTBI of 5 minutes
  - Completed August 2018
- Colby test lap
  - Industrial area with limited traffic
  - 2 km loop, one stop sign
- Bathurst test circuit
  - All residential, two lane roads
  - 9.6 km sequence
  - 32 separate scenarios encountered

Bathurst





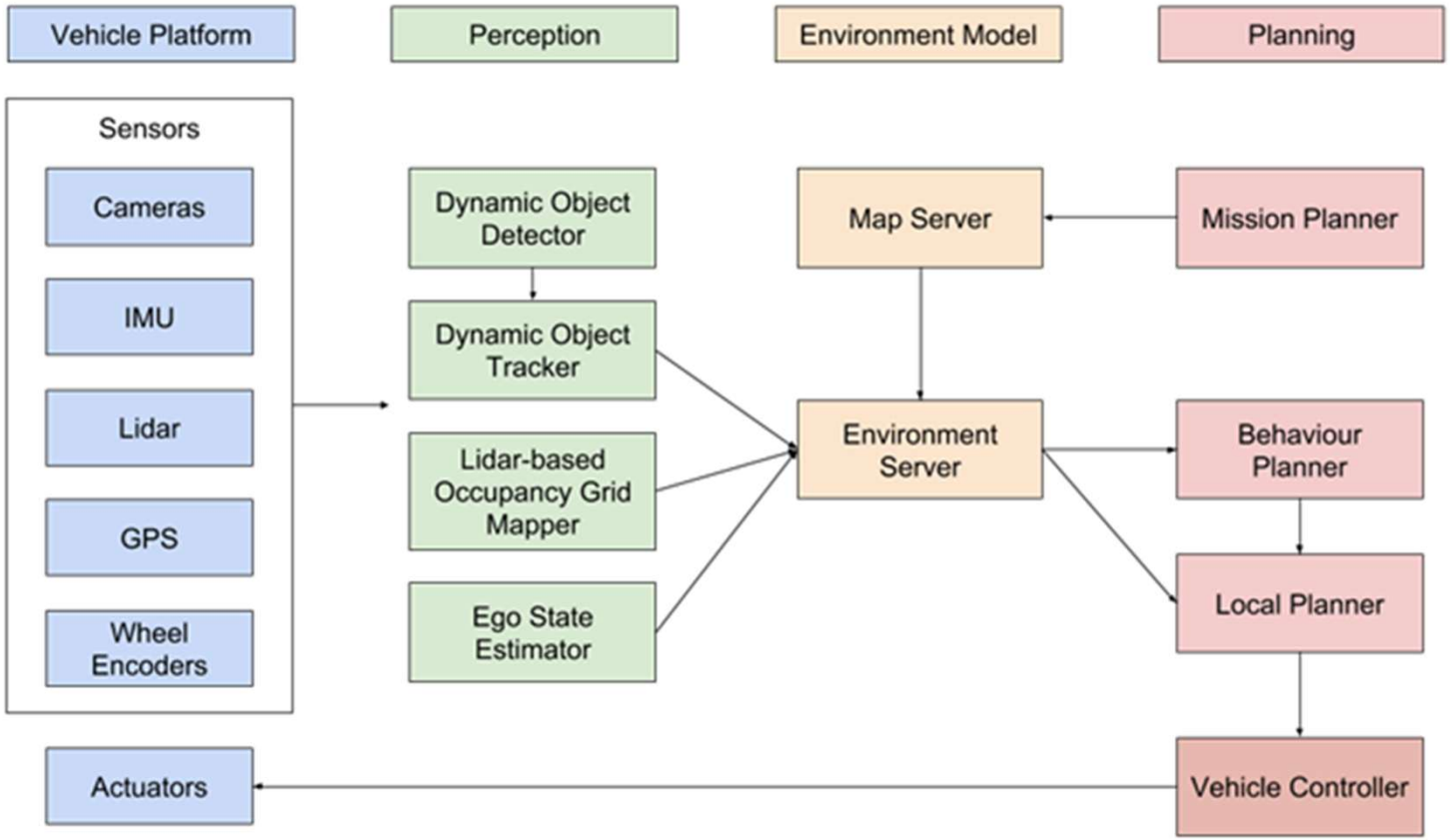




Full Bathurst Drive

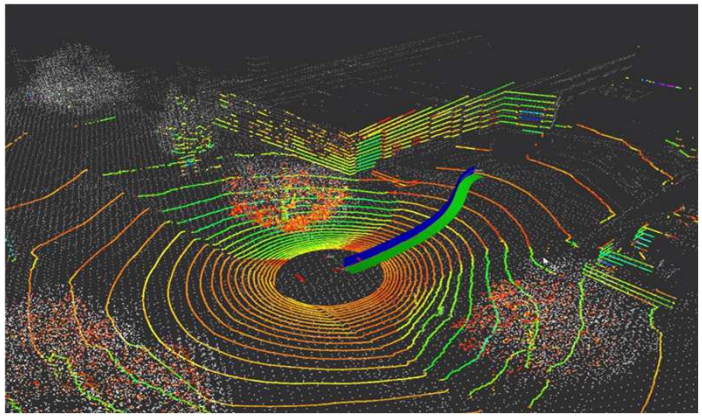


Autonomy System Architecture

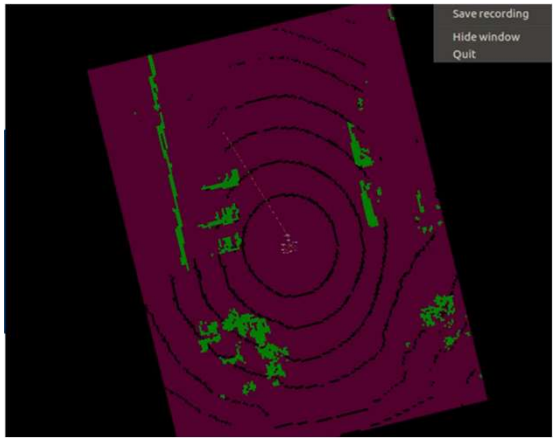


Autonomoose Perception

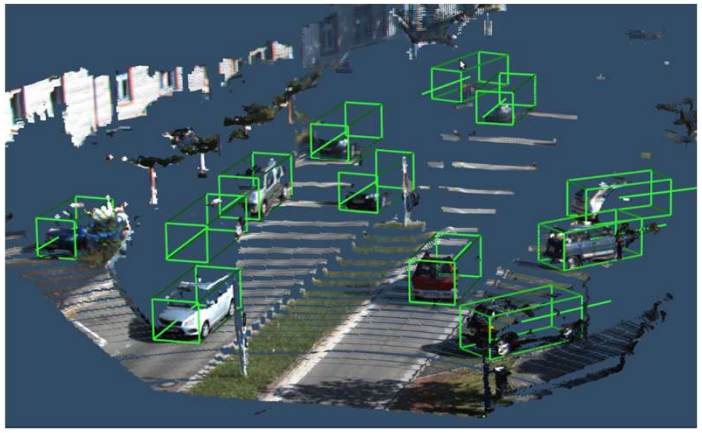
Localization & Mapping



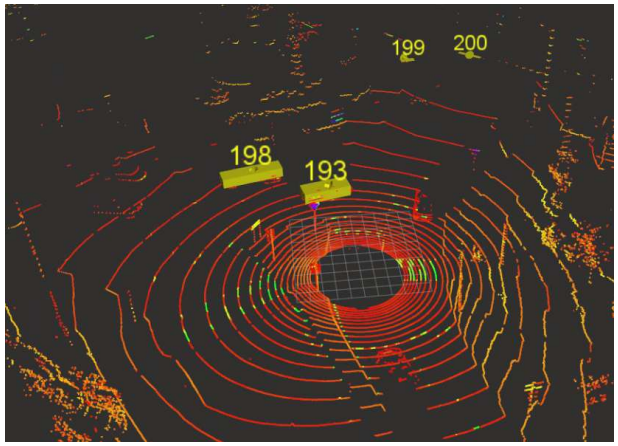
Occupancy Mapping



Static & Dynamic Object Detection

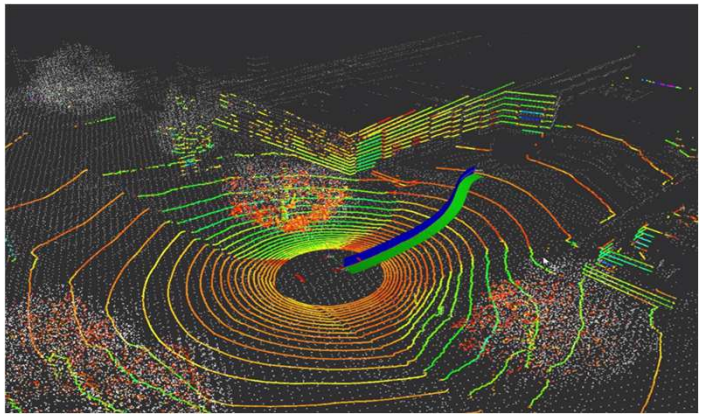


Dynamic Object Tracking

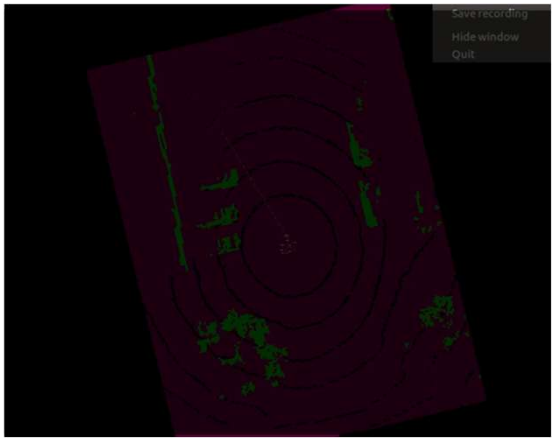


Autonomous Vehicle Perception

Localization & Mapping



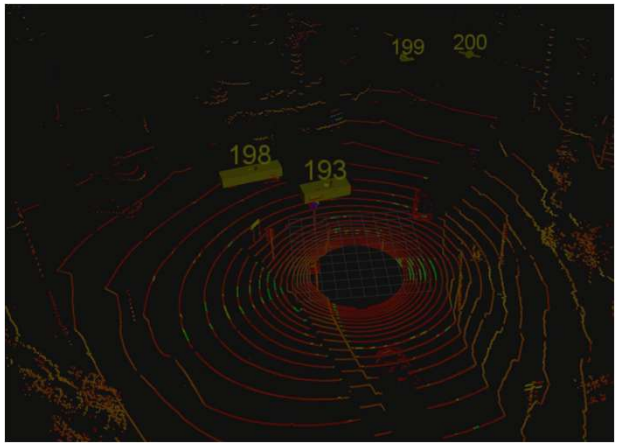
Occupancy Mapping



Static & Dynamic Object Detection

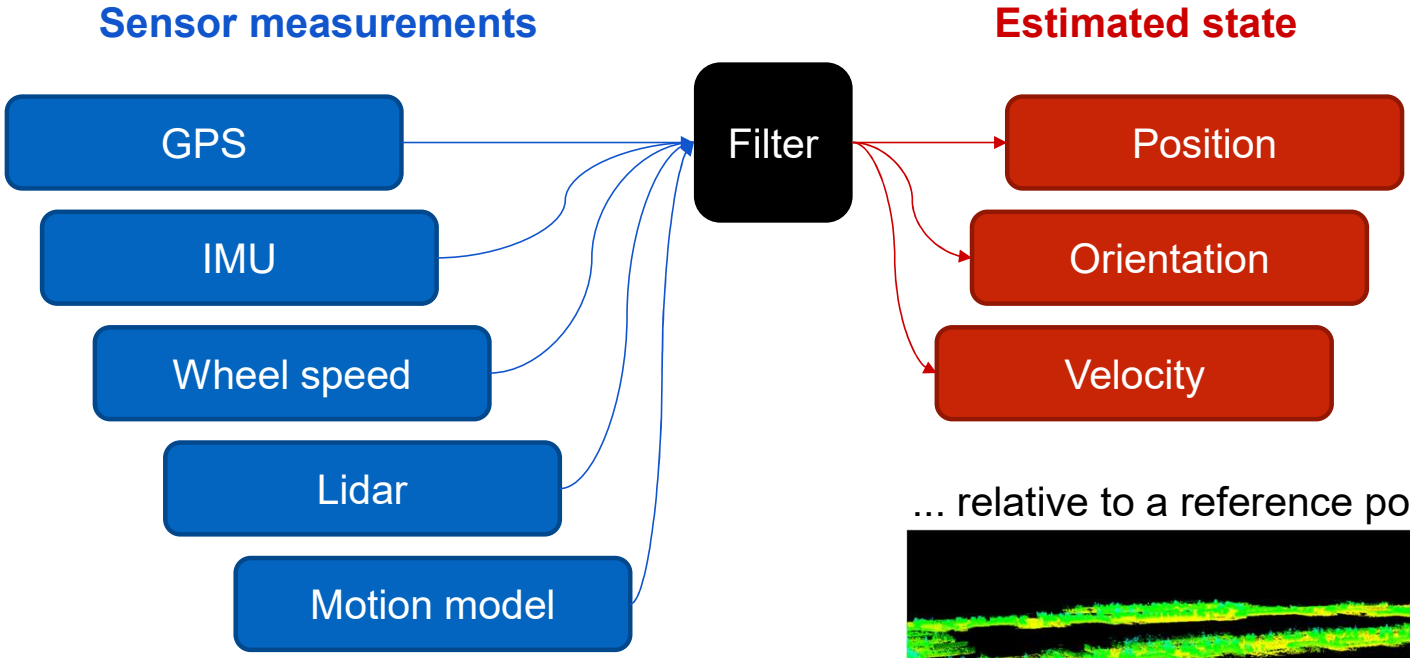


Dynamic Object Tracking

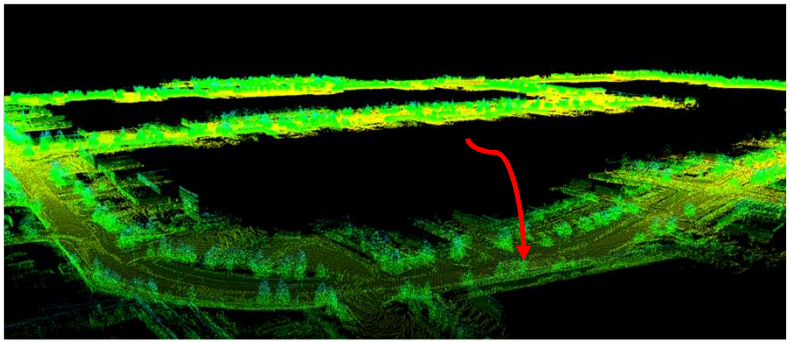


Localization through Sensor Fusion

- Fuse all available sensor measurements in real time to estimate state …



... relative to a reference point and fixed map



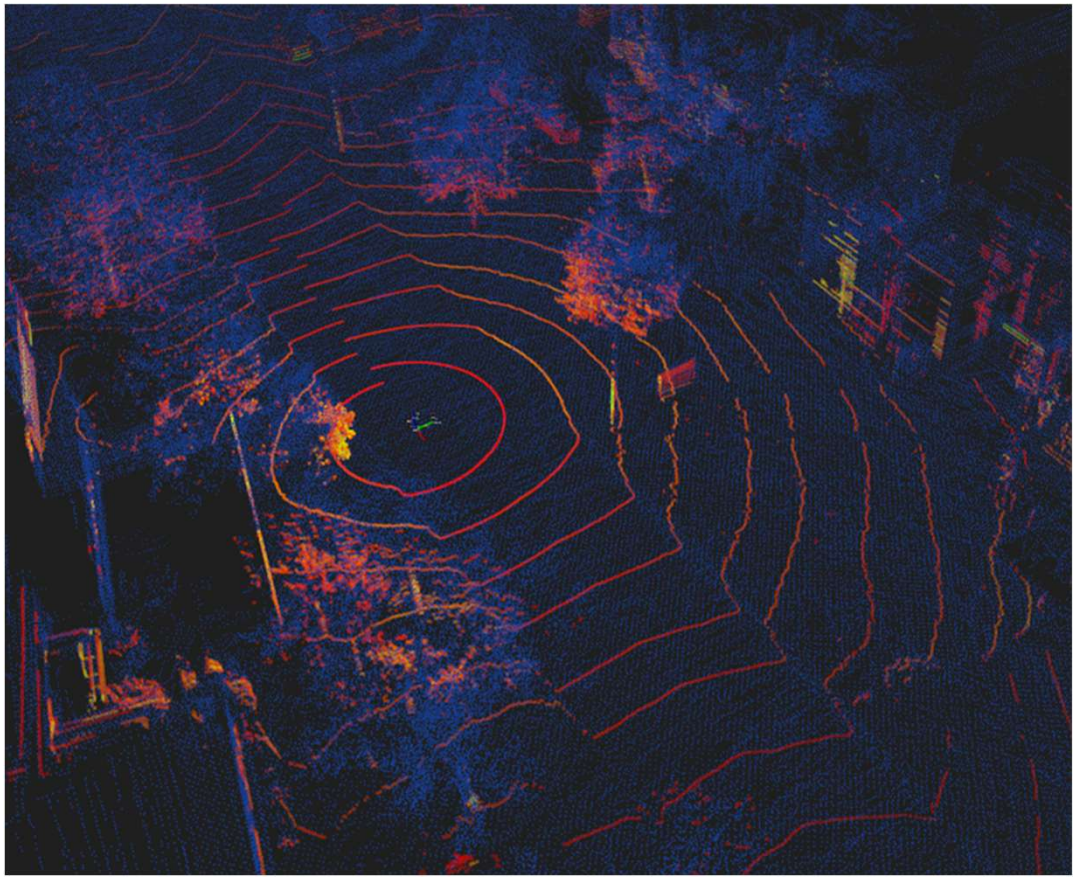
Lidar Map Construction

- Data collected on multiple runs of the environment
- Full pose graph optimization
  - SE(3) formulation
  - Integrated extrinsic calibration
    - GPS-LIDAR especially important
  - RTK-GPS and lidar filtering incorporated

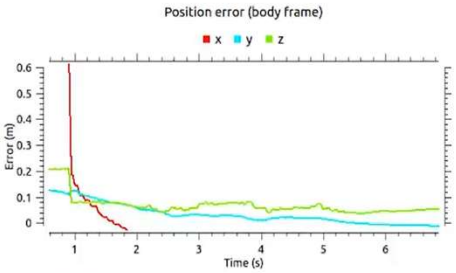


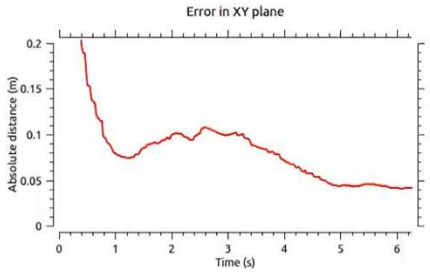
Lidar Localization Measurements

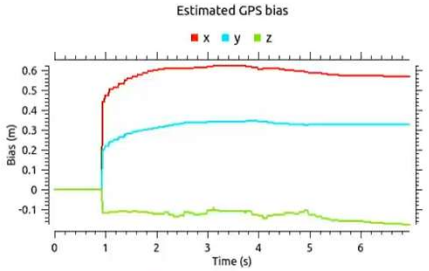
- Match lidar scan to a pre-built map
- ICP matching
- Initialization
  - Need an initial estimate of pose in the map from other sensors
- Motion correction
  - Need an estimate of the vehicle motion during scan capture as well

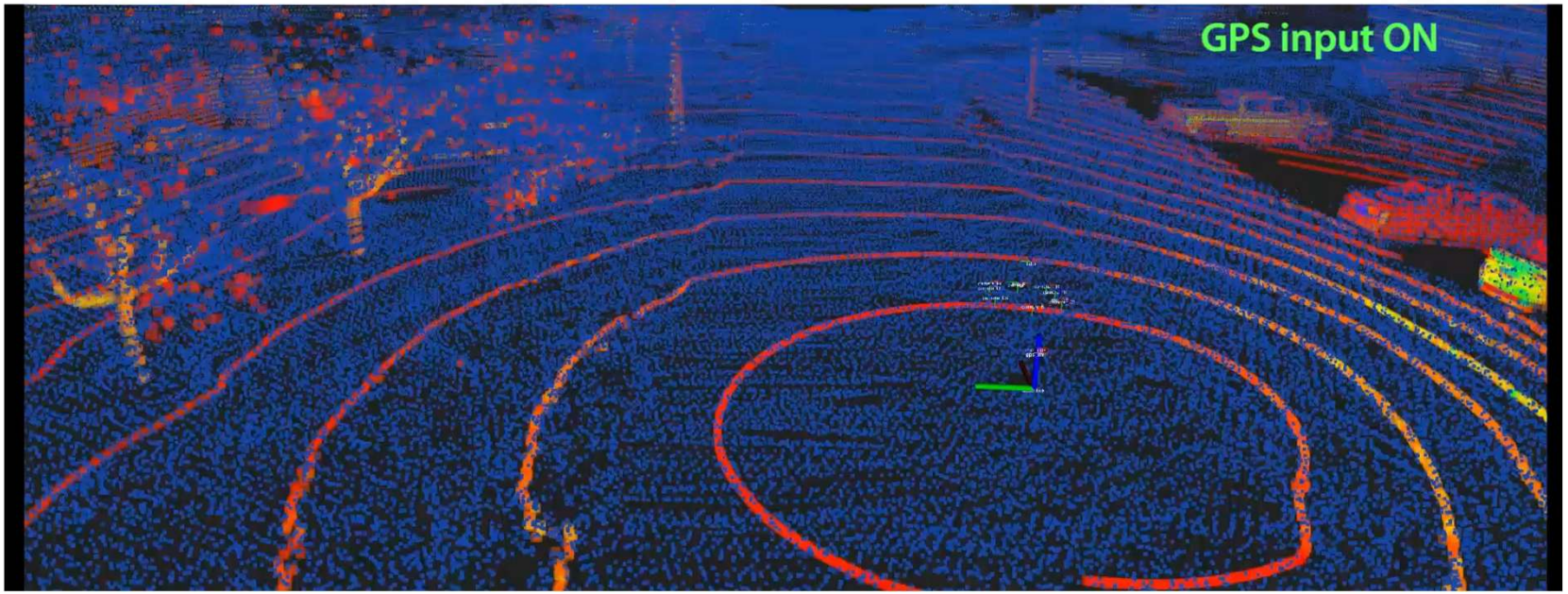


Localization Performance



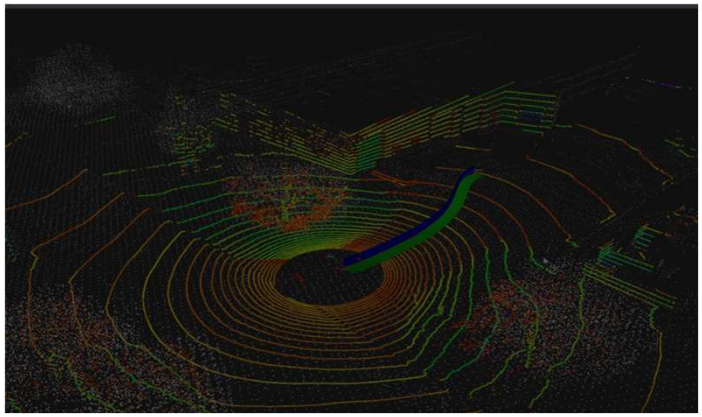




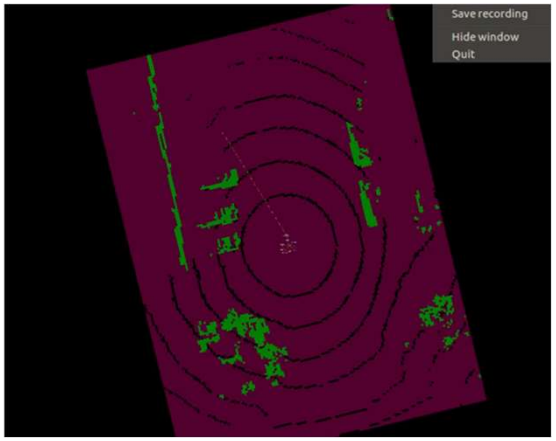


Autonomous Vehicle Perception

Localization & Mapping



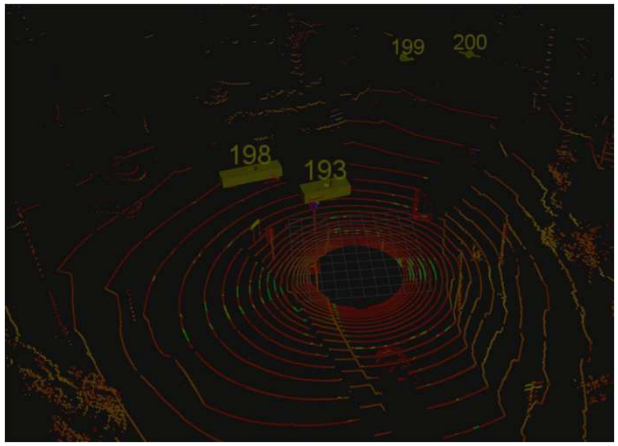
Occupancy Mapping



Static & Dynamic Object Detection

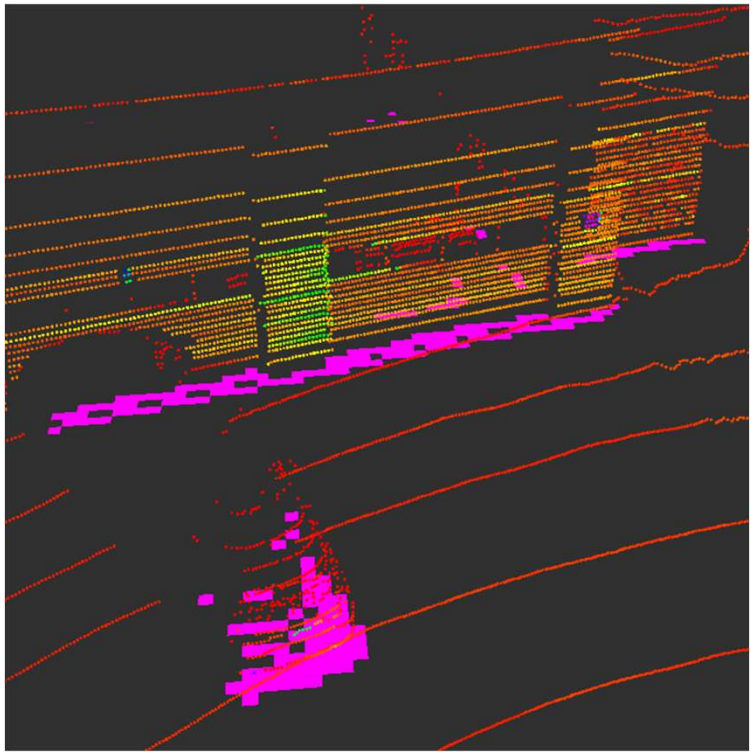


Dynamic Object Tracking

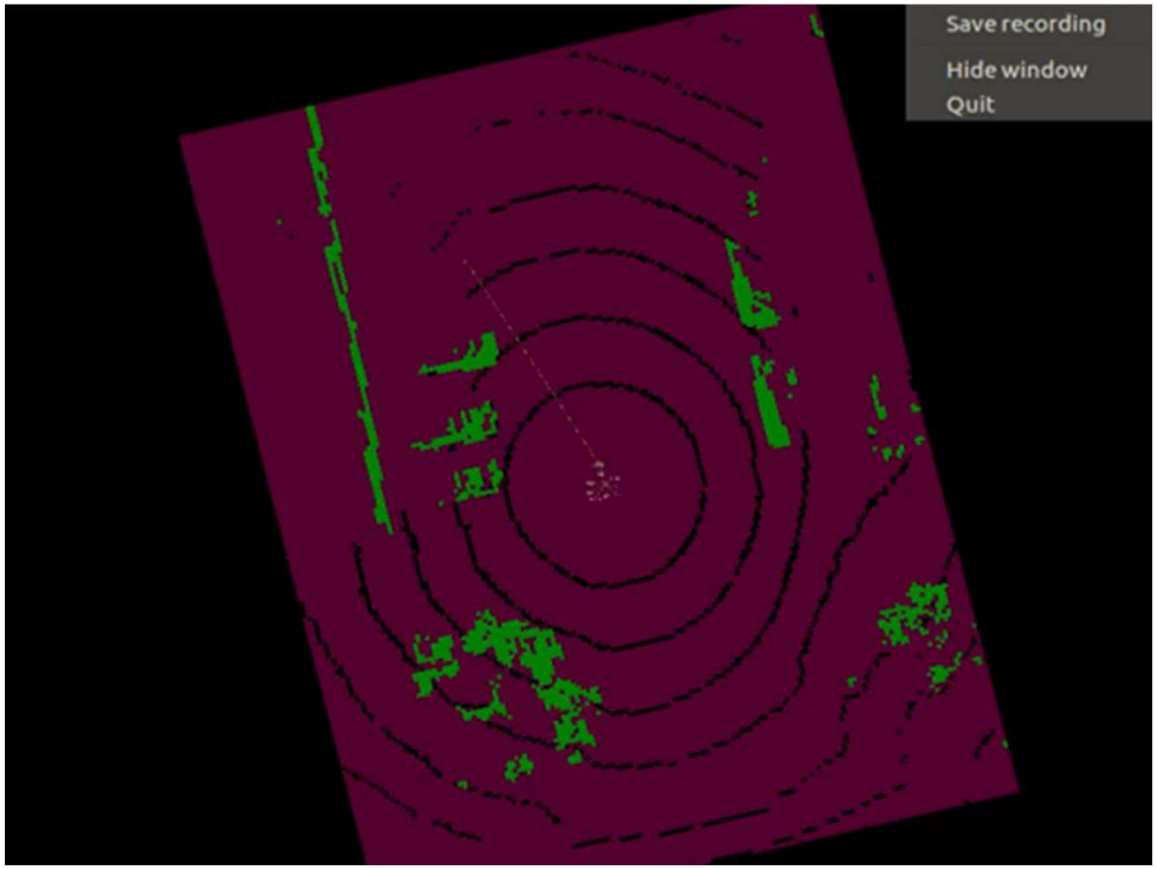


Occupancy Grid Mapping

- Create a two dimensional map of the environment from 3D Lidar data with tracked objects removed
  - Crop and Voxelize
  - Transform to base frame
  - Remove ground plane points and outliers
  - Track occupancy via logit updates
  - Find, track, predict and remove moving objects

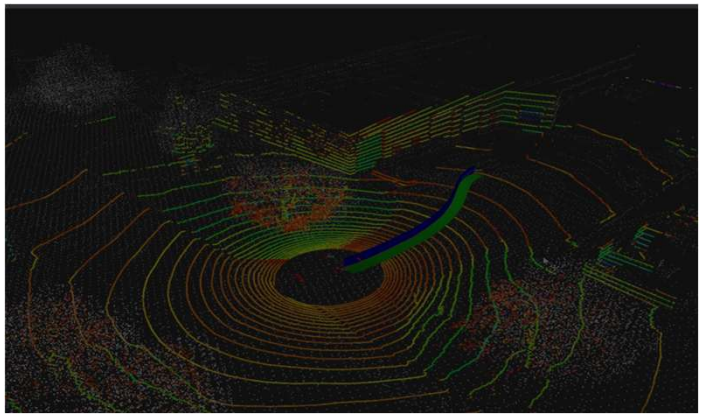


Final Occupancy Grid Map

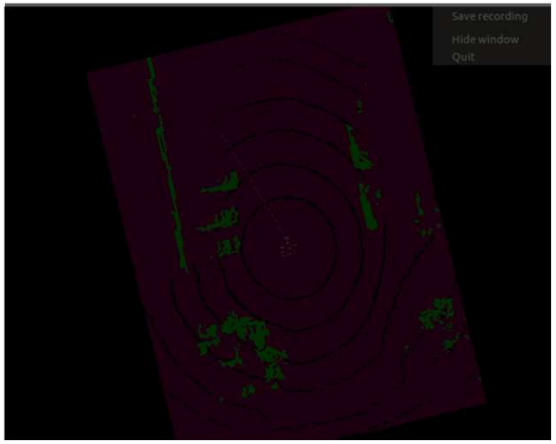


Autonomous Vehicle Perception

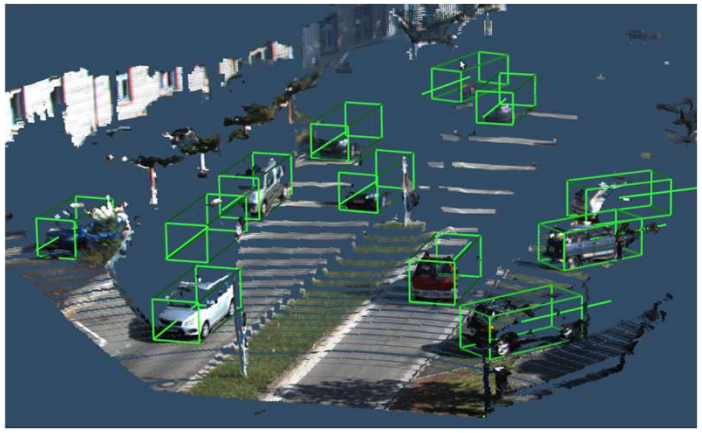
Localization & Mapping



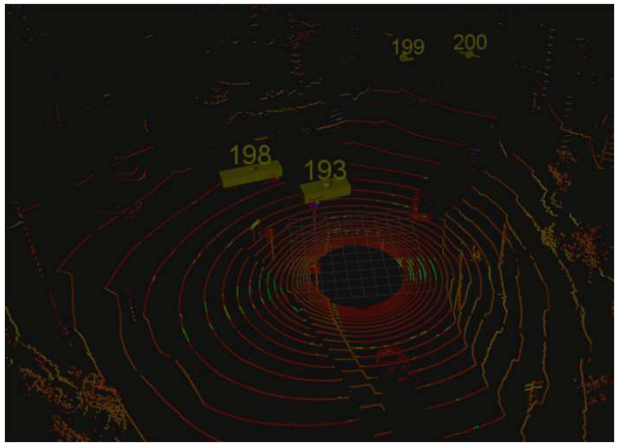
Occupancy Mapping



Static & Dynamic Object Detection

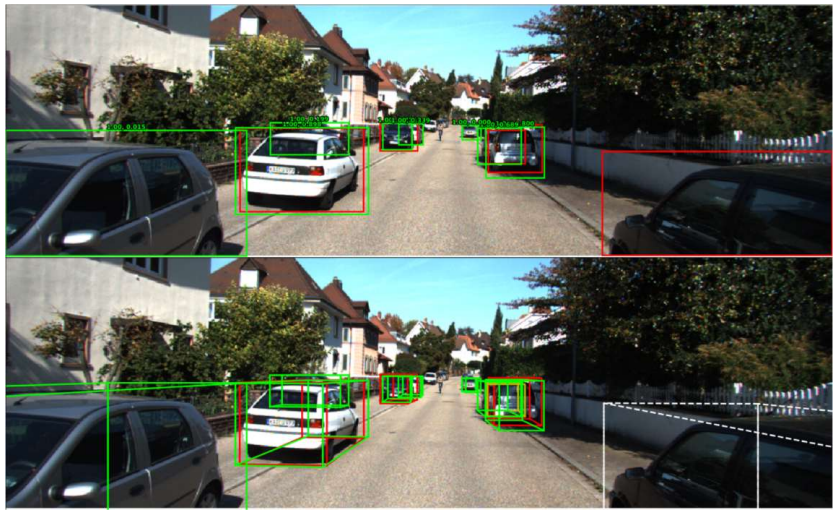


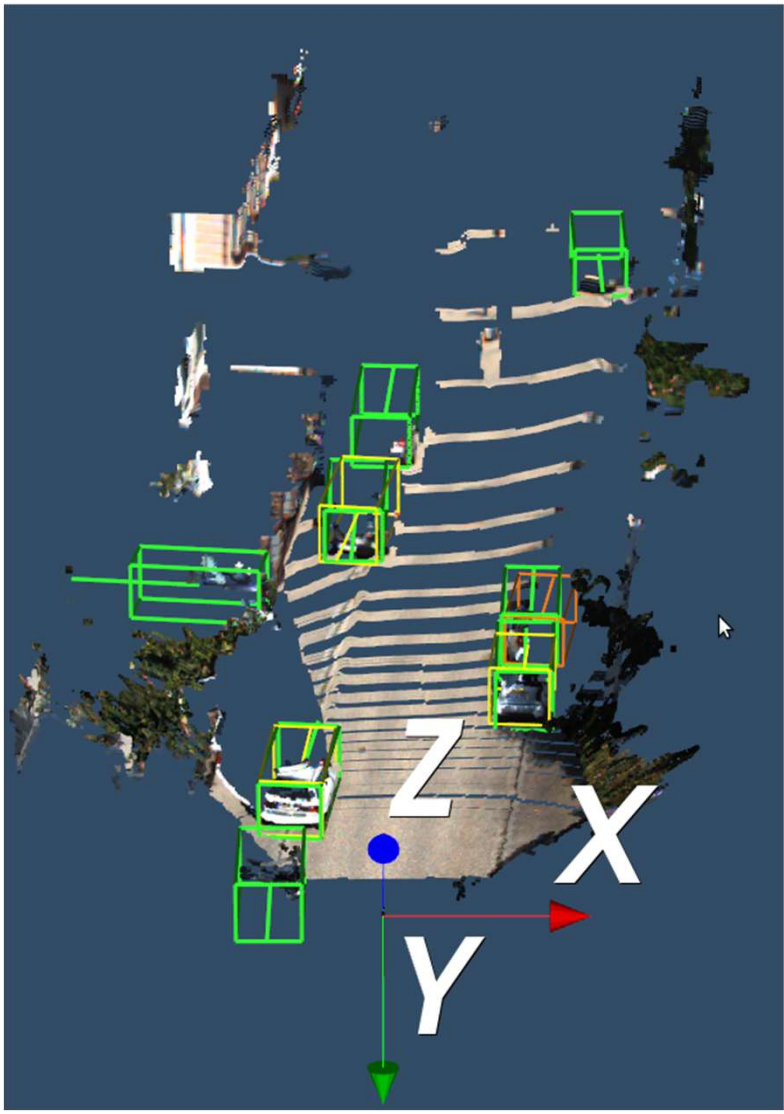
Dynamic Object Tracking



3D object detection and datasets

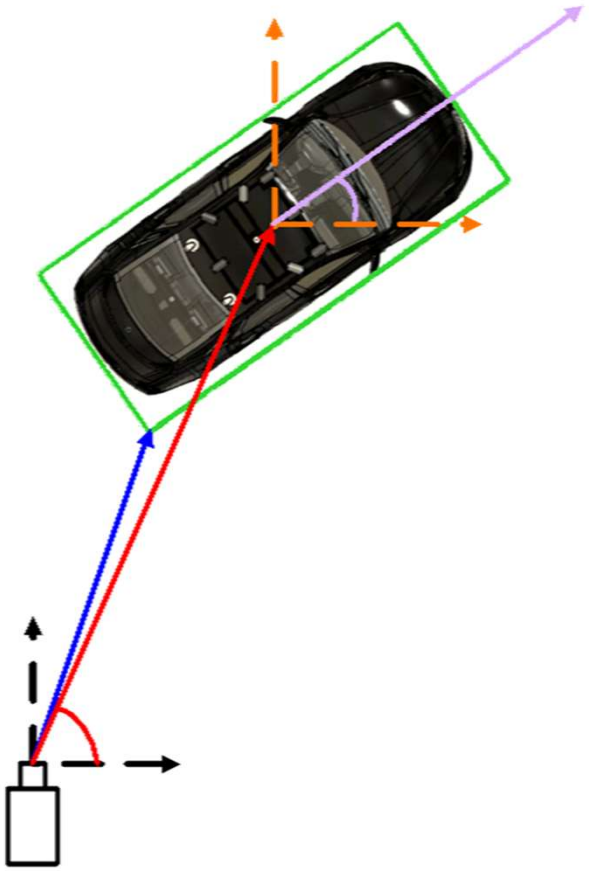
- Multi-channel deep learning approach that combines depth and RGB information
- 3D position, orientation and extent far more challenging than 2D
  - Best algorithms at 70% mean AP (detection box accuracy)



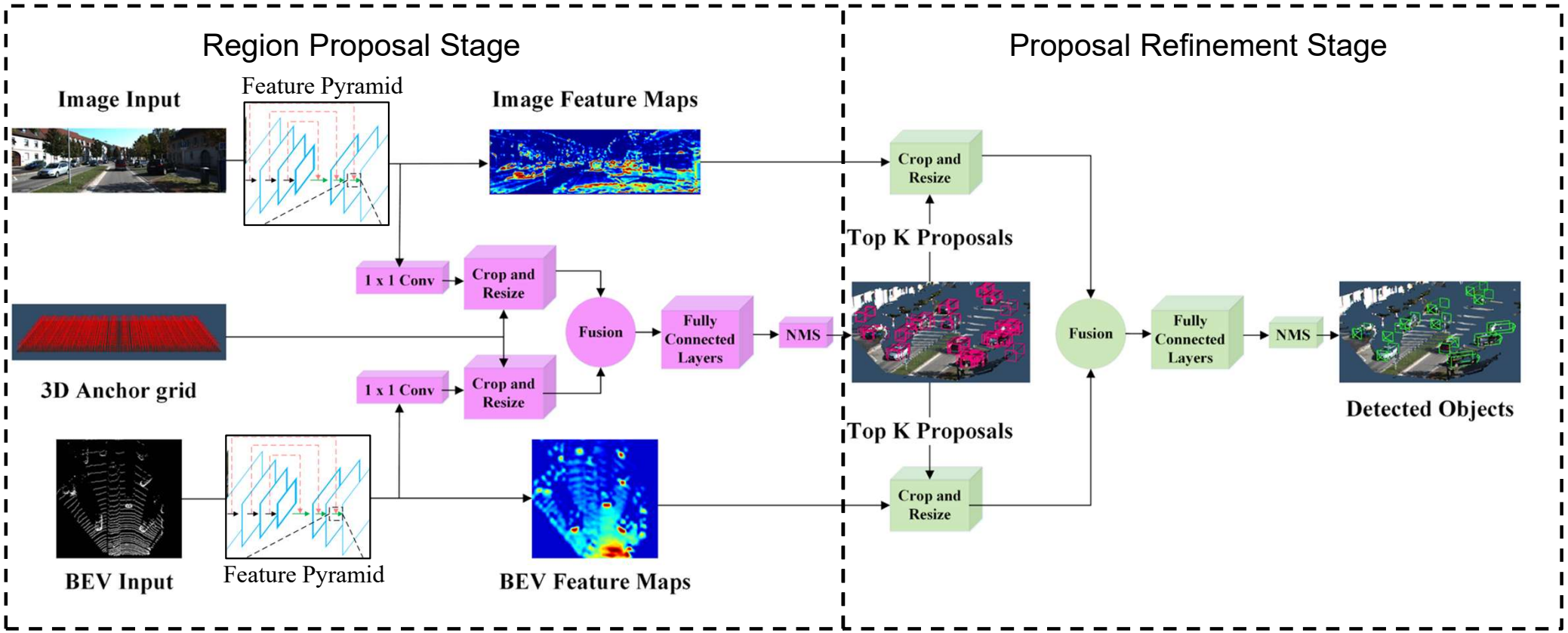


3D Object Detection Objectives

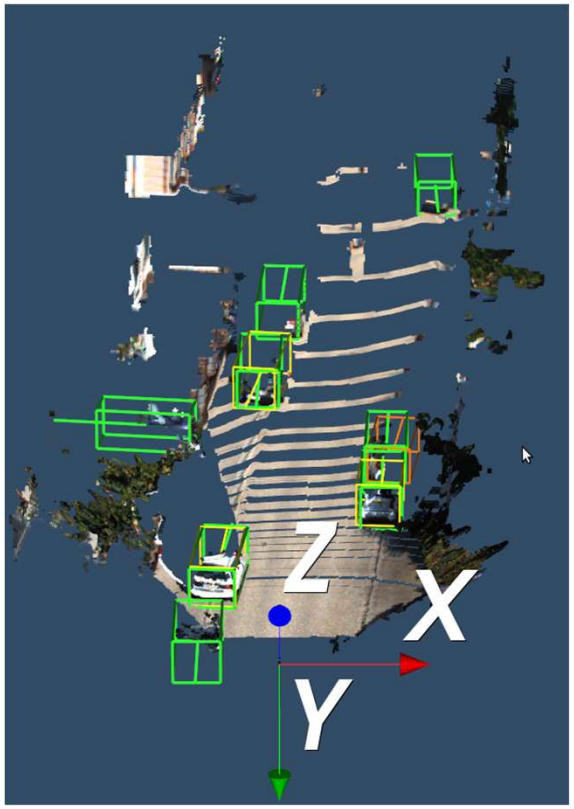
- Class identification
  - Car, pedestrian, bicycle
- Centroid Estimation
  - X,Y,Z position in camera frame
- Bounding box extent
  - Intersection over Union with ground truth
- Vehicle Orientation
  - Forward direction



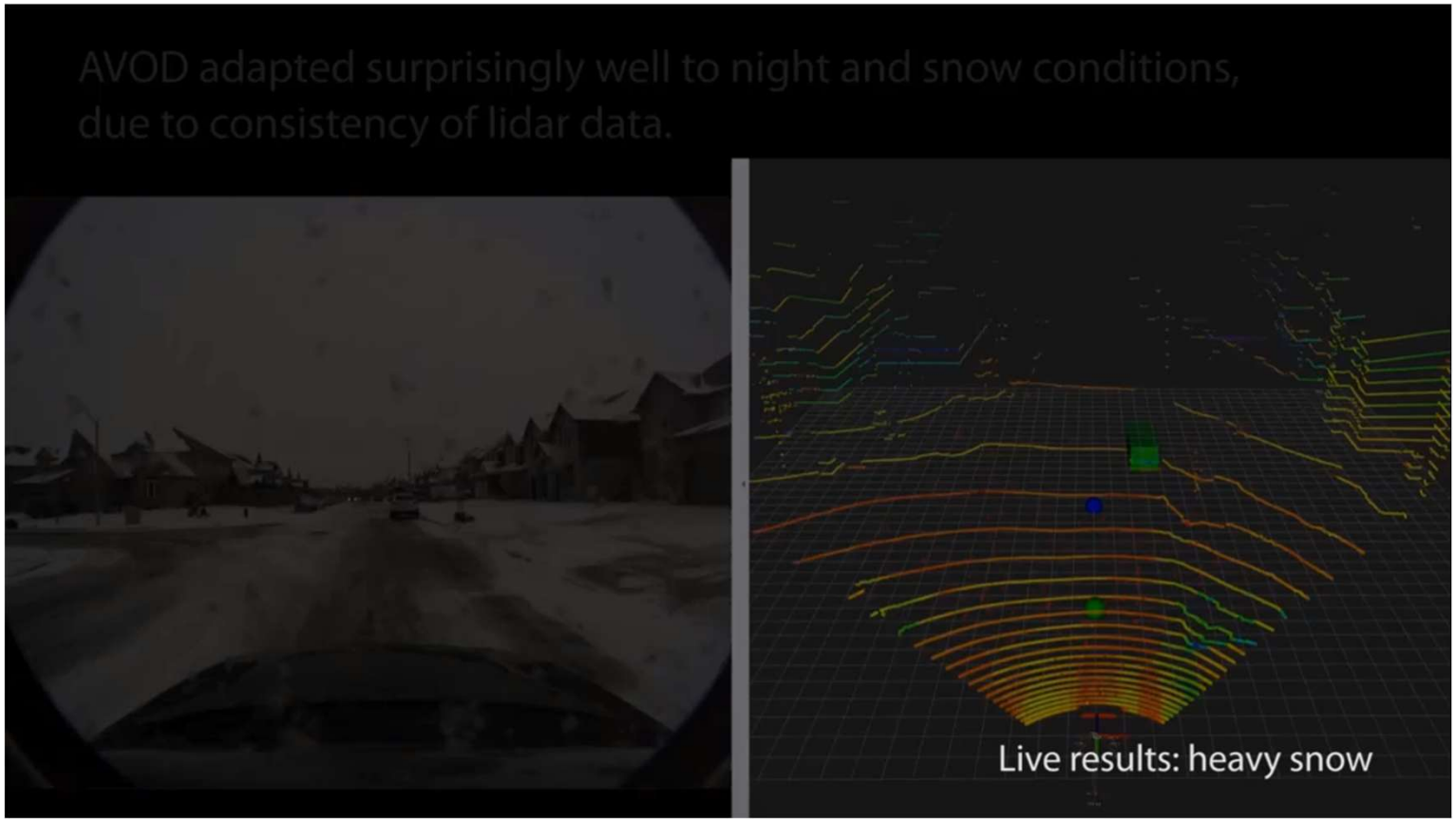
AVOD-FPN Meta-Architecture



3D Object Detection Results



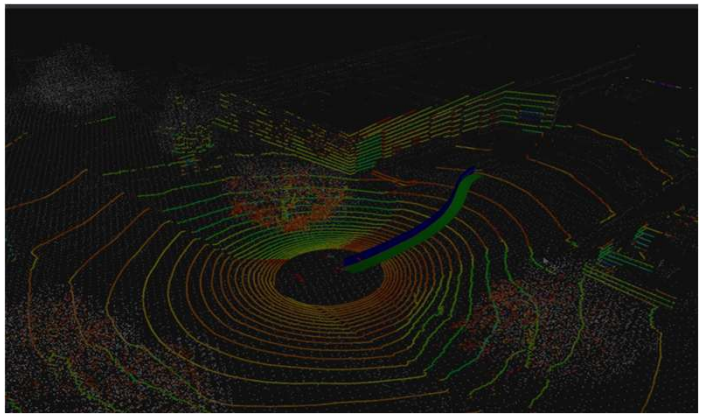
Detection in Snow



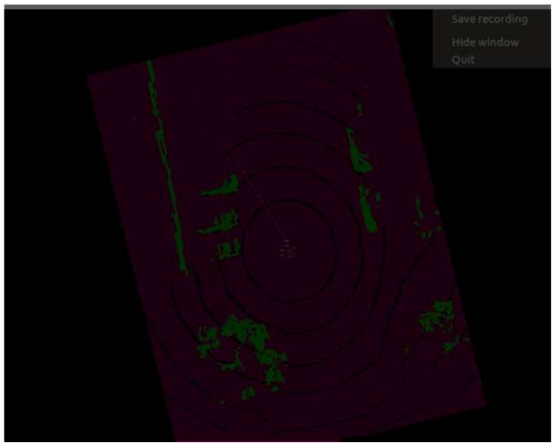
https://www.youtube.com/watch?v=Q1f-s6_yHtw

Autonomous Vehicle Perception

Localization & Mapping



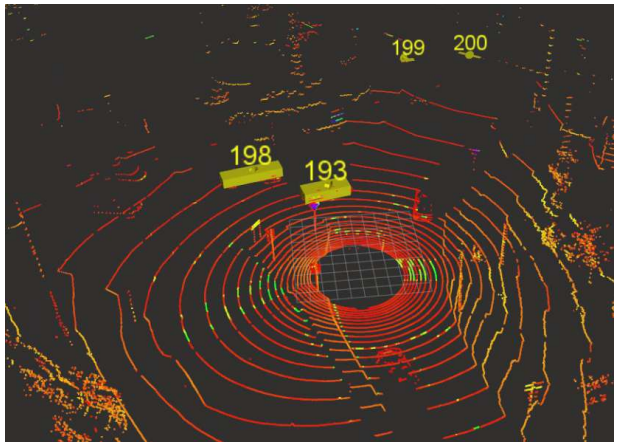
Occupancy Mapping



Static & Dynamic Object Detection

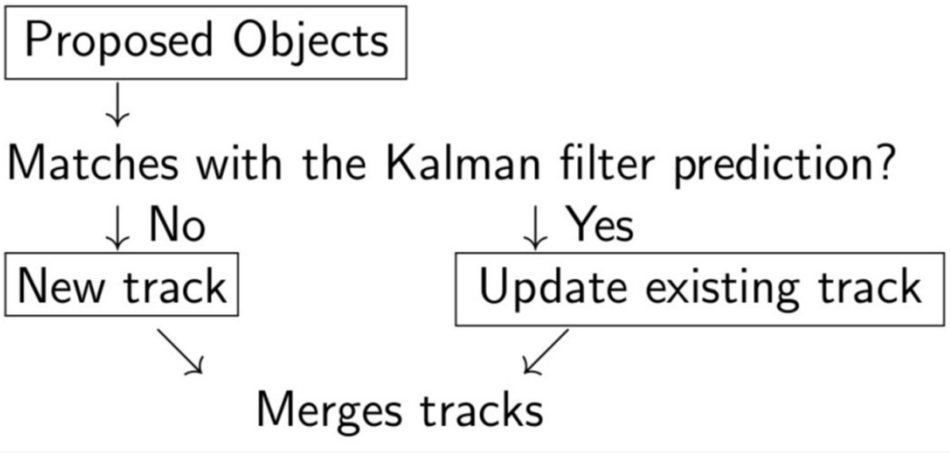


Dynamic Object Tracking

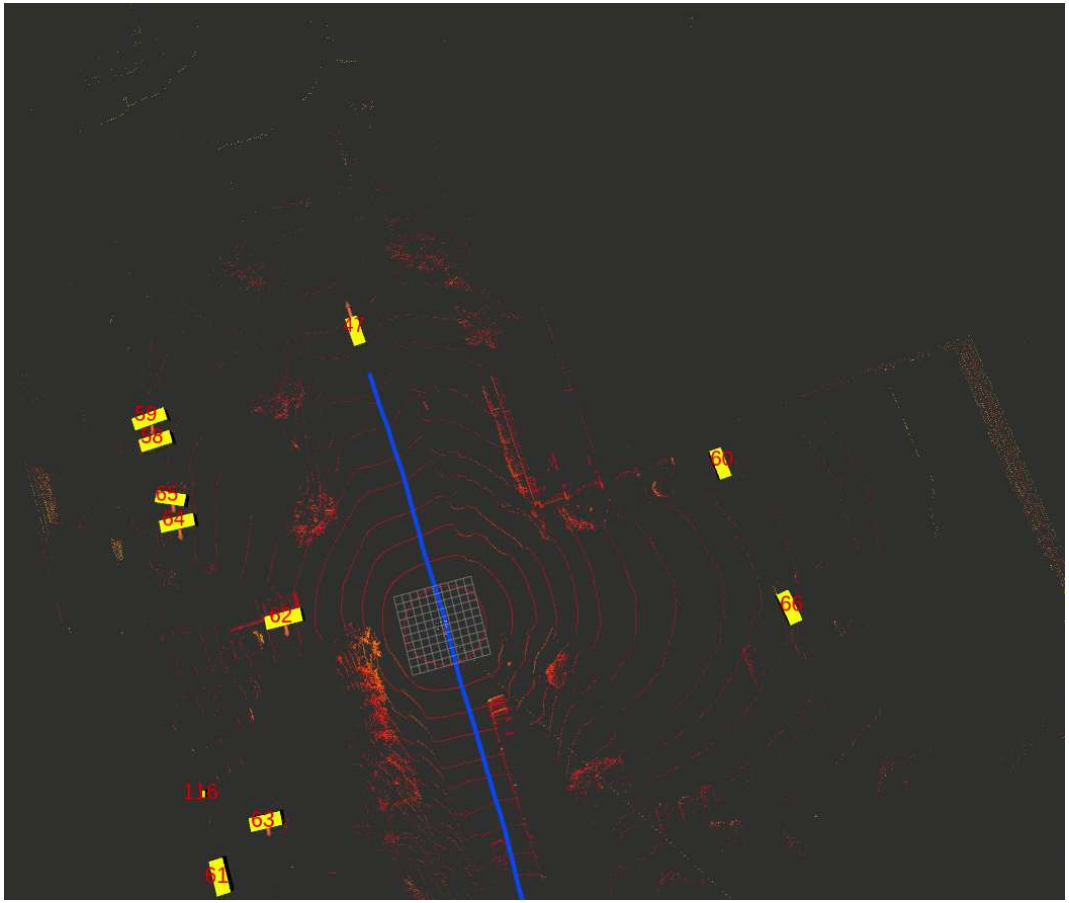


Dynamic Object Tracking

- Following the tracking-by-detection paradigm, tracker processes all outputs of a object detector.
- Vehicles and Pedestrians are treated separately.
- Includes two parts: motion estimation and object identification
  - The motion of each track is estimated by a 2D Cartesian space Kalman Filter.
  - Object identification and correspondence based on windowed matching



Multi-Object Tracking



Dynamic World Perception

- Self-driving Zoox
- Obelix robot
- Multi-UAVs

The current era – A wild ride for robotics

- Getting rebranded as Physical AI or Embodied AI has led to huge investment
- Bigger models and more data are making new things possible, but rarely live up to the demo video claims and are still not as general as we need
- LLMs, VLMs, MLLMs, VLAs, World Models, World-Action Models are everywhere
- Imitation Learning has replaced Reinforcement Learning as the go-to method to make progress on robotic manipulation tasks.
- Open world perception and agentic reasoning are steadily improving too.

Key Takeaways

- This course is evolving, all contributions and suggestions welcome

- Perception grows in complexity with the challenges of the environments
  - Ego motion
  - Static environments
  - Dynamic environments

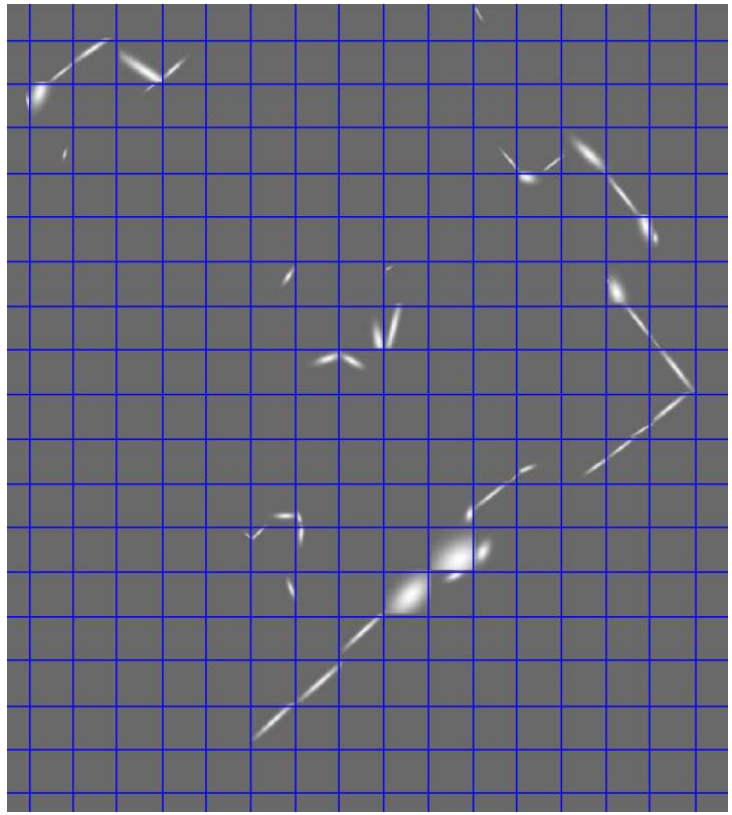
- Core concepts to be developed
  - Ego State: Sensor modeling and pre-processing
  - Static World: Point clouds and registration, point features and matching
  - Dynamic World: Identifying objects, tracking, predicting and scene understanding

Backup Slides

Normal Distributions Transform

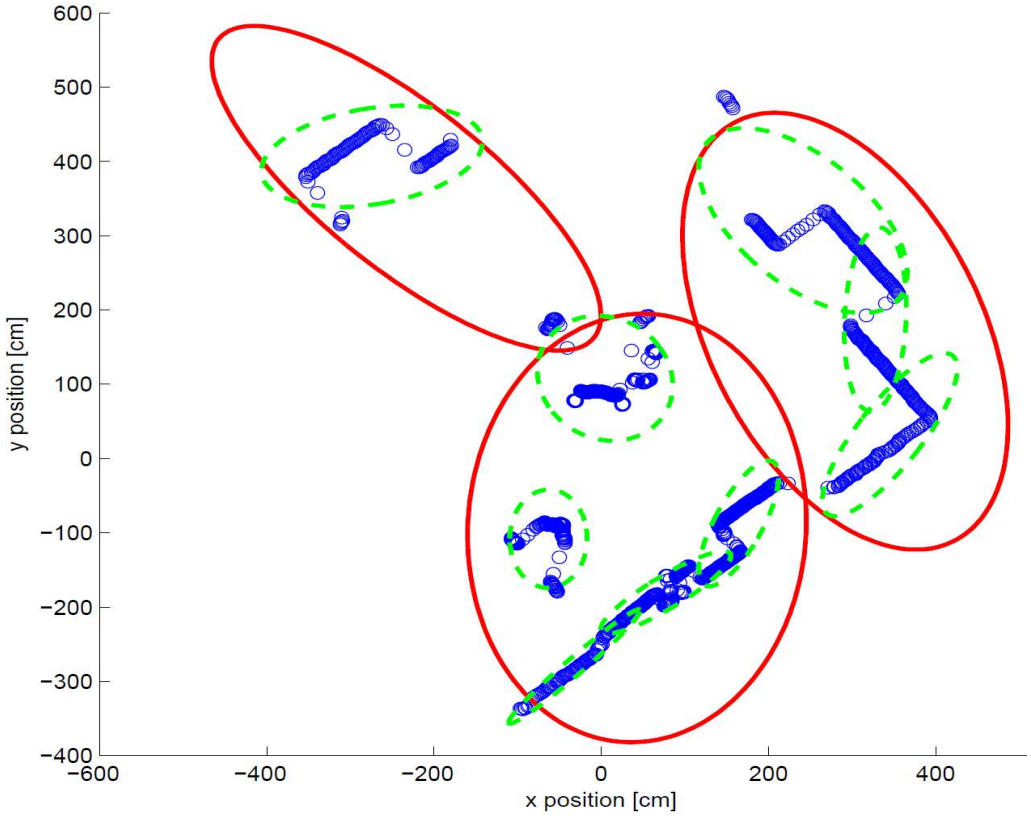
- The likelihood of having measured point y within cell c_i

$$\rho(y) = \exp\left(-\frac{(y-\mu_{c_i})^T \Sigma_c^{-1} (y-\mu_{c_i})}{2}\right)$$



Multi-Scale K-Means NDT

- Clustering of the scan at multiple scales
  - Led to stronger convergence properties, more accurate registrations
  - Many variations have since been proposed



Experiments

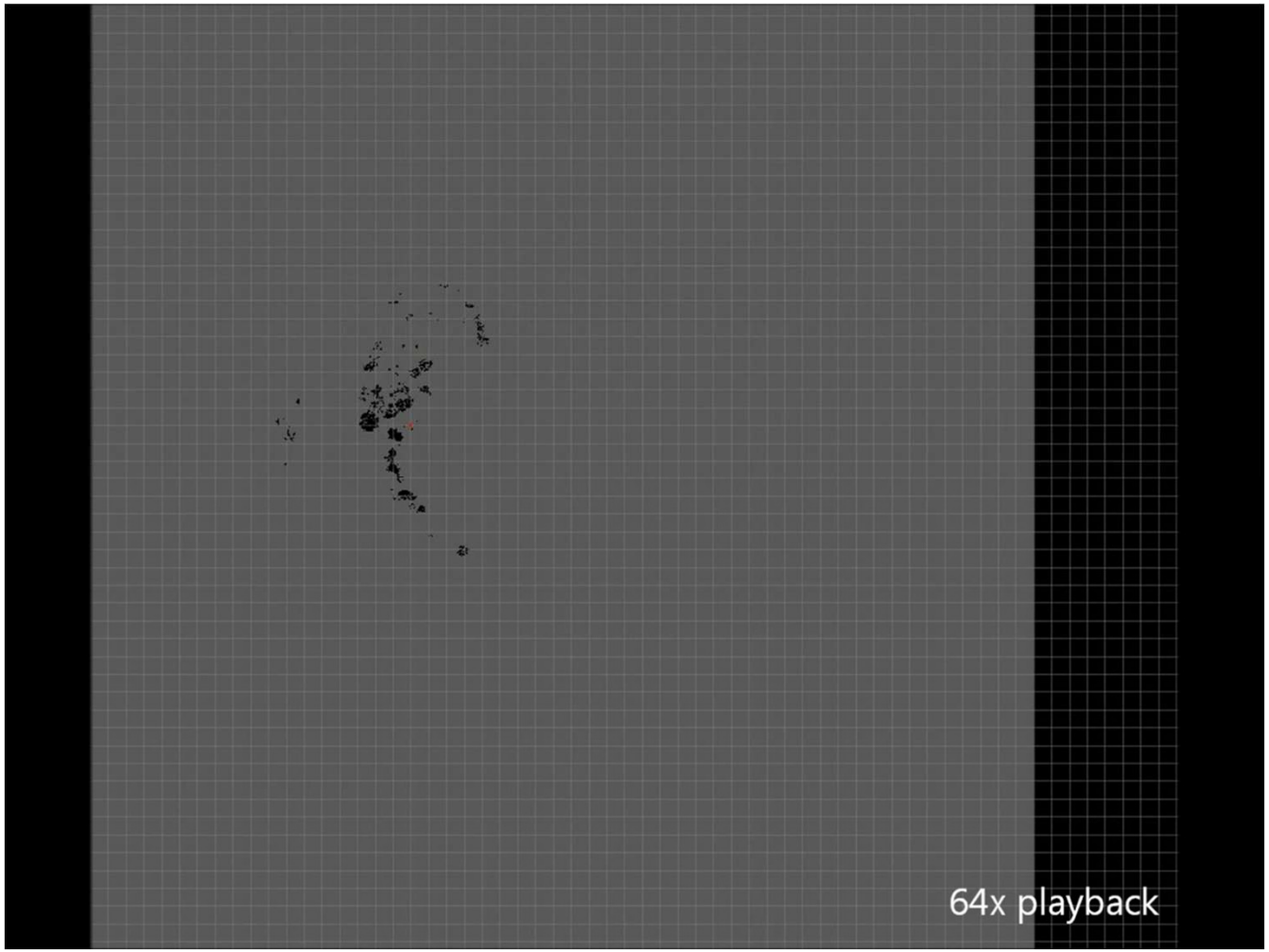
- Approach is tested using sweep path (important for coverage) and with large loop closures (important for navigation and obstacle avoidance)
- Data collected at Waterloo park (400m x 400m largest area)
- Ground truth collected using Robotic Total Station



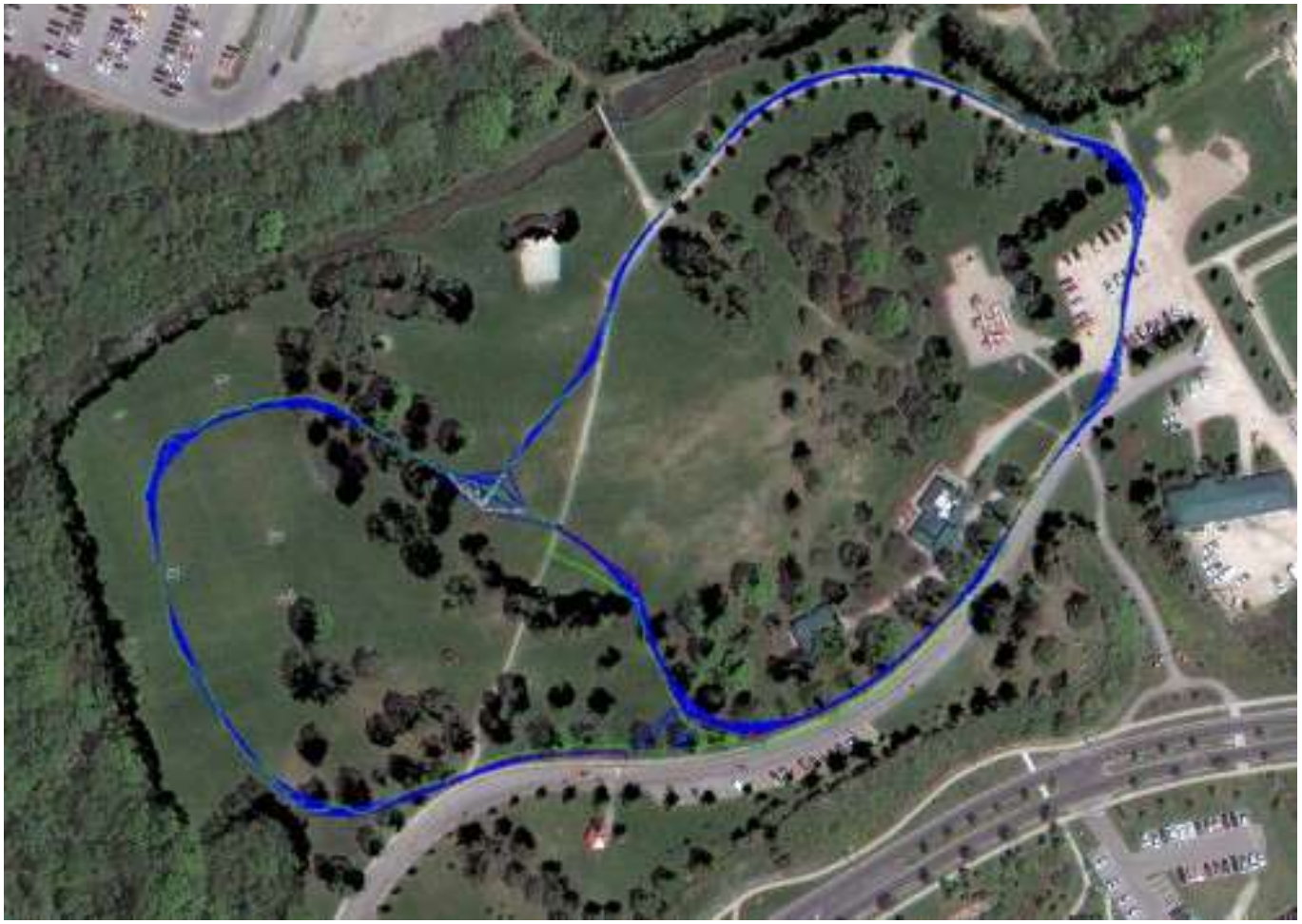




Results (Waterloo Park)



Results (Waterloo Park)



Multi-Camera Parallel Tracking and Mapping

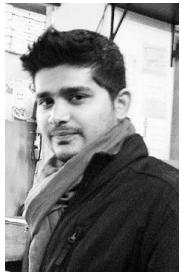

Inna Sharf


Adam Harmat

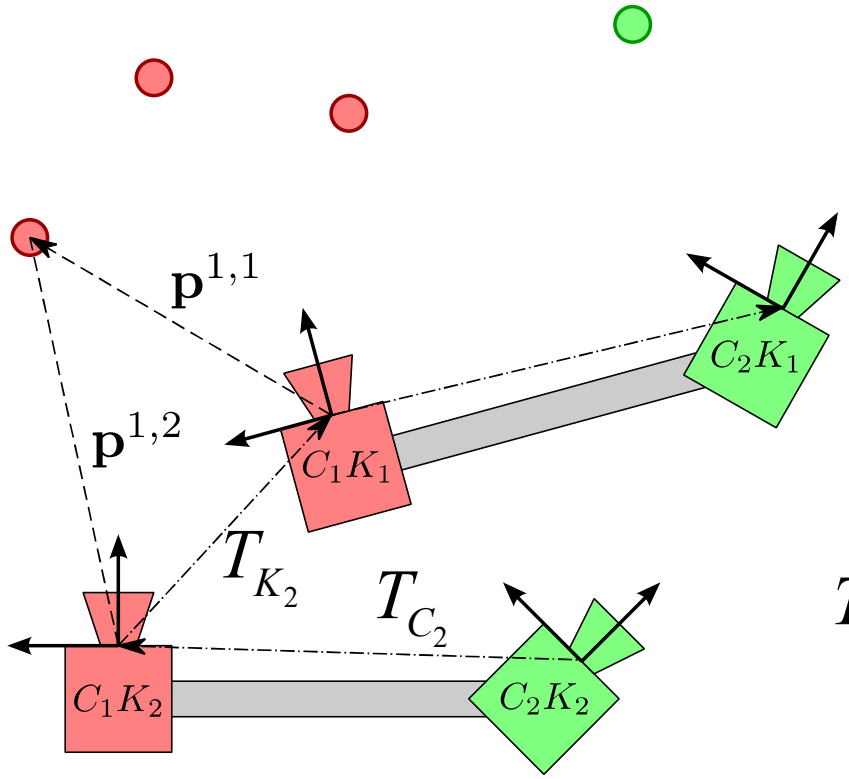

Mike Tribou


David Wang


Arun Das


Jason Rebello

Multi-Camera PTAM



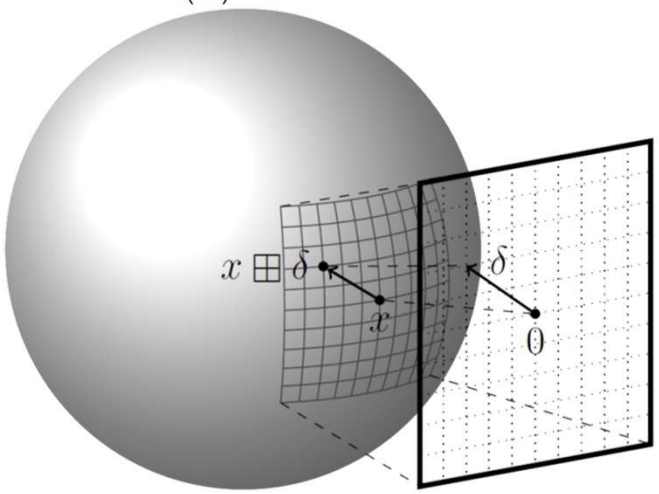
$$T_{K_2} = \begin{bmatrix} R_{K_2} & t_{K_2} \\ 0 & 1 \end{bmatrix}$$

$$T_{C_2} = \begin{bmatrix} R_{C_2} & t_{C_2} \\ 0 & 1 \end{bmatrix}$$

State Representation and Updates

- ⊞-Manifolds for singularity and constraint free updates
  - Define ⊞ operator which respects manifold topology
  - Perform optimization updates in local real vector space
  - Store poses as homogeneous transformations

- Keyframe pose and camera baseline using transforms on SE(3)
  - Exponential map from minimal representation of update on ℝ6 to SE(3)
  - Only addition is needed for graph optimization



Map Feature Points

- Spherical feature position parametrization
  - Defined in anchor keyframe where first observed
  - Focuses uncertainty in one parameter, unlike Cartesian
    - No overlap between cameras required, no IMU required
  - Avoids singularity in inverse depth parametrization at 180°
    - Suited for wide-FOV cameras

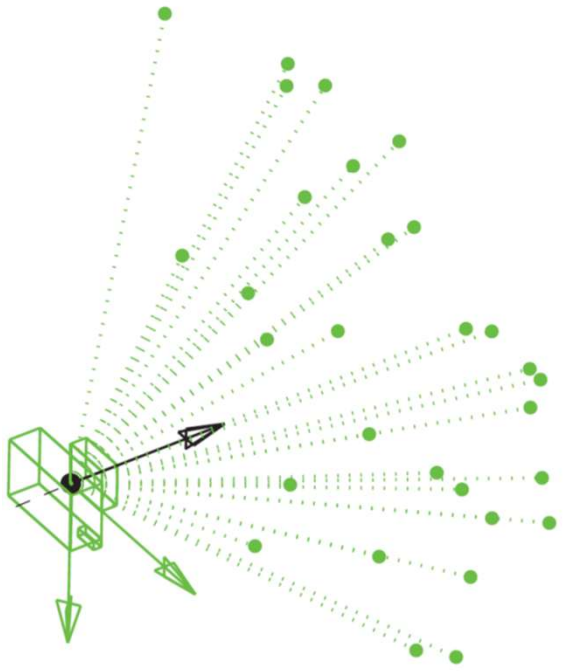
- For a point j in i^{th} keyframe seen by camera k

$$p_j^{i,k} = \begin{bmatrix} \theta_j \\ \phi_j \\ r_j \end{bmatrix}_{K_i}$$

Azimuth

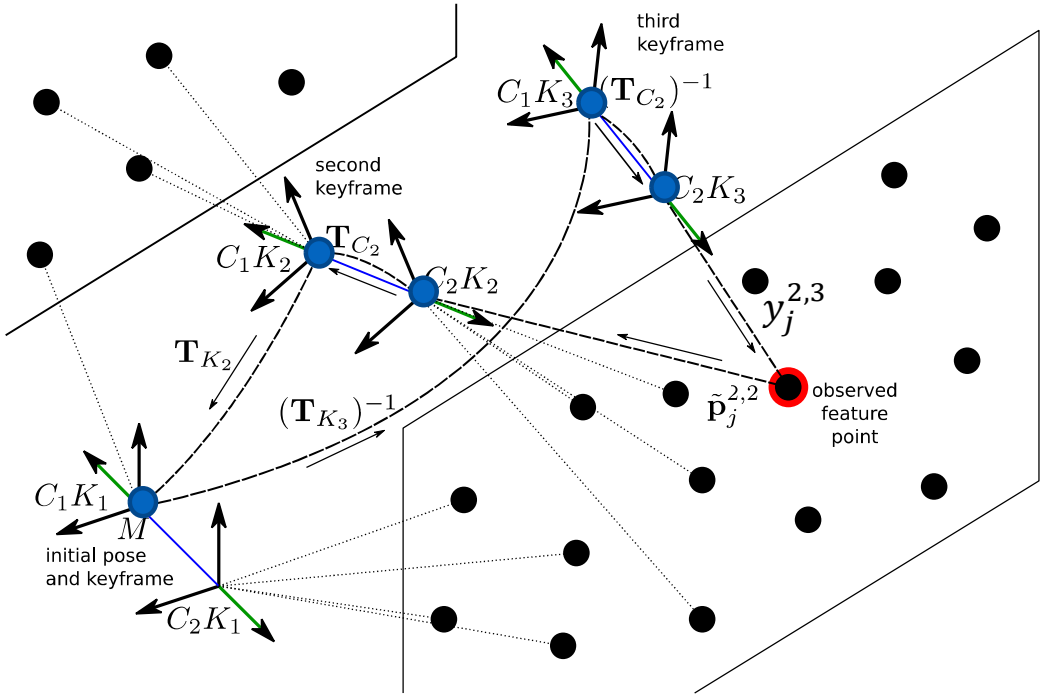
Elevation

Depth



Multi-Camera PTAM

- Features observed in multiple keyframes can be related to the feature anchor keyframe through the following sequence of transformations
  - Feature j observed in C_2 from keyframes K_2 (anchor) and K_3



$$p_j^{2,2}$$

$$T_{C_2} p_j^{2,2}$$

$$T_{K_2} T_{C_2} p_j^{2,2}$$

$$T_{K_3}^{-1} T_{K_2} T_{C_2} p_j^{2,2}$$

$$T_{C_2}^{-1} T_{K_3}^{-1} T_{K_2} T_{C_2} p_j^{2,2}$$

$$y_j^{2,3} = \pi_2 \left(K_2 T_{C_2}^{-1} T_{K_3}^{-1} T_{K_2} T_{C_2} p_j^{2,2} \right)$$

Multi-Camera PTAM

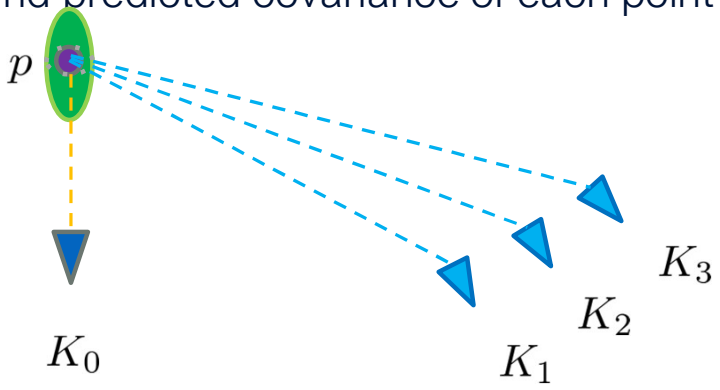
- Measurement equation – non-overlapping camera viewpoints
  - Features defined in first observed camera frame
  - Map defined in initial camera 1 frame

$$y_j^{2,3} = h\left(x_{K_2}^r, x_{K_3}^r, p_j^{2,2}\right) + \varepsilon_j^{2,3}$$

$$= \pi_2\left(K_2 T_{C_2}^{-1} T_{K_3}^{-1} T_{K_2} T_{C_2} p_j^{2,2}\right) + \varepsilon_j^{2,3}$$

Keyframe Selection for Map Maintenance

- A buffer of keyframes is maintained $\Phi = \{M_{k_0}, M_{k_0+1}, M_{k_0+2}, \ldots, M_k\}$
- For each possible keyframe, find predicted covariance of each point tracked by the tracker.



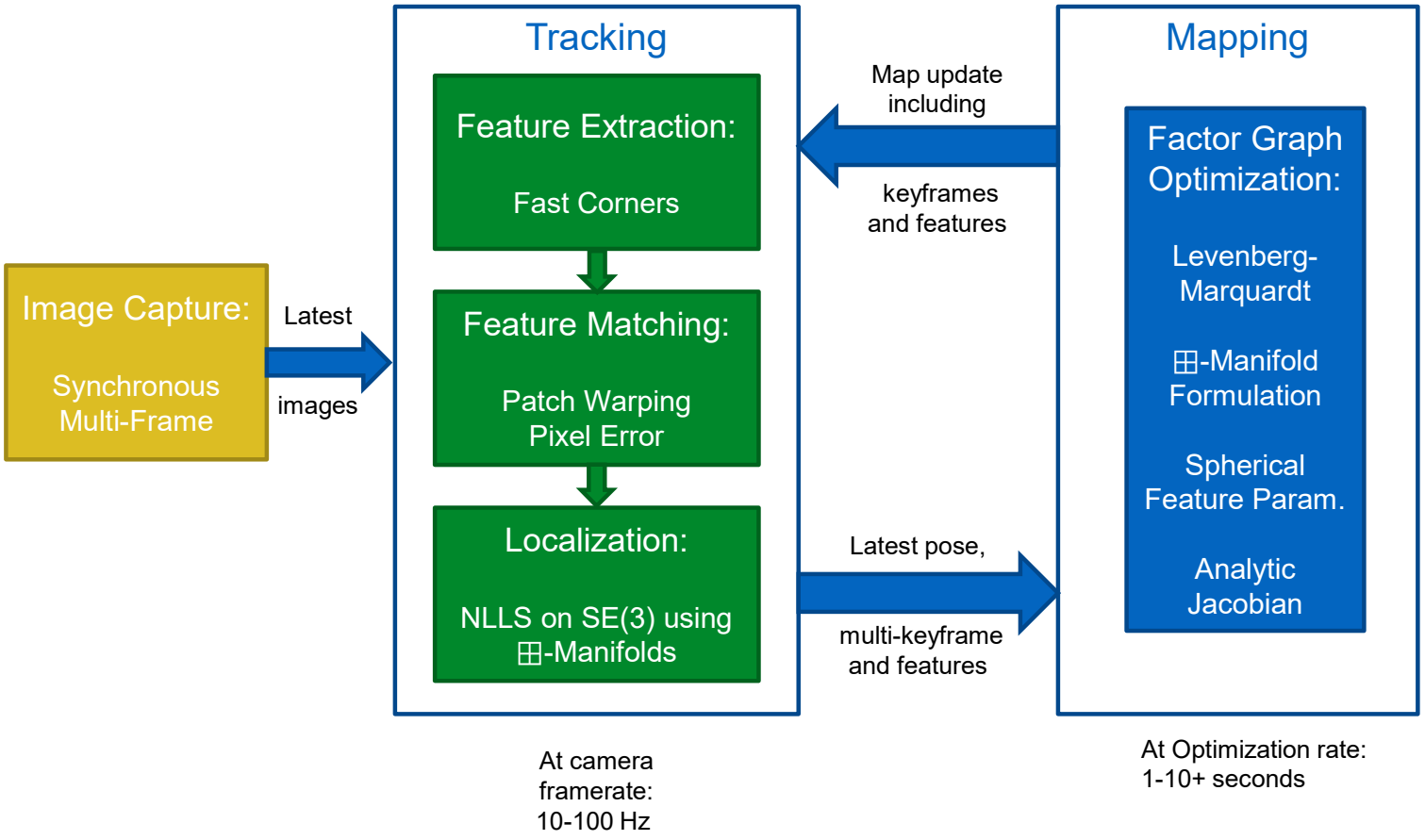
- The predicted point covariance is calculated as

$$J_{p_j} = \left.\frac{\partial h_j}{\partial p}\right|_{\hat{p}_j} \qquad \bar{\Sigma}_j = (\mathbf{I} - \Sigma_j \boldsymbol{J}_{\boldsymbol{p}_j}^T (\boldsymbol{J}_{\boldsymbol{p}_j} \Sigma_j \boldsymbol{J}_{\boldsymbol{p}_j}^T + \mathrm{R})^{-1} \boldsymbol{J}_{\boldsymbol{p}_j}) \Sigma_j$$

- Insert the keyframe which maximizes the total sum of point entropy reduction

$$M^* = \arg\max_{M_i \in \Phi} \sum_{p \in M_i} h_e(\hat{p}_j) - h_e(p_j) \qquad h_e(p_j) = \frac{1}{2} \ln\left((2\pi e)^n \left|\Sigma_j\right|\right)$$

MCPTAM Architecture

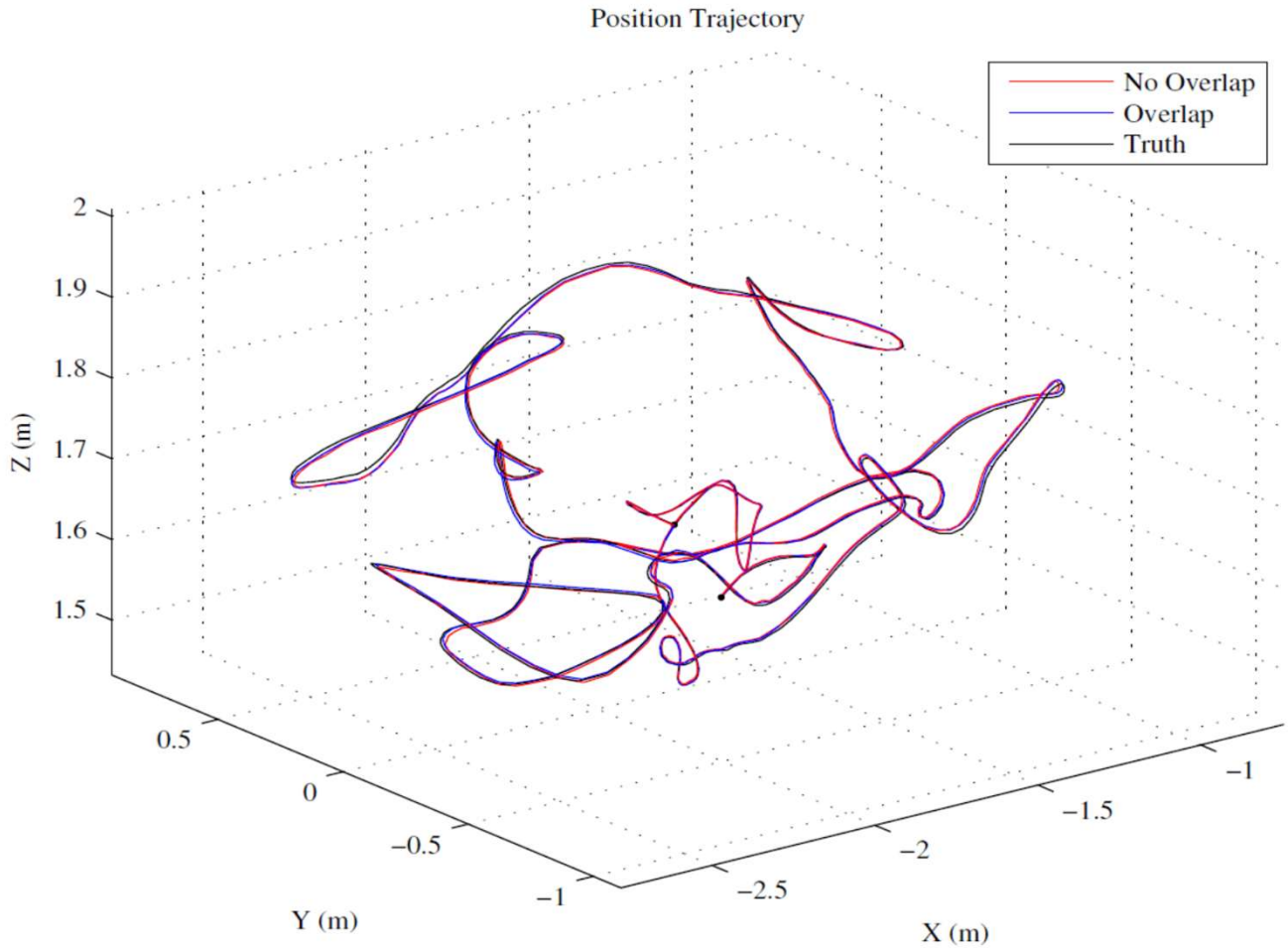


Code available at https://github.com/wavelab/mcptam

MCPTAM Indoor Results – Stable Map



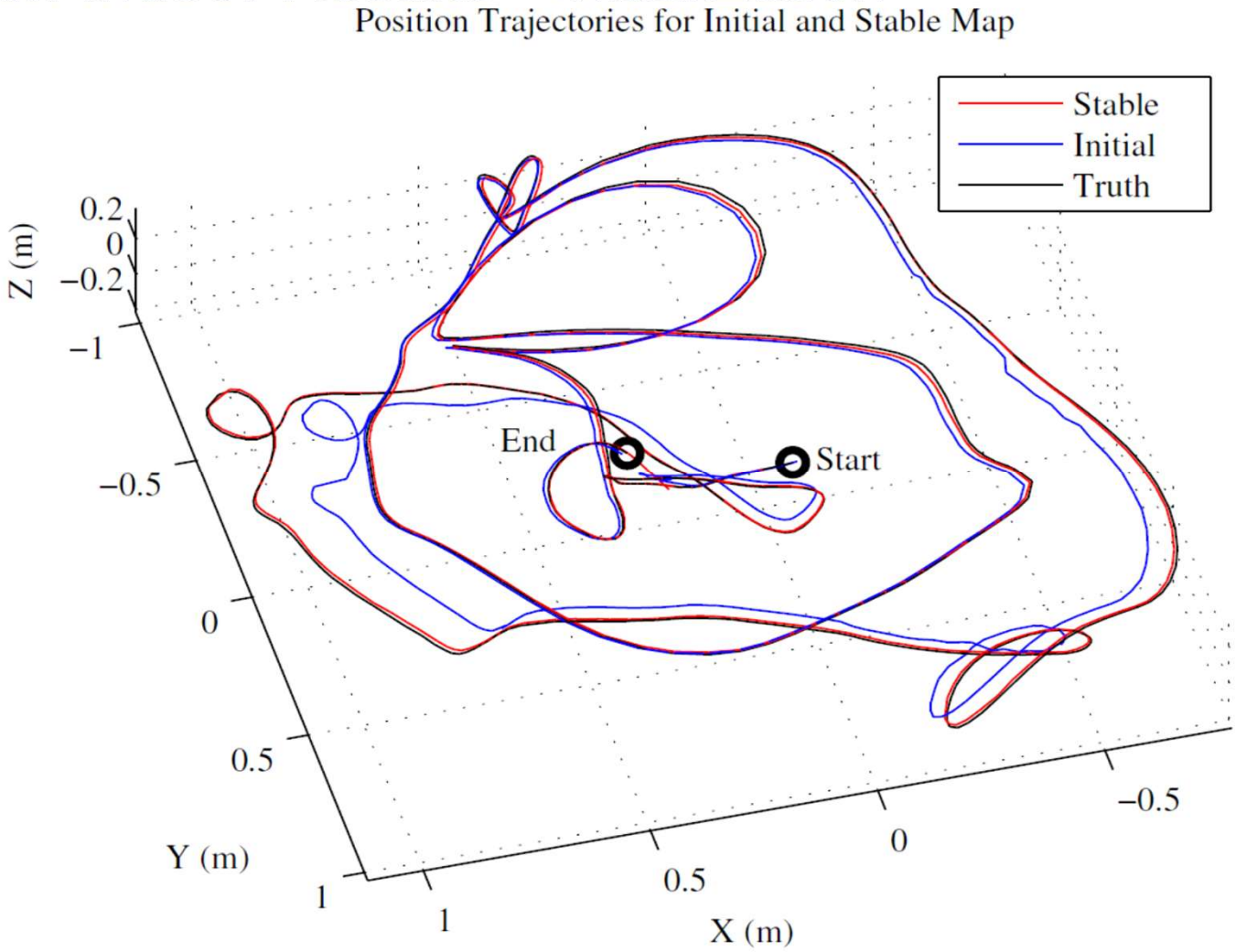
MCPTAM Indoor Results – Stable Map



MCPTAM Indoor Results - Initialization



MCPTAM Indoor Results - Initialization

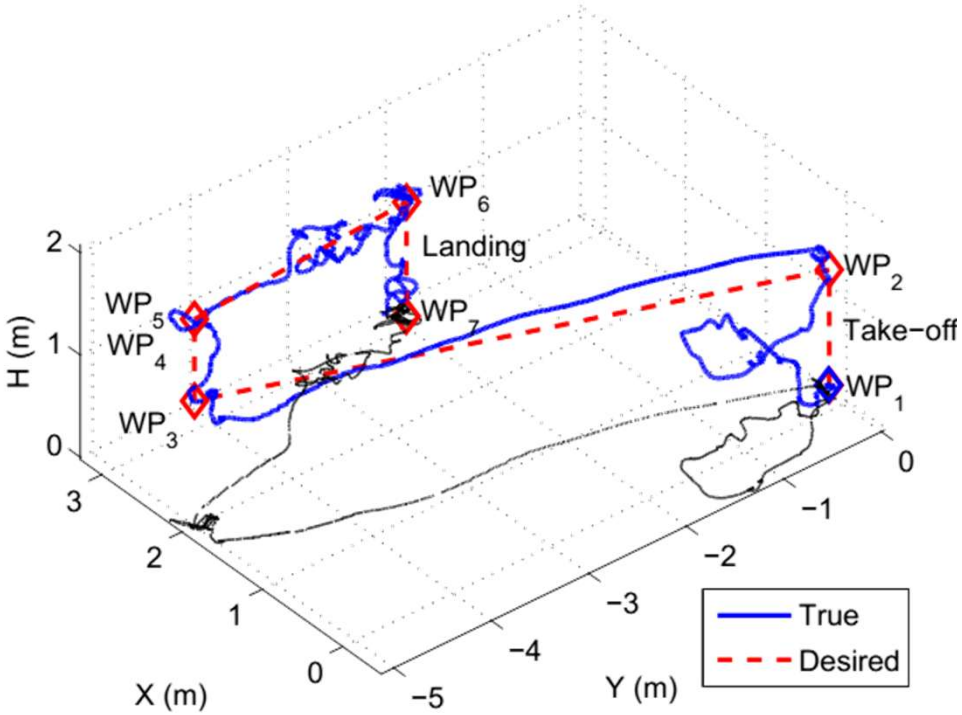


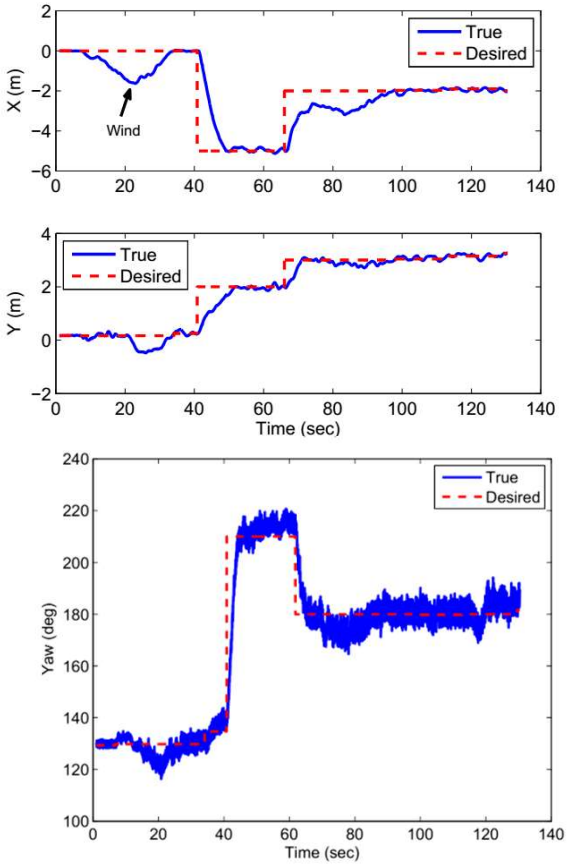
MCPTAM Outdoor Flight



Vehicle Control Using MCPTAM at 7 Hz

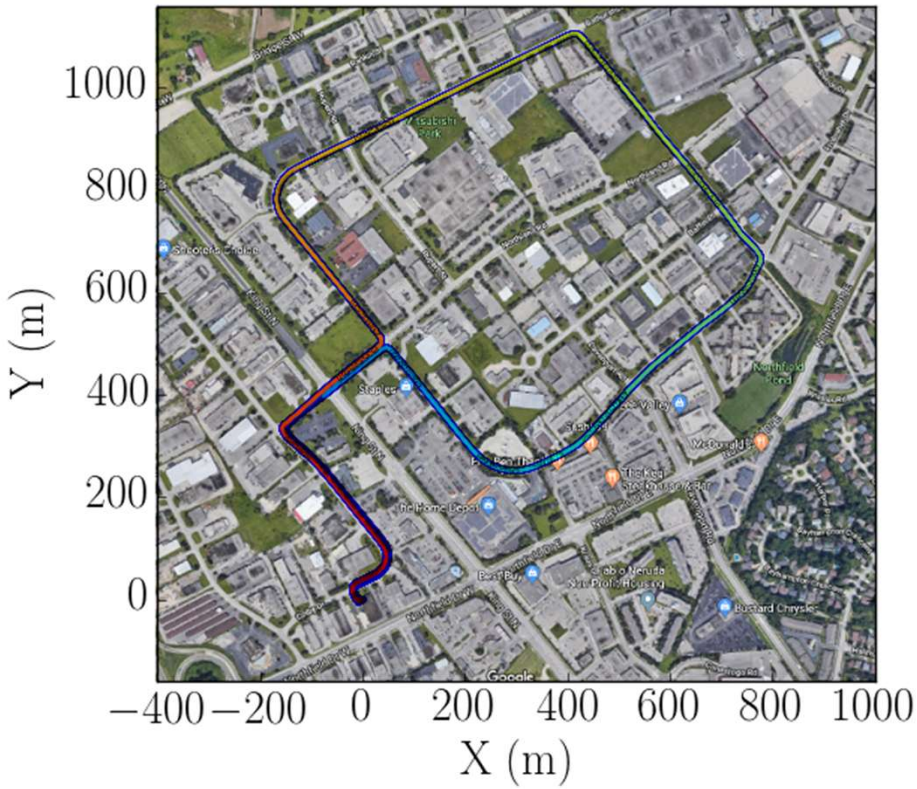
- Closed loop outdoor control using MCPTAM on Dragonflyer X8

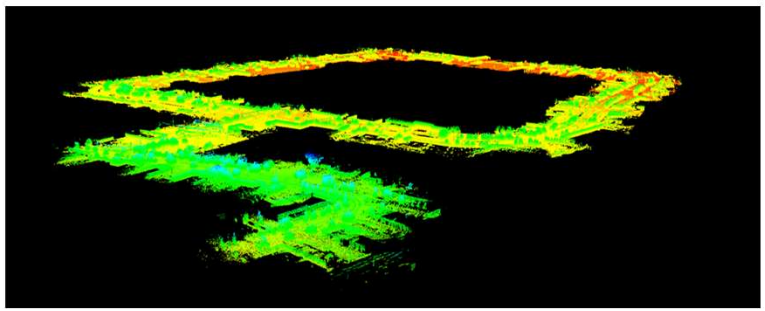




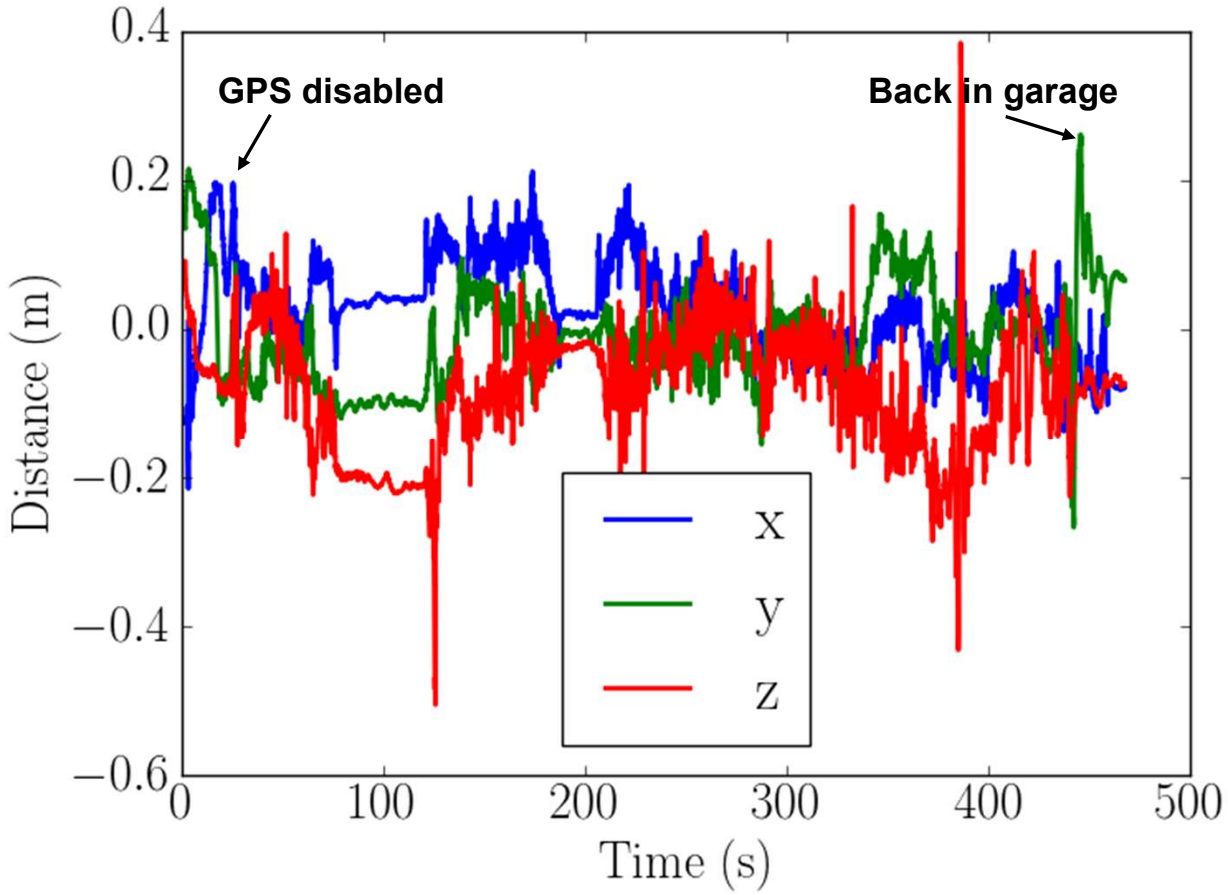
Evaluation

- 4 km loop, 10 m elevation range
- A map of the route was constructed on Ju 25
- Localizer was run on data from July 27
- GPS input was disabled shortly into the ru
- Ground truth prepared from July 27 data, using Inertial Explorer RTK solution with fixed base station



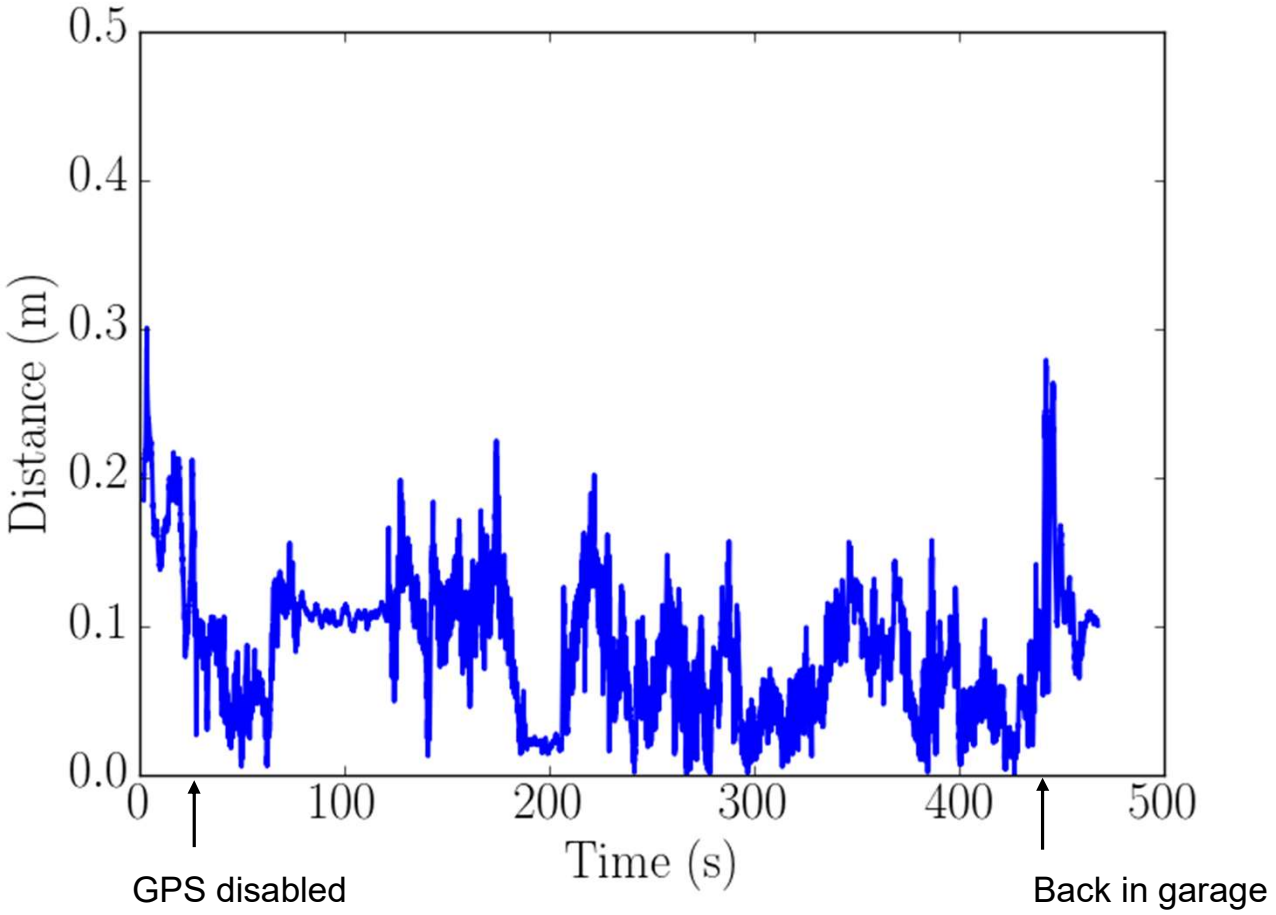


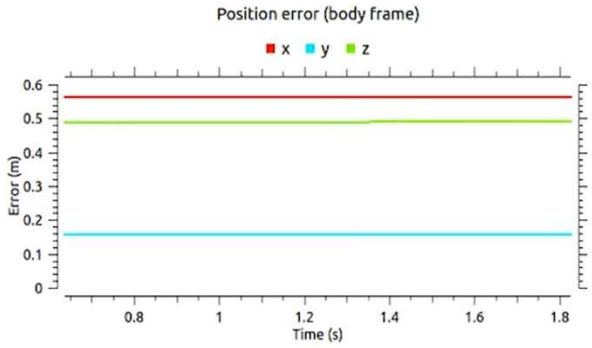
Translation error in the body frame

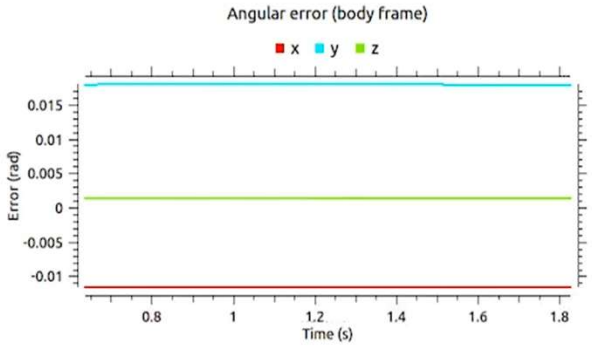


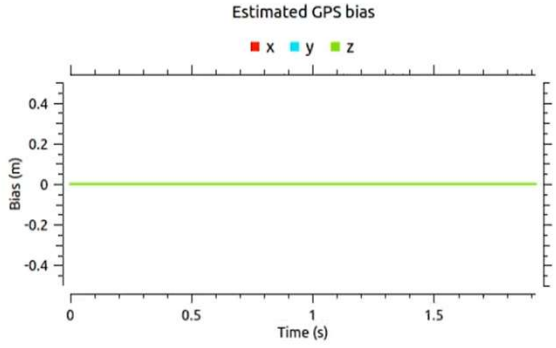
Horizontal translation error

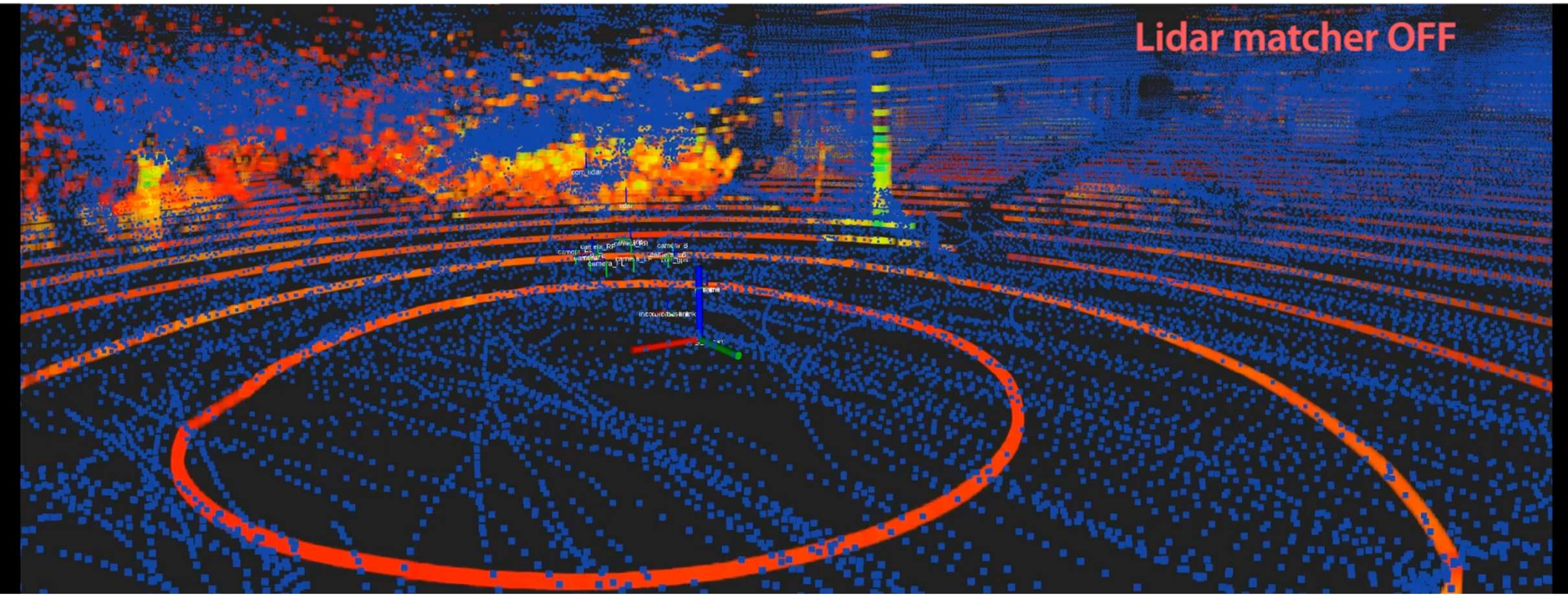
- RMSE between the marks is 0.090 m




Position error (body frame)
x y z
Error (m)
0.6
0.5
0.4
0.3
0.2
0.1
0
0.8
1
1.2
1.4
1.6
1.8
Time (s)

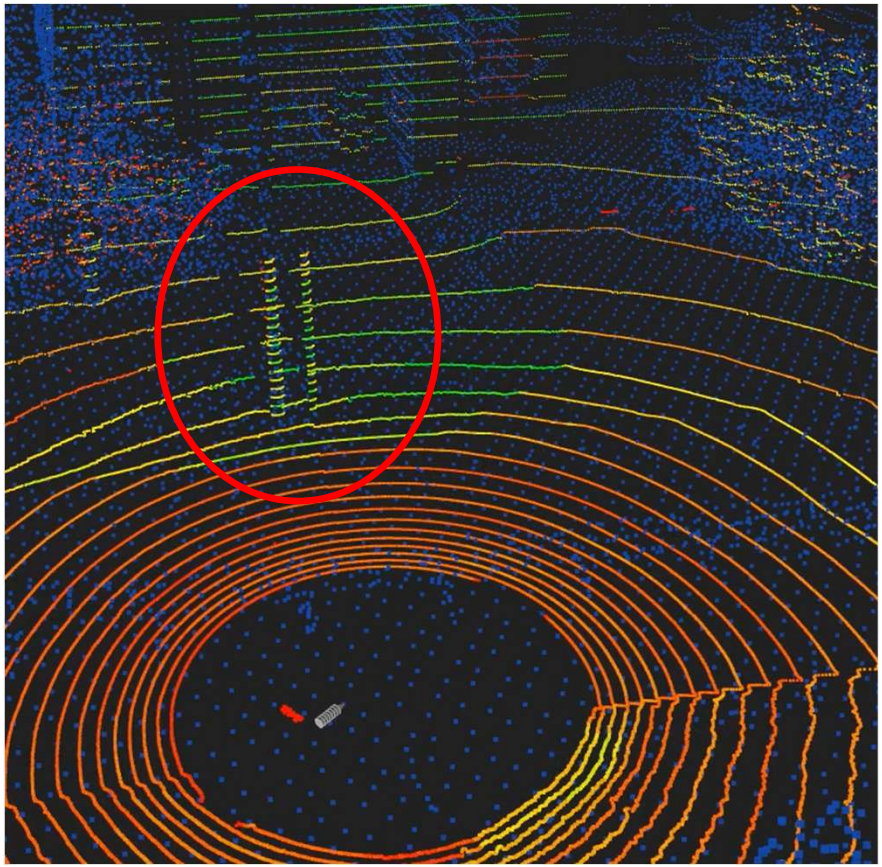

Angular error (body frame)
x y z
Error (rad)
0.015
0.01
0.005
0
-0.005
-0.01
0.8
1
1.2
1.4
1.6
1.8
Time (s)


Estimated GPS bias
x y z
Bias (m)
0.4
0.2
0
-0.2
-0.4
0
0.5
1
1.5
Time (s)

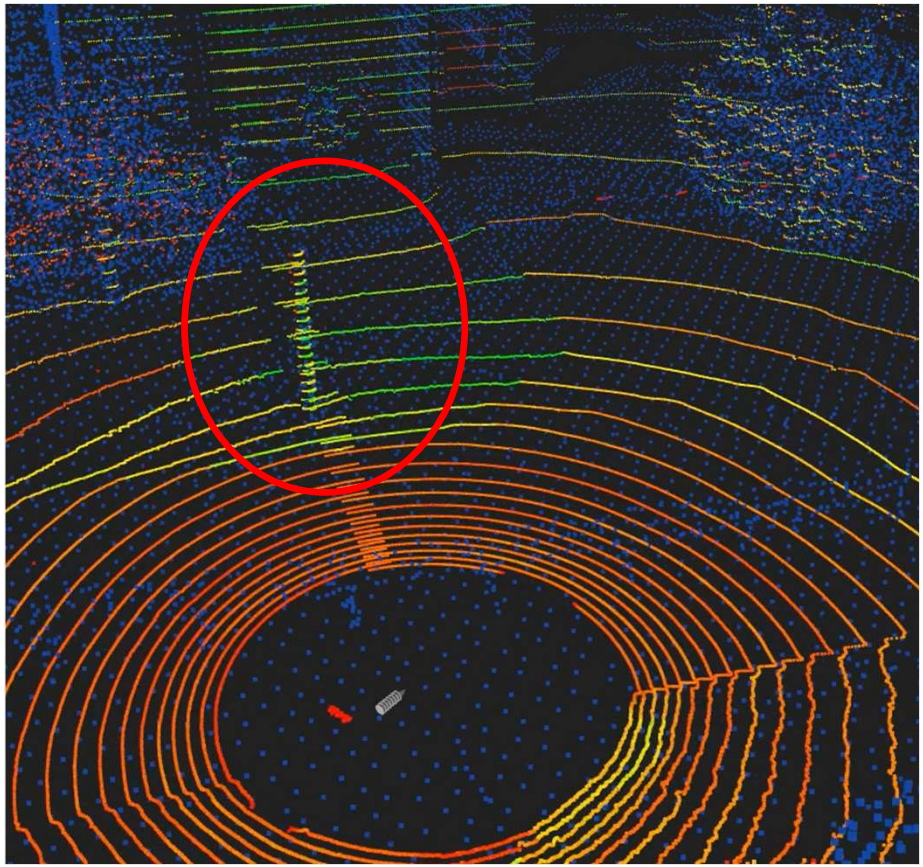

Lidar matcher OFF

Vehicle motion correction

without motion correction

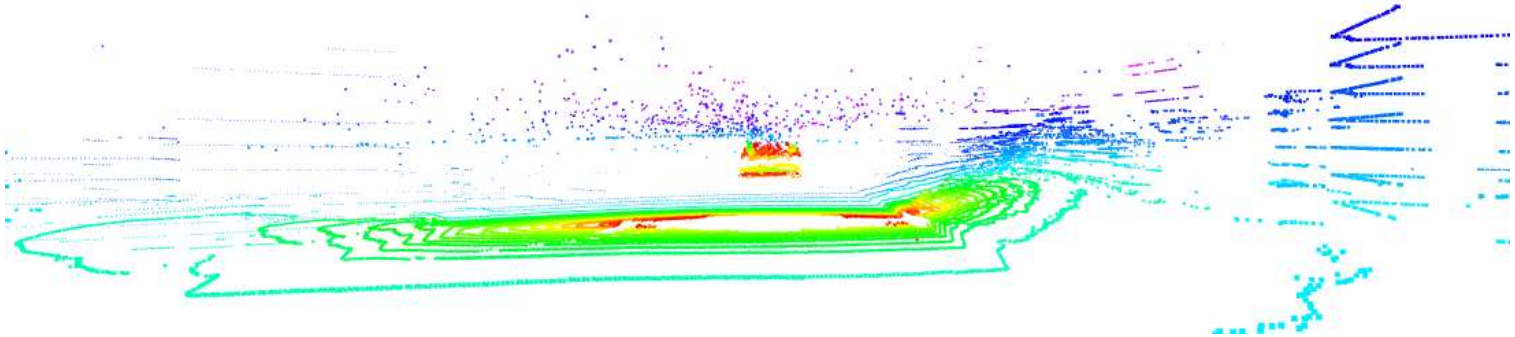


with motion correction

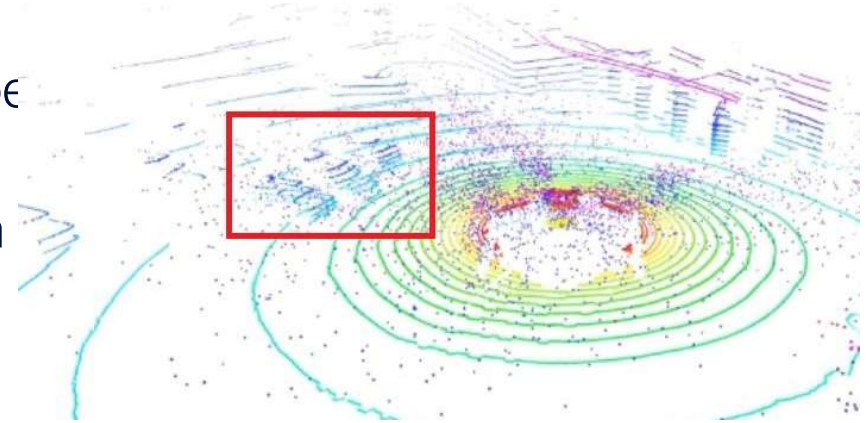


Real-Time Removal of Snow in Lidar Data

- Lidar captures falling snow, causing noisy point clouds

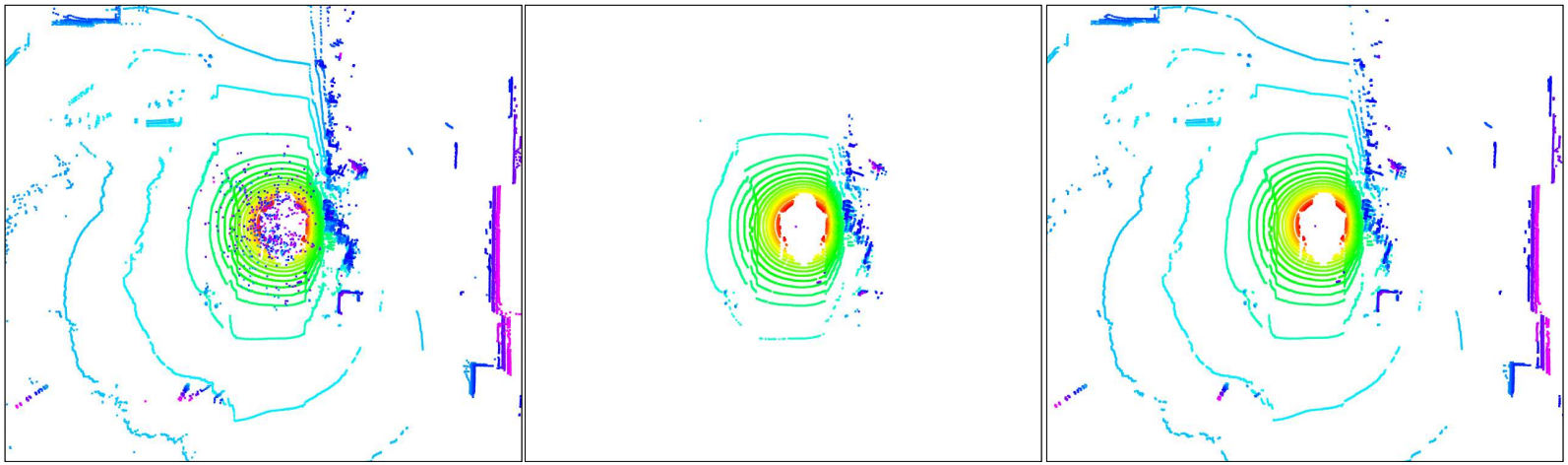


- Noisy point clouds can hinder the ability to pe
  autonomous driving tasks
  (e.g. mapping, localization, object detection



Real-Time Removal of Snow in Lidar Data

- Results



Raw Point Cloud

Radius Outlier Removal (ROR)

DROR Filter

Filtering of Snowfall in Lidar Point Clouds

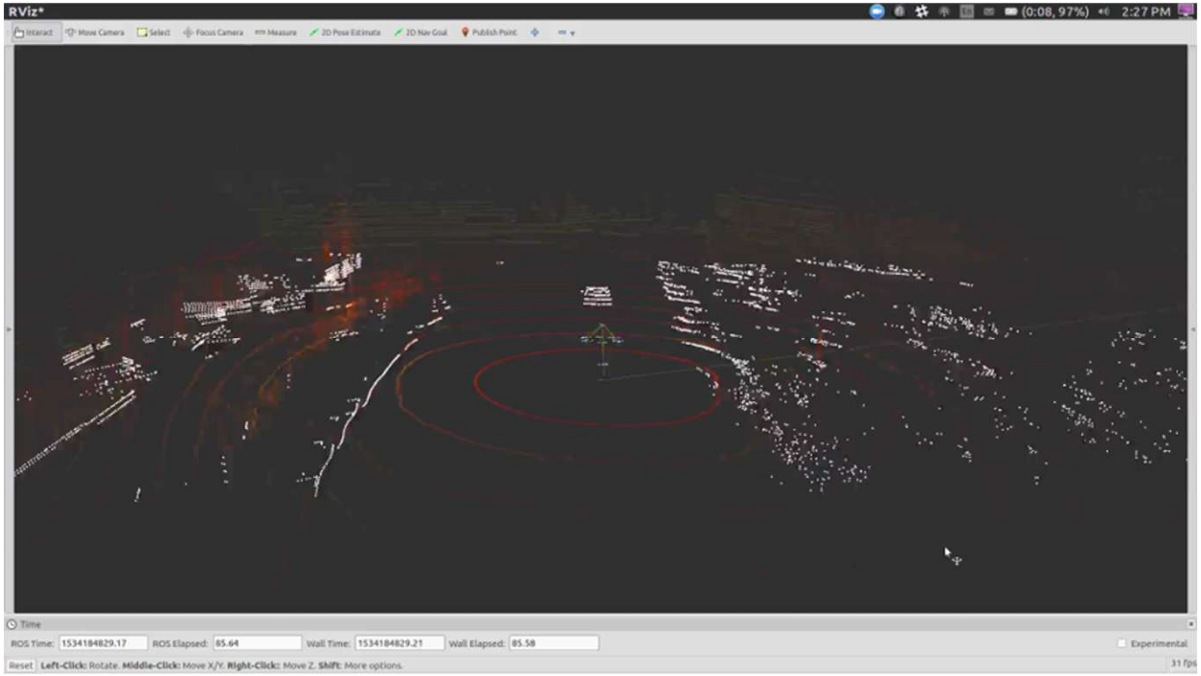
Winter 2018 Results





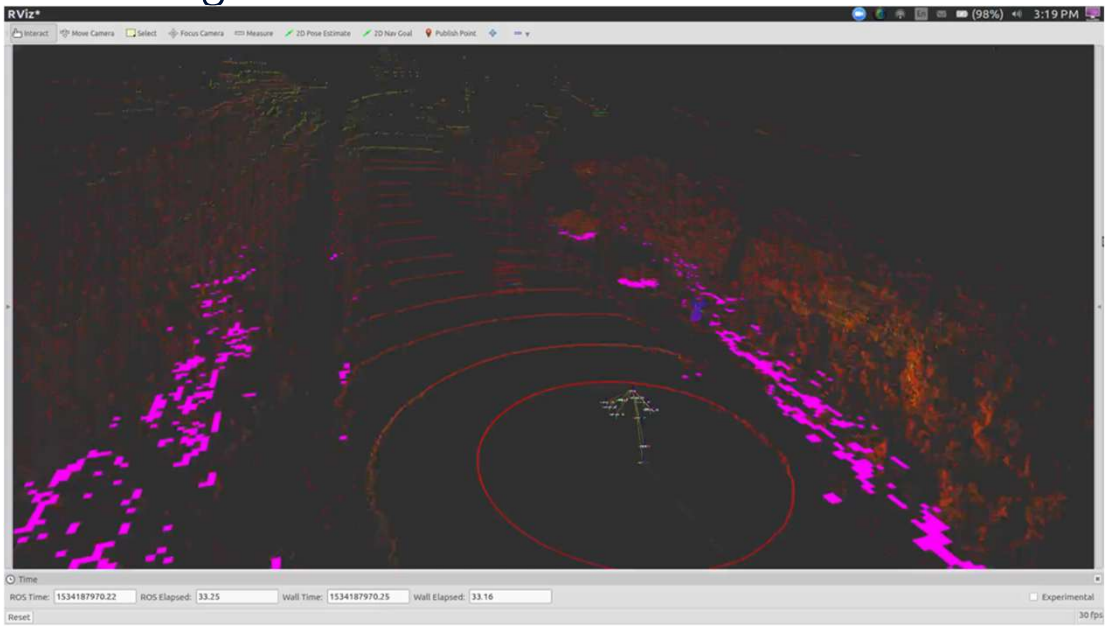
Preparing the obstacle Point Cloud

- Crop & Voxelize
- Transform from Lidar to base link frame
- Ground segmentation implementation used to create obstacle point cloud
- Outliers removed through radius outlier removal



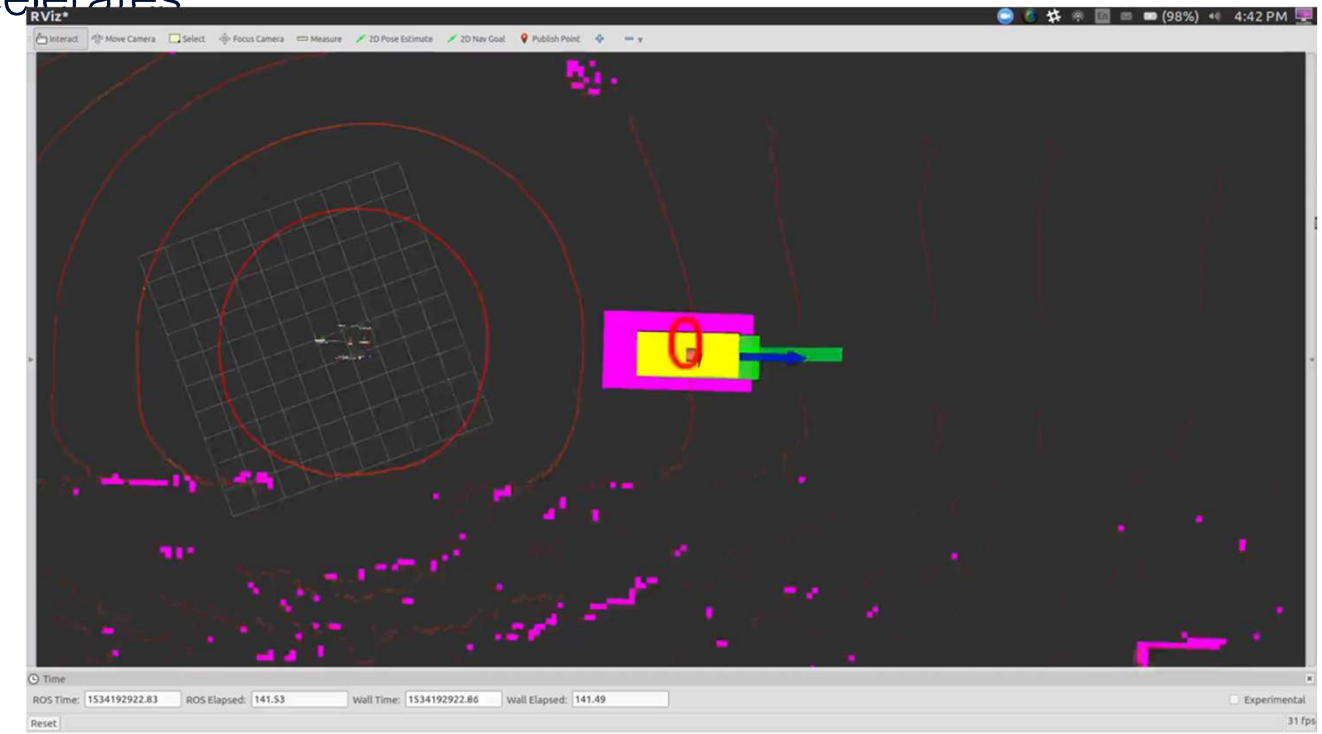
Updating the occupancy grid

- Transfer old map cell values into the new map cells
  - Old map cell values are faded
- Probabilistically update map cells with given pointcloud
- Map stored is in log odds



Tracked object removal

- Find cells occupied by the tracked object and set the values to 0
- Store previous car states to better remove cells when a track object accelerates or decelerates



Input processing

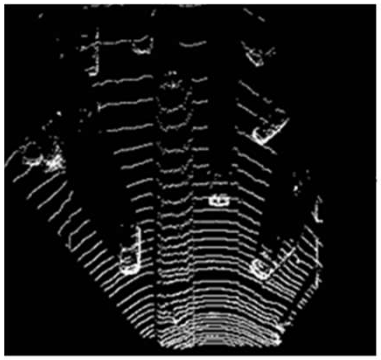
- Kitti Dataset Images
  - RGB images at 1392 X 512
  - Uniform lighting
  - City driving
  - Summer conditions

- Kitti Lidar Data
  - Velodyne HDL-64E
  - Converted to 6 channel Bird's Eye View Map
    - 0.1 m grid cells with 5 horizontal slices
    - Maximum point height per horizontal slice
    - 6th channel is grid cell point density

Image Input



BEV Input



Second stage regression

- Second stage performs classification, bounding box offset and orientation regression
  - Orientation encoded as [cos ϴ, sin ϴ]
- Bounding box representation had a significant impact on regression performance
  - 4% improvement from 8 to 4 corner + heights.
  - Ambiguity in corner ordering likely culprit
  - Anchors designed based on class statistics to minimize second stage effort

